

Chapter 21

Peptide Mapping by Thin-Layer Chromatography and High Voltage Electrophoresis

Keith Gooderham

*MRC Clinical and Population Cytogenetics
Unit, Western General Hospital, Crewe
Road, Edinburgh, United Kingdom*

Introduction

Thin-layer chromatography and electrophoresis, either separately or in combination, provide a simple, high resolution technique for producing peptide maps. Recently, thin-layer methods have tended to be increasingly overshadowed by the development of high performance liquid chromatography (HPLC). Although thin-layer methods are generally not as sensitive as HPLC (requiring milli- or microgram amounts of protein instead of the nanogram or less quantities used in HPLC), they are still preferred for many applications. Thin-layer peptide

mapping not only has the advantage of being relatively inexpensive, but it is also a very simple technique and it is often possible to analyze a large number of different samples in a single experiment. In addition, a number of different stains can be used for the detection of specific amino acids and these considerably increase the value of this method.

Although thin-layer methods are probably most widely used in comparative peptide mapping experiments (*see*, for example, Figs. 1 and 2), they also find a variety of other applications including. (i) monitoring enzyme and chemical cleavages of proteins; (ii) the preparative isolation of peptides for microsequencing studies (1), (iii) assaying peptide purity following preparative paper chromatography/electrophoresis, ion-exchange and gel filtration chromatography, and so on—a single band on a thin-layer plate, together with a single *N*-terminal amino acid, usually being a good indication that the sample contains only one peptide.

Two different methods are used in thin-layer peptide mapping experiments, namely high voltage electrophoresis and chromatography. In high voltage electrophoresis experiments, peptides are fractionated primarily upon the basis of charge, whereas the hydrophobic properties of the peptides are more important for chromatographic separations. Both methods may be used independently to produce one-dimensional peptide maps or alternatively these methods may be combined to produce two-dimensional peptide maps. In the latter case, it is usually necessary to first produce a series of one-dimensional peptide maps in order to establish the optimum separation conditions for the peptides under investigation. Once these conditions have been established, they should be rigorously adhered to in a given series of experiments, in order to ensure that comparable peptide maps are obtained, and ideally all of the samples should be prepared at the same time and as far as possible worked up in parallel.

Materials

1. Preprepared thin-layer cellulose plates (20 cm × 20 cm × 0.1 mm) can be obtained from a number of suppliers.

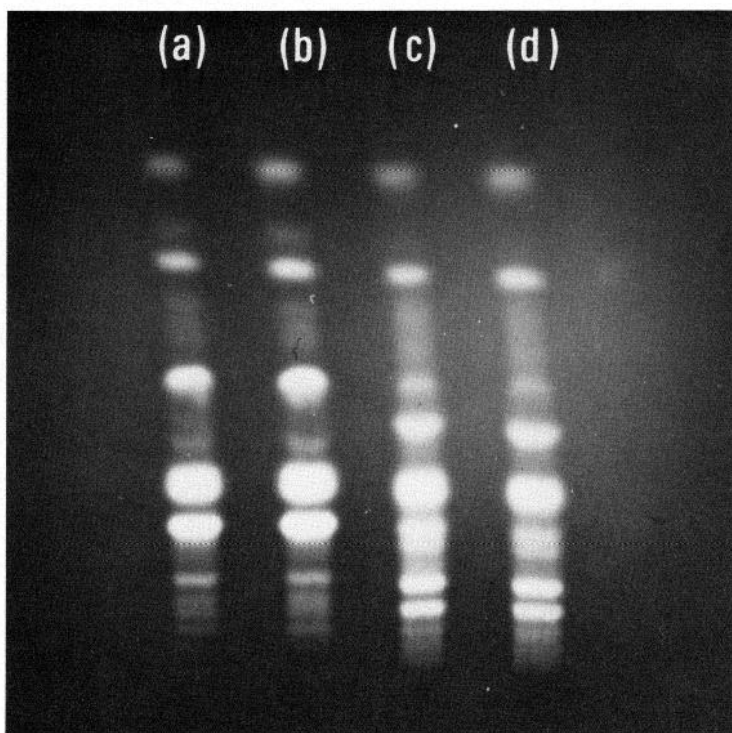


Fig. 1. Comparative one-dimensional tryptic peptide maps of pig and calf thymus HMG 1 and HMG 2 chromosomal proteins after staining with phenanthraquinone. Protein samples (a) pig HMG 1; (b) calf HMG 1; (c) pig HMG 2; and (d) calf HMG 2 at 5 mg/mL in 0.2M ammonium bicarbonate solution were digested with trypsin at an enzyme to substrate ratio of 1:50 (w/w) for 48 h at 37°C. Peptides were then separated by ascending thin-layer chromatography in a mixture of butan-1-ol, acetic acid, water and pyridine (15:3:10:12, by volume) and arginine containing peptides were detected by staining with phenanthraquinone. HMG 1 and 2 proteins are two very closely related chromosomal proteins [~80% homology (11)]. However, when tryptic peptides from these proteins are separated by one-dimensional thin-layer chromatography followed by specific staining for arginine-containing peptides the two proteins are clearly shown to be different.

For thin-layer chromatography, either glass or aluminum supports are equally acceptable, but for thin-layer high voltage electrophoresis only glass-backed plates can be used.

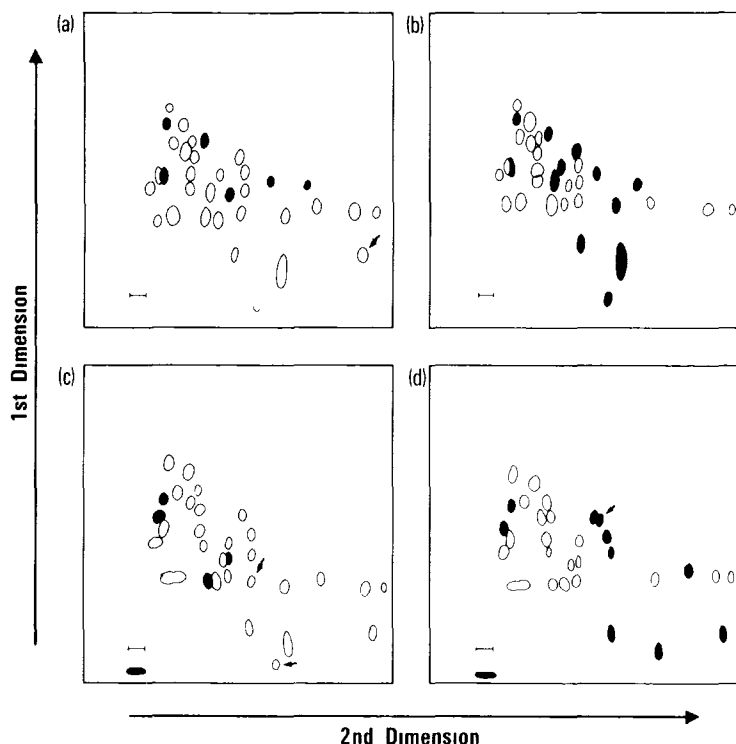


Fig. 2. Comparative two-dimensional tryptic peptide maps of pig and calf thymus HMG 1 and HMG 2 chromosomal proteins. Protein samples (a) pig HMG 1, (b) calf HMG 1, (c) pig HMG 2, and (d) calf HMG 2 at 5 mg/mL in 0.2M ammonium bicarbonate solution were digested with trypsin at an enzyme to substrate ratio of 1:50 (w/w) for 48 h at 37°C. Peptides were then separated by thin-layer high voltage electrophoresis in pH 6.5 buffer at 2.5 kV for 25 min (1st dimension). After drying the thin-layer plate the peptides were separated (2nd dimension) by ascending chromatography in a mixture of butan-1-ol, acetic acid, water, and pyridine (15:3:10:12, by volume). The peptides were detected by staining with ninhydrin-cadmium acetate solution and their positions were recorded on tracing paper. Although the primary sequences of HMG 1 and 2 proteins are very similar [80% homologous (11)] when the peptide maps for these proteins are examined [i.e., compare (a) with (c) and (b) with (d)] marked differences between these proteins are observed (peptides unique to either HMG 1 or 2 are shown as solid spots). Similarly, although the primary sequences of these proteins are clearly very conserved [compare (a) and (b) and also (c) and (d)] species specific differences (shown by arrows) are clearly apparent using this method (cf., Fig. 1).

2. Chromatography tanks capable of accepting two or more plates can be obtained through most laboratory supply houses. Ideally these tanks should be made of glass in order to avoid any possible reaction between the tank and the various solvents which will be used. In order to ensure an air-tight seal between the tank and the lid, the rim of the tank should be lightly coated with silicone grease.
3. High voltage electrophoresis apparatus and power supplies are available from a number of suppliers. Probably the most versatile of these is the Model L24 and associated power supply manufactured by Shandon Southern Instruments. All of the procedures described below are intended for use with this instrument, but they should be equally suitable for use with other instruments with little or no modification.
4. Aerosol spray guns can be obtained from most laboratory supply houses.
5. Wherever possible, electrophoresis buffers are prepared with analytical grade reagents and distilled water is used throughout. All of these buffers should be stored at room temperature and they can be kept for up to 6 months without any apparent deterioration. Some suitable electrophoresis buffers are listed below.
 - (a) pH 6.5 buffer is prepared by mixing together pyridine, acetic acid, and water in the following proportions 25:1:225 by volume.
 - (b) pH 4.8 buffer is prepared by mixing together pyridine, acetic acid, and water in the following proportions 3:3:394 by volume.
 - (c) pH 3.5 buffer is prepared by mixing together pyridine, acetic acid, and water in the following proportions 1:10:189 by volume.
 - (d) pH 2.0 buffer is prepared by mixing together acetic acid, formic acid (98% v/v), and water in the following proportions 8:2:90 by volume
6. Wherever possible, all chromatography solvents are prepared with analytical grade reagents and distilled water is used throughout. All solutions should be stored at room temperature and they can be kept for up to 6 months without any apparent deterioration. Some suitable chromatography solvents are listed below:

- (a) Butan-1-ol:acetic acid:water:pyridine ("BAWP") are mixed together in the following proportions 15:3:10:12 by volume.
 - (b) Butan-1-ol:acetic acid:water are mixed together in a separating funnel in the following proportions 4:1:5 by volume. After allowing the phases to separate, the lower phase is poured off and discarded while the upper phase is retained.
 - (c) Butan-1-ol:urea:acetic acid are mixed together in the following proportions 4.5:1 (v/w/v). (This solution should be freshly prepared for each experiment.)
- 7 For preparing stains, all stock solutions are prepared with analytical grade reagents wherever possible and distilled water is used throughout. Unless otherwise indicated, all solutions should be stored in the dark and can be kept for up to 6 months. Some commonly used stains are listed below. Stains (a) to (d) are general purpose stains whereas stains (e) to (g) are stains for specific amino acids.
- (a) Ninhydrin:cadmium acetate (2). Two stock solutions are prepared:

Solution A. 6 g of cadmium acetate is dissolved in a mixture of 600 mL of water and 300 mL acetic acid.

Solution B. A 0.5% (w/v) solution of ninhydrin in acetone is prepared.

The stain is made by mixing 15 mL of Solution A with 85 mL of Solution B and is used immediately.
 - (b) Ninhydrin:butan-1-ol 0.5% (w/v) solution of ninhydrin in butan-1-ol is prepared.
 - (c) Fluorescamine (3). A 0.001% (w/v) solution of fluorescamine in a mixture of acetone and pyridine (99:1 v/v) is prepared.
 - (d) *o*-Phthaldehyde (OPA) (4). A 3 g quantity of boric acid is dissolved in approximately 90 mL of water and adjusted to pH 10.5 with concentrated potassium hydroxide solution. To this solution, 0.05 g of OPA dissolved in 1 mL of ethanol plus 0.05 mL of 2-mercaptoethanol, is added. The solutions are then mixed and made up to 100 mL with water. Finally 0.3 mL of a 30% (w/v) solution of Brij is added.

- (e) Ehrlich stain for tryptophan (5). A 1% (w/v) solution of *p*-dimethylaminobenzaldehyde in acetone containing 10% (v/v) hydrochloric acid is prepared. This solution is then stored at 4°C and can be kept for up to 2 months.
- (f) Pauly stain for histidine and tyrosine (5). The following solutions are prepared as required.
 - Solution C*. 5% (w/v) sodium nitrite
 - Solution D*. 1% (w/v) sulfanilic acid in 1M hydrochloric acid
 - Solution E*. 15% (w/v) sodium carbonate in water
- (g) Phenanthraquinone stain for arginine (6). The following solutions are prepared.
 - Solution F*. A 0.02% (w/v) phenanthrenequinone solution in ethanol is prepared. This solution should be stored in the dark and can be kept for up to 3 months.
 - Solution G*. A 10% (w/v) NaOH solution in 60% (v/v) ethanol is prepared just before the stain is to be used.The stain is then prepared by mixing equal volumes of Solutions F and G and is used immediately.

Method

N.B. Disposable plastic gloves should be worn in all of the subsequent steps, not only to prevent the thin-layer plates from becoming contaminated, but also as a protection against the various stains and solvents used during the course of this work. A fume hood should also be used wherever possible and particularly when drying or spraying plates.

Tryptic Digest

1. The protein sample is dissolved at a concentration of between 2 and 10 mg/mL in 0.2M ammonium bicarbonate. A freshly prepared solution of trypsin (DCC-treated, bovine trypsin) at a concentration of 1 mg/mL in water is then added to this solution, giving a final enzyme-to-substrate ratio of 1:50.

2. The sample is then mixed and incubated at 37°C for 48 h.
3. The resulting digest is either stored at -20°C until required or taken straight on to the next step.

Thin-Layer Electrophoresis

1. An origin line is carefully drawn across the center of the plate using a soft pencil. The samples are then spotted on to the plate using a 10 or 25 μL syringe as 1-cm long bands along this line, leaving a 1.5–2.0 cm gap between each sample. Ideally the syringe needle should never touch the surface of the plate or the cellulose thin layer will be damaged. Where large sample volumes (i.e., greater than 10 μL) have to be loaded, a hair drier can be used to accelerate the drying of the sample.
2. Once the samples are dry the plate should be uniformly saturated with electrophoresis buffer, applied as an aerosol from a spray gun. If the surface of the plate becomes too wet at this stage, i.e., if pools of buffer form on the surface of the plate, the excess buffer should be carefully blotted off with a sheet of filter paper.
3. The plate is then placed, sample side uppermost, on top of a sheet of cellulose acetate covering the cooling platen of the electrophoresis apparatus. The electrode wicks (Whatman No. 1 filter paper or similar) saturated in electrophoresis buffer are then positioned so as to cover about 1 cm of the top and bottom edge of the plate. Again excess buffer is blotted off with spare sheets of filter paper. (If there is too much free buffer present at this stage, a short-circuit is likely to occur once the run starts)
4. The upper insulating sheet of cellulose acetate is then placed on top of the filter paper wicks and the thin-layer plate and the lid of the chamber closed. In a preliminary experiment only a very short run is required, ~10–15 min at 2 kV.
5. The plate is then removed and dried in a stream of warm air in a fume hood and the peptide separations are examined after staining with one of the nonspecific stains described below.
6. Having determined the basic pattern of the peptide separation it is then possible to optimize the separation

by moving the origin towards either the anode or cathode as appropriate and increasing the length of the run proportionally. Alternatively, should the first buffer system fail to give a satisfactory separation, one of the other buffer systems given in the Materials section should be tried. (The electrophoresis buffer should be replaced at least every 2 weeks.)

Thin-Layer Chromatography

1. An origin line is drawn as for high voltage electrophoresis, but instead of being across the middle of the plate the line should be 2 cm from one edge of the plate. The samples are then loaded as described previously.
2. The plate is then placed in a chromatography tank containing sufficient solvent to cover about 1 cm of the plate, but not enough to risk covering the samples or they will be washed off. The chromatography is then allowed to proceed, usually overnight, until the solvent reaches the top of the plate. (The solvent in the chromatography tank should be replaced every 2 weeks in order to ensure that the best separations are obtained.)
3. At the end of the run the chromatogram is removed from the tank and dried in a stream of warm air in a fume hood and stained.

Two-Dimensional Peptide Mapping

Having established the optimum conditions for both the electrophoretic and the chromatographic separation of the peptides, it is then possible to extend the analysis to two dimensions. Although this means that only one sample can be analyzed per plate it is frequently justified in terms of the increased resolution which can be obtained (compare Figs 1 and 2).

The procedure for producing two-dimensional peptide maps is essentially as described in the previous two sections. Theoretically either electrophoresis or chromatography may be used for the first dimension separation, but generally better results are obtained where electrophoresis is used for the first dimension and chromatography in the second dimension.

1. The sample is loaded along a 1 cm origin line, parallel to the bottom of the plate and 2 cm from the left-hand side of the plate. The precise position of the origin line from the bottom of the plate should be determined by a trial one-dimensional separation as described above.
2. After running the sample in the first dimension the plate is thoroughly dried in a stream of warm air
3. The plate is then rotated through 90° with the former left-hand edge of the plate now forming the bottom of the plate and the peptides are separated by ascending chromatography as described previously.

Diagonal Peptide Mapping

Two-dimensional thin-layer peptide mapping experiments rely on a combination of two different fractionation techniques in order to obtain the best possible separation of peptides. If instead of using two different separation methods, the same system is used in both the first and second dimensions the peptides will form a single diagonal line across the plate. Although this diagonal separation does not normally offer any advantage over a simple one-dimensional separation it can be exploited for the identification of cysteine containing peptides (7):

1. The sample is loaded on to a thin-layer plate and the peptides are then separated by thin-layer high voltage electrophoresis or chromatography as described previously.
2. At the end of the run the plate is thoroughly dried in a stream of warm air
3. While the plate is drying, a solution of performic acid is prepared by mixing together 95 mL of 98% (v/v) formic acid and 5 mL of hydrogen peroxide. This solution is then placed in a shallow dish, which is placed at the bottom of a chromatography tank.
4. The dry plate is placed in this tank and left for 2 h, after which time the plate is removed from the tank and carefully dried once more.
5. The thin-layer plate is then rotated through 90° , i.e., the former left-hand edge of the plate becomes the new

bottom edge of the plate and the peptides are run in the second dimension, using the same separation system as before.

6. At the end of the run the plate is again dried and stained (*see* below). The majority of the peptides will be seen to lie along a single diagonal line. However, peptides that were previously joined together by S—S bridges will now have different mobilities owing to the intervening performic acid oxidation step and the resulting cysteic acid containing peptides will fall off this line.

Stains

All of the following stains (*see* also Chapter 4) are applied as aerosols from a distance of 30–50 cm, ensuring that the plate is uniformly saturated with reagent. It is important, however, that the plates do not become too wet or the peptide separations will become blurred

- (a) Ninhydrin cadmium acetate. After spraying the plate it is left to dry for 5 min before being placed in an oven at 110°C for 5–10 min. The majority of the peptides will appear as red spots against a pale pink background. However, peptides with either glycine or threonine *N*-terminal amino acids will initially appear as yellow spots that gradually turn red over the course of 1–2 d, while peptides with an *N*-terminal valine frequently stain weakly if at all.
- (b) Ninhydrin butan-1-ol. Peptides are detected as described in (a) above and appear as blue or purple spots against a light purple background. (The lack of contrast between the nonspecific background staining and the peptides can make these separations more difficult to photograph in comparison to peptides stained with ninhydrin–cadmium acetate reagent). Unlike the ninhydrin–cadmium acetate reagent this stain also has the disadvantage of reacting with the side chains of lysine residues that may give a false impression of the abundance of peptides containing a large number of lysine residues

- (c) Fluorescamine. After spraying the plate it is left to dry for 15–20 min (if the plate is too wet much of the fluorescence will be quenched) before examining the plate under a long wavelength (366 nm) ultraviolet light. As fluorescamine only reacts with primary amines peptides with an *N*-terminal proline residue will not be detected, but these peptides can be detected by staining with ninhydrin once the fluorescamine staining pattern has been recorded. The failure to detect proline residues is, however, more than compensated for by the fivefold increase in sensitivity compared to that obtained with ninhydrin.
- (d) *o*-Phthaldehyde. After spraying the plate it is left to dry for 15–20 min (again if the plate is too wet much of the fluorescence will be quenched) before examining the plate under a long wave length (366 nm) ultraviolet light. Peptides are visible by their intense blue fluorescence (visible for up to 2 h) and as little as 10 pmol of a single peptide may be detected, making this by far the most sensitive of the general purpose stains.
- (e) Ehrlich reagent for tryptophan. After spraying the plate it is allowed to dry and tryptophan-containing peptides are seen as bright purple spots that gradually fade over a number of hours. This stain may be used on a plate that has previously been stained with ninhydrin as the hydrochloric acid in the Ehrlich's reagent will bleach the ninhydrin-stained peptides.
- (f) Pauly reagent for histidine and tyrosine. Solutions C and D are left on ice for 20 min. Equal volumes of these two solutions are then mixed together and the plate is immediately sprayed with this solution. After allowing the plate to dry (~20 min) the plate is then sprayed with Solution E and again left to dry. Histidine and tyrosine-containing peptides will appear as either red or weak brown spots, respectively.
- (g) Phenanthraquinone stain for arginine. After staining the plate is thoroughly dried (~20 min). Arginine-containing peptides may then be visualized by examining the plate under a long wavelength (366 nm) ultraviolet light; the stained peptides will remain brightly fluorescent for at least 18 h. After recording the distribution of these peptides they can be stained

with either ninhydrin–cadmium acetate solution or with the Pauly reagent if required. In the case of the Pauly reagent, the plate will already be sufficiently basic following the phenanthraquinone staining to permit the Pauly reagent to be used without needing to use Solution E.

Notes

1. Although trypsin is a very effective protease it will only cleave those peptide bonds that are exposed on the surface of the protein. In proteins that contain disulfide bridges some of these potential cleavage sites will be buried within the protein molecule and consequently will be protected from the trypsin. In proteins where this is a problem this can be overcome by reducing the cystines and then blocking the SH groups by S-carboxymethylation as described by Crestfield et al (8). (See also p. 35.)
2. In the majority of peptide mapping experiments peptides are produced by digestion with trypsin. This enzyme is a specific protease, which cleaves peptide bonds on the C-terminal side of arginine and lysine residues. These amino acids are relatively abundant in most proteins, accounting for approximately 11.5% of the total amino acids in an “average” protein (9). However, a number of other specific enzymatic and chemical cleavages are available (for examples, see 10) and these can be successfully used where the amino acid or sequence analysis for a particular protein indicates that they will produce a better range of peptides.
3. The choice of which electrophoresis buffer to use is largely empirical. Peptides containing large numbers of acidic and basic residues are usually best resolved in the more basic buffers, while good separations of peptides rich in neutral amino acids are generally obtained with the more acidic buffer systems.
4. In the majority of chromatography experiments the “BAWP” solvent will produce excellent results, however, should this prove unsatisfactory, one of the other solvent systems should be tried.

5. A number of the stains described in this chapter are now commercially available, in ready to use, aerosol cans. Further details about these stains can be found in the catalogs of many of the leading chemical manufacturers (see also Chapter 4 for other available stains)

References

- 1 Powers, D. A., Fishbein, J. C., and Place, A. R. (1983) Thin-layer peptide mapping with sequencing at the nanomole level. In *Methods in Enzymology* **91**, Hirs, C. H. W. and Timasheff, S. N. eds. pp. 466–486. Academic Press, New York.
- 2 Heilman, J., Barrolier, J., and Watzke, E. (1957) Beitrag zur Aminosäurebestimmung auf Papierchromatogrammen. *Hoppe-Seyler's Zeit.* **309**, 219–220.
- 3 Felix, A. M., and Jimenez, M. H. (1974) Usage of fluorescamine as a spray reagent for thin-layer chromatography. *J. Chromatogr.* **89**, 361–364.
4. Gardner, W. S., and Miller III, W. H. (1976) Reverse-phase liquid chromatographic analysis of amino acids after reaction with o-phthalaldehyde. *Anal. Biochem.* **101**, 61–65.
- 5 Bennett, J. C. (1967) Paper chromatography and electrophoresis; special procedure for peptide maps. In *Methods in Enzymology*, **XI**, Enzyme structure, pp. 330–339 (Hirs, C. H. W. ed.) Academic Press, New York.
- 6 Yamada, S., and Itano, H. A. (1966) Phenanthrenequinone as an analytical reagent for arginine and other monosubstituted guanidines. *Biochim. Biophys. Acta* **130**, 538–540.
7. Brown, J. R., and Hartley, B. S. (1966) Location of disulphide bridges by diagonal paper electrophoresis. The disulphide bridges of bovine chymotrypsinogen A. *Biochem. J.* **101**, 214–228.
- 8 Crestfield, A. M., Moore, S., and Stein, W. H. (1963) The preparation and enzymatic hydrolysis of reduced and S-carboxymethylated proteins. *J. Biol. Chem.* **238**, 622–627.
- 9 Dayhoff, M. O., Hunt, L. T., and Hurst-Calderone, S. (1978) *Atlas of Protein Sequence and Structure* **5**, Supplement 3, pp. 363–373 (ed. Dayhoff, M. O.) National Biomedical Research Foundation, Washington, DC.
- 10 Croft, L. R. (1980) *Handbook of Protein Sequence Analysis*. 2nd edition, Wiley, New York.
11. Walker, J. M., Gooderham, K., Hastings, J. R. B., Mayes, E., and Johns, E. W. (1980) The primary structures of non-histone chromosomal proteins HMG 1 and 2. *FEBS Lett.* **122**, 264–270.

Chapter 22

In Situ Peptide Mapping of Proteins Following Polyacrylamide Gel Electrophoresis

Keith Gooderham

*MRC Clinical and Population Cytogenetics
Unit, Western General Hospital, Crewe
Road, Edinburgh, United Kingdom*

Introduction

Polyacrylamide gel electrophoresis is a simple, yet versatile, high resolution technique for the analysis of complex mixtures of proteins. However, this is not to say that this method is without problems. For example, where a series of different protein samples are run on a gel, many of the proteins will have the same mobility and it is frequently impossible to be certain if these bands represent the same protein or whether they simply share similar mobilities. Conversely the samples may contain degradation products or structurally related proteins of differing mobilities

In this chapter a simple *in situ* peptide-mapping technique, based on a method originally developed by Cleveland et al (1) is described. Briefly the method is as follows: Proteins are separated by discontinuous sodium dodecyl sulfate polyacrylamide slab gel electrophoresis (SDS PAGE) and stained with Coomassie brilliant blue (see Chapter 6). After destaining, bands containing polypeptides of interest are cut out of the gel and equilibrated against Tris pH 6.8 buffer. The gel slices are then ready for re-electrophoresis and together with a suitable protease are loaded onto a second SDS polyacrylamide slab gel and run into the stacking gel at a low current. As the proteins and protease migrate through the low percentage polyacrylamide pH 6.8 stacking gel, the proteins are digested by the protease and the resulting peptides are then resolved in the lower pH 8.8 separating gel.

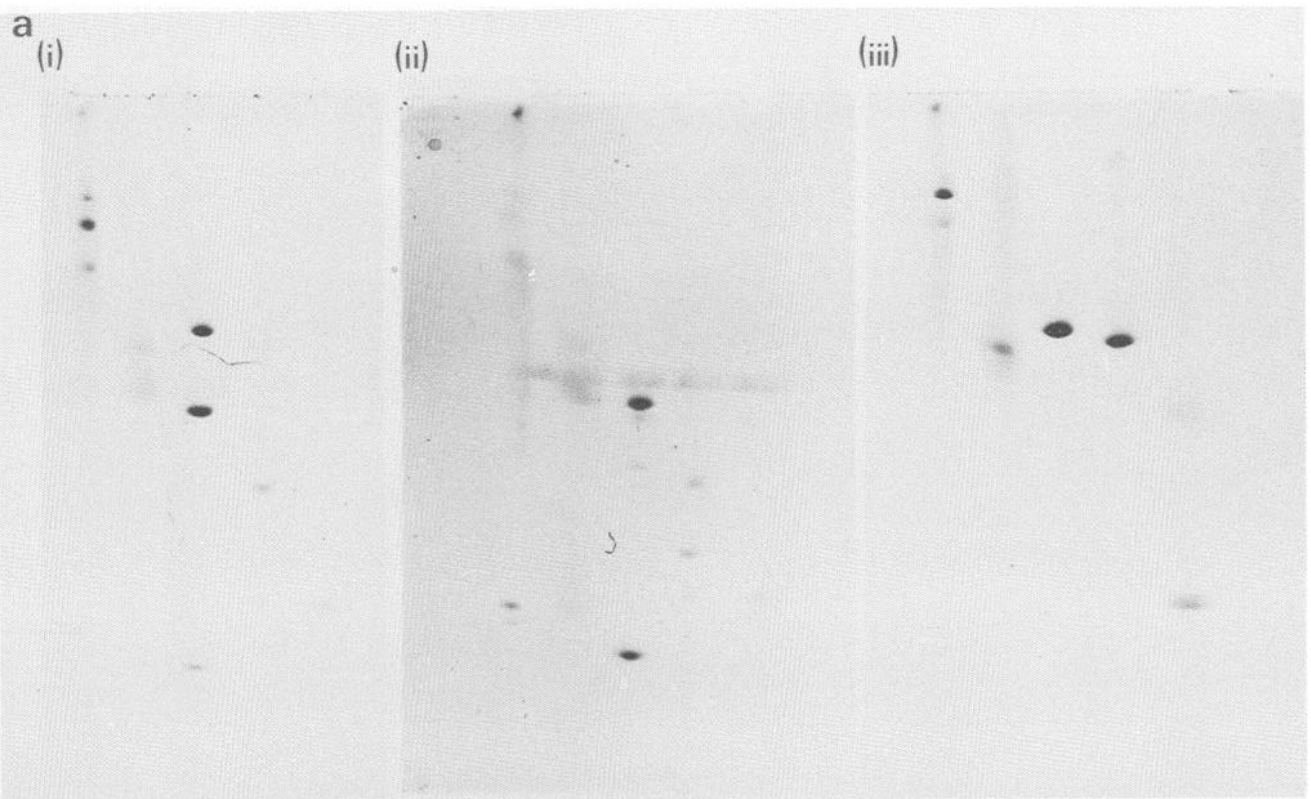
Peptides are then detected by standard methods of staining, autoradiography, and so on. The recent development of high-sensitivity silver staining methods for the detection of proteins following polyacrylamide gel electrophoresis (see Chapter 13) has proved to be a very valuable method when used in conjunction with the peptide-mapping technique described here. Silver staining methods are between 20 and 100 times more sensitive than traditional Coomassie staining methods (see Fig. 1), while not significantly less sensitive than fluorography following labeling with tritium. Silver staining also has the advantage of not requiring any special sample preparation, and the actual staining procedure takes approximately the same amount of work as is involved in preparing a gel for fluorography.

In situ peptide mapping therefore requires very little extra work or equipment and yet considerably extends the information potential of both one- and two-dimensional polyacrylamide gels. There are a large number of possible applications of this method, the most obvious and important of which are: (i) comparison of comigrating proteins (see, for example, Fig. 1); (ii) identification of structurally related proteins; (iii) identification of degradation products; and (iv) detection of different domains, e.g., antigen-binding sites, following protein blotting (see Chapter 20). In addition, although this method has been most widely

used for the analysis of proteins separated by SDS PAGE, it is a very versatile technique, and with suitable modifications, it can be used with almost any combination of gel systems (2)

Materials

1. Gel apparatus. Both the primary protein separation and the secondary peptide-mapping gels are run in a vertical slab gel apparatus based on the design originally described by Studier (3). The spacers and comb for the primary gel are 1 mm thick and a 14-well comb with 1.5 cm deep and 1 cm wide wells is used. The spacers and combs for the secondary gel are 1.5 mm thick and a 20-well comb with 6 mm wide and 20 mm deep wells is used. The increased thickness of the secondary gel is necessary in order to accommodate the gel slices that swell considerably during the staining and equilibration steps. Similarly the increased depth of the secondary gel wells is required in order to accommodate the gel slices while leaving sufficient room for the addition of the protease buffer.
2. Equilibration buffer. The following solutions are mixed together. 1.25M Tris-HCl, pH 6.8 (10 mL), 10% SDS (1 mL, electrophoresis grade), 0.1M Na₂EDTA, pH 7.0 (1 mL) and made up to a final volume of 100 mL with water.
3. Overlay buffer. The following solutions are mixed together. 1.25M Tris-HCl, pH 6.8 (10 mL), 10% SDS (1 mL, electrophoresis grade), 0.1M Na₂EDTA, pH 7.0 (1 mL), glycerol (20 mL), 1% (w/v) Bromophenol blue (1 mL) and made up to a final volume of 100 mL with water.
4. Protease buffer. The following solutions are mixed together: 1.25M Tris-HCl, pH 6.8 (10 mL), 10% SDS (1 mL, electrophoresis grade), 0.1M Na₂EDTA, pH 7.0 (1 mL), glycerol (10 mL), 1% (w/v) Bromophenol blue (1 mL) and made up to a final volume of 100 mL with water.
5. Proteases. A number of different proteases have been used for peptide mapping experiments with SDS



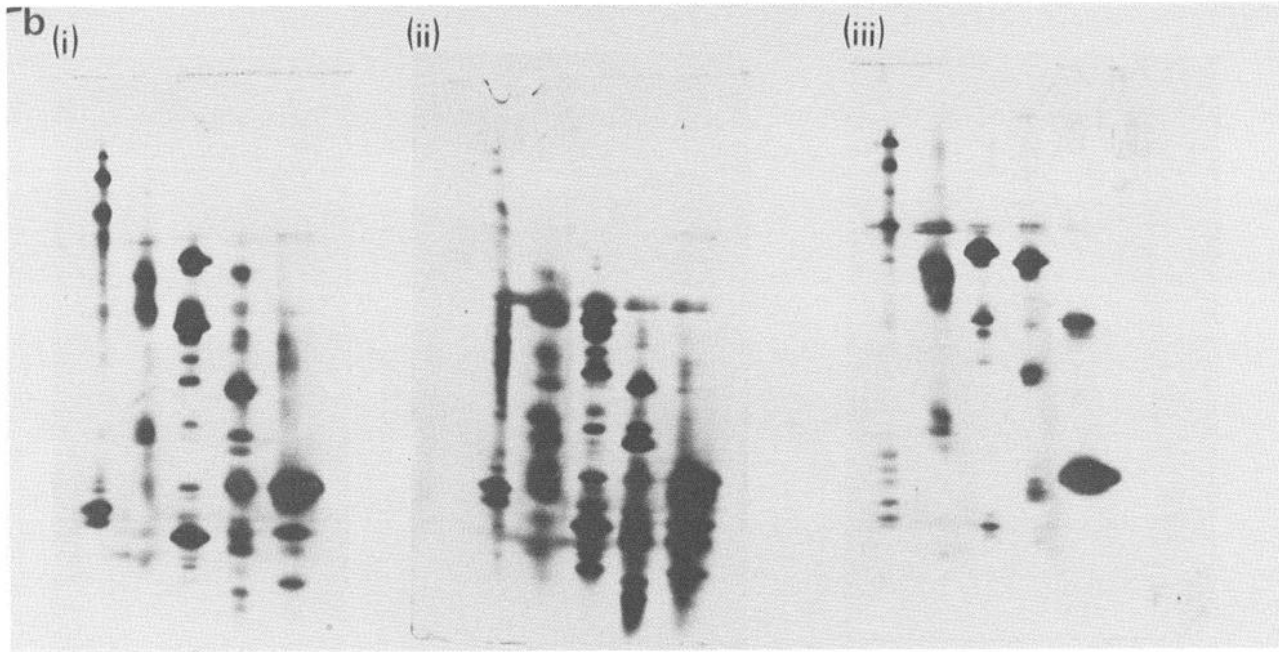


Fig. 1. In situ peptide maps of five standard proteins. The effect of protease concentration and a comparison of Coomassie and silver stains. Six peptide maps, each containing an identical panel of proteins (from left to right: phosphorylase a, 100,000 d (daltons); ovalbumin, 43,000 d; 3-phosphoglyceric phosphokinase, 43,000 d; actin, 43,000 d; and β -lactoglobulin, 35,000 d. 2.5 μ g of each protein was loaded) are shown. The proteins were initially run on two 15% (w/v) polyacrylamide SDS gels (not shown). After staining with Coomassie brilliant blue the individual protein bands were cut out and loaded onto two 1.5 mm thick peptide mapping gels [15% (w/v) polyacrylamide] and subjected to *in situ* peptide mapping with (i) 0.05 μ g V8 protease; (ii) 0.5 μ g V8 protease; and (iii) no protease. The resulting peptides were then visualized by either (a) Coomassie brilliant blue or (b) by silver staining (see Chapter 13).

polyacrylamide gels. Of these, the most frequently used are *Staphylococcus aureus* V8 protease, chymotrypsin, and papain. V8 protease is relatively expensive and is most conveniently prepared as a 1 mg/mL stock solution in water and stored in aliquots at -20°C . Each aliquot may be frozen and thawed several times without any apparent loss of activity. The other two enzymes are freshly prepared as 1 mg/mL stock solutions for each experiment. Where other gel systems are to be used, e.g., acid gels, it may not be possible to use these enzymes, but either pepsin or one of the chemical cleavage techniques (*see* Note 6) may well prove to be suitable alternatives.

Unless otherwise indicated, all stock solutions and buffers are prepared from analytical grade reagents and are made up in deionized double-distilled water. With the exception of the SDS and Bromophenol blue stock solutions, which are stored at room temperature, all other solutions and buffers are stored at 4°C and are stable for at least three months.

Method

1. Proteins are separated by polyacrylamide slab gel electrophoresis (SDS PAGE or any other suitable system), followed by staining with Coomassie brilliant blue R 250 (Chapter 6).
2. The gel should then be photographed in order to produce a permanent record of the proteins under investigation. Bands containing proteins of interest are then cut out of the gel and placed in a 10 mL screw-top plastic test tube containing 5 mL of equilibration buffer. The tubes are then laid on a shaker table and gently agitated. After 1 h the buffer is replaced with a further 5 mL of fresh buffer and the equilibration continued for a further hour.
3. The buffer is then drained off and the gel slices can then conveniently be stored at -20°C until required.
4. After thawing, the gel slices are placed (short-end down) in the sample wells of an extra thick (1.5 mm) SDS polyacrylamide gel with a 5 cm stacking gel. (This is twice the normal size of stacking gel, but is necessary

in order to allow sufficient time for the protease to digest the proteins, as well as to permit the efficient "stacking" of the samples.)

5. The gel slices are covered with overlay buffer followed by 10 μL of protease buffer containing 0.05 μg of protease.
6. The samples are then run at a constant current of 20 mA until the marker dye reaches the top of the separating gel when the current is increased to 40 mA for the remainder of the run. The total run time is about 6 h.
7. Peptides are then stained overnight in Coomassie brilliant blue R250 or fixed and subsequently visualized by silver staining (Chapter 13), fluorography (Chapter 17), and so on

Notes

1. The versatility of the peptide mapping technique makes it suitable for use with a wide range of different gel systems. The peptide mapping gel can be either the same or different to the one used for the initial protein separation, providing that a suitable intervening equilibration step is included. However, for most applications the SDS PAGE system (4) is compatible with the widest range of enzymes, as well as offering the best resolution of the resulting peptides. The percentage of acrylamide used in the separating gel is determined by the molecular weight of the proteins under investigation as well as by the size of the resulting peptides. As a rough guide, an 8% (w/v) gel should be used for proteins of 90,000 daltons or greater, whereas for smaller proteins either 15% (w/v) or 20% (w/v) gels will produce the best results. Alternatively, where a wide range of different sized peptides are generated a gradient gel can be used (5).
2. In Cleveland's original paper (1), a very brief staining step (15 min) is used prior to the removal of the protein bands and their being carried on to the equilibration step. However, this is not necessary and the proteins can be completely stained before being removed from the gel, with the result that minor proteins can be detected more readily. This also allows the protein sepa-

ration to be photographed and therefore avoids the need to run a duplicate gel. However, by exposing the proteins to the very acidic staining and destaining solutions for several days, peptide bonds between aspartic and proline residues may be hydrolyzed [see Fig. 1a (iii) and 1b (iii)]. The extent to which this occurs depends upon a number of factors, including the length of the staining and destaining steps, as well as the temperature at which the gel is stored. Therefore, in order to obtain comparable peptide maps, it is important that all of the proteins should, wherever possible, be taken from the same gel, or where this is not practical, the gels should be processed in exactly the same way.

3. Once the bands containing the proteins of interest have been cut out of the gel they can be stored at -20°C immediately, or at the end of the first stage of the equilibration step, or upon completing this step. The gel slices should be carefully drained before freezing or they may disintegrate when thawed. Once frozen, the samples can be stored for as long as three months.
4. In the methods section above, an enzyme concentration of $0.05\text{ }\mu\text{g/sample}$ is recommended. At this concentration all three enzymes will produce acceptable peptide maps over a wide range of protein concentrations ($0.1\text{--}5\text{ }\mu\text{g}$), assuming that the protein contains suitable cleavage sites. However, in some cases, it may be worthwhile to try a range of different enzyme concentrations in order to obtain a better distribution of peptides.
5. When a series of different gel slices contain varying amounts of protein, this can be overcome by initially running a trial gel and measuring the concentration of the proteins either by densitometry (Chapter 14) or alternatively by overnight extraction of the Coomassie stained protein with 25% (v/v) pyridine. The optical densities of these extracts are then measured at 595 nm (6) and the sample volumes adjusted accordingly so as to produce bands of equal intensity. However, this method will generally tolerate quite a wide range of enzyme and substrate concentrations and any differences that do occur tend to be quantitative rather than qualitative (see Fig. 1).

6. In addition to using the various proteolytic enzymes described above, a number of chemical cleavage methods have been described. These include *N*-chlorosuccinimide (7), cyanogen bromide (8), hydroxylamine (8), and formic acid (9). All of these methods involve rather extreme reaction conditions that are unlikely to be compatible with the gel system used for the peptide separation. The proteins are therefore cleaved in the gel slices and prior to the equilibration step. The sole purpose of the secondary gel is then to separate the peptides and consequently the size of the stacking gel can be reduced to 2.5 cm while proportionally increasing the size of the separating gel.
7. Where marker proteins are required, either to identify the starting protein or as molecular weight standards, they should be taken from the same gel as the other proteins. If this is not done and they are simply loaded directly onto the peptide mapping gel in solution they will travel faster than the other proteins. This is because these proteins are delayed in their migration by first having to travel out of the gel slices before entering the mapping gel.
8. A wide range of methods has been used for detecting peptides produced by this technique and they all have their respective advantages and disadvantages. Generally, Coomassie brilliant blue is not sufficiently sensitive (see Fig. 1) and we have favored the use of one of the high-sensitivity silver stains (see Fig. 1 and Chapter 13). The major disadvantage of this technique is that it is nonspecific, staining not only the peptides, but also the protease. Much cleaner results can be obtained using either radioactively labeled proteins or, alternatively, proteins that have been labeled with dansyl chloride, but both of these require advance planning and preparation.

References

- 1 Cleveland, D W , Fischer, S G., Kirschner, M W , and Laemmli, U K (1977) Peptide mapping by limited proteolysis in sodium dodecyl sulfate and analysis by gel electrophoresis *J Biol Chem* **252**, 1102-1106

- 2 Spiker, S (1980) Slab gel designed for enzymatic digestion of proteins in polyacrylamide gel slices and direct resolution of peptides *J. Chromatog.* **198**, 169–171
- 3 Studier, W F (1973) Analysis of bacteriophage T7 early RNAs and proteins on slab gels *J Mol. Biol* **79**, 237–248.
- 4 Laemmli, U K. (1970) Cleavage of structural proteins during the assembly of the head of bacteriophage T4. *Nature* **227**, 680–685.
- 5 Tijssen, P , and Kurstak, E. (1979) A simple and sensitive method for the purification and peptide mapping of proteins solubilized from densonucleosis virus with sodium dodecyl sulfate. *Anal. Biochem* **99**, 97–104
- 6 Fenner, C , Traut, R R., Mason, D T., and Wilkeman-Coffelt, J (1975) Quantification of Coomassie blue stained proteins in polyacrylamide gels based on analyses of eluted dyes *Anal Biochem.* **63**, 595–602
- 7 Lischwe, M. A., and Ochs, D. (1982) A new method for partial peptide mapping using *N*-chlorosuccinimide/urea and peptide silver staining in sodium dodecyl sulfate polyacrylamide gels *Anal Biochem* **127**, 453–457.
8. Lam, K. S., and Kasper, C B (1979) Electrophoretic analysis of three major nuclear envelope polypeptides Topological relationship and sequence homology. *J Biol Chem* **254**, 11,713–11,720
- 9 Sonderegger, P , Jaussi, R , Gehring, H , Brunschweiler, K., and Christen, P. (1982) Peptide mapping of protein bands from polyacrylamide gel electrophoresis by chemical cleavage in gel pieces and re-electrophoresis *Anal. Biochem* **122**, 298–301.

Chapter 23

The Dansyl Method for Identifying N-Terminal Amino Acids

John M. Walker

*School of Biological and Environmental
Sciences, The Hatfield Polytechnic,
Hatfield, Hertfordshire, England*

Introduction

The reagent 1-dimethylaminonaphthalene-5-sulfonyl chloride (dansyl chloride, DNS-Cl) reacts with the free amino groups of peptides and proteins as shown in Fig. 1. Total acid hydrolysis of the substituted peptide or protein yields a mixture of free amino acids plus the dansyl derivative of the *N*-terminal amino acid, the bond between the dansyl group and the *N*-terminal amino acid being resistant to acid hydrolysis. The dansyl amino acid is fluorescent under UV light and is identified by thin-layer chromatography on polyamide sheets. This is an extremely sensitive method for identifying amino acids and in partic-

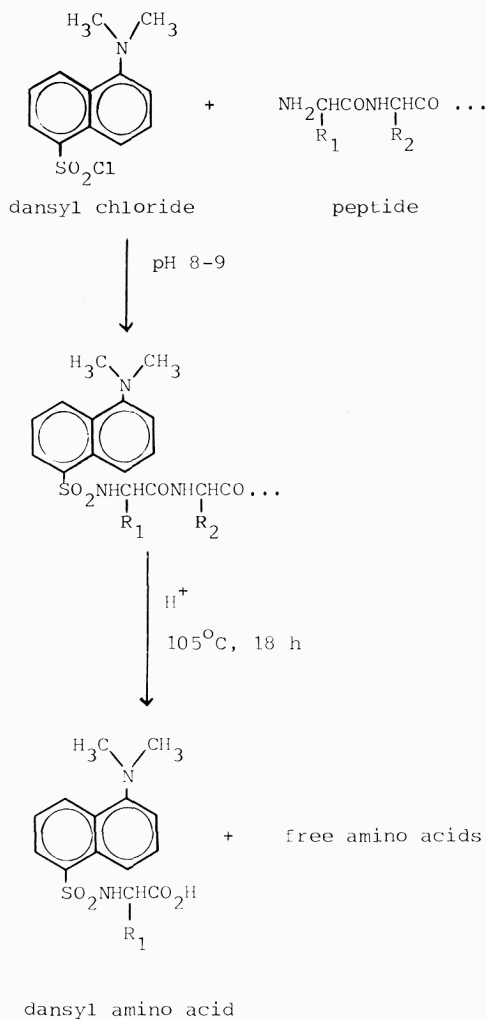


Fig. 1. Reaction sequence for the labeling of *N*-terminal amino acids with dansyl chloride.

ular has found considerable use in peptide sequence determination when used in conjunction with the Edman degradation (see Chapter 24). The dansyl technique was originally introduced by Gray and Hartley (1), and was developed essentially for use with peptides. However, the method can also be applied to proteins (see Note No. 12).

Materials

1. Dansyl chloride solution (2.5 mg/mL in acetone). Store at 4°C in the dark. This sample is stable for many months. The solution should be prepared from concentrated dansyl chloride solutions (in acetone) that are commercially available. Dansyl chloride available as a solid invariably contains some hydrolyzed material (dansyl hydroxide).
2. Sodium bicarbonate solution (0.2 M, aqueous). Store at 4°C. Stable indefinitely, but check periodically for signs of microbial growth.
3. 5N HCl (aqueous)
4. Test tubes (50 × 6 mm) referred to as "dansyl tubes"
5. Polyamide thin layer plates (7.5 × 7.5 cm). These plates are coated on both sides, and referred to as "dansyl plates". Each plate should be numbered with a pencil in the top corner of the plate. The origin for loading should be marked with a pencil 1 cm in from each edge in the lower left-hand corner of the numbered side of the plate. The origin for loading on the reverse side of the plate should be immediately behind the loading position for the front of the plate, i.e., 1 cm in from each edge in the lower *right-hand* corner.
6. Three chromatography solvents are used in this method.

Solvent 1. Formic acid:water, 1:5:100, v/v
Solvent 2. Benzene:acetic acid, 9:1, v/v
Solvent 3. Ethyl acetate:methanol:acetic acid, 2:1:1, v/v/v
7. An acetone solution containing the following standard dansyl amino acids: Pro, Leu, Phe, Thr, Glu, Arg (each approximately 50 µg/mL)
8. A UV source, either long wave (265 µm) or short wave (254 µm)

Method

1. Dissolve the sample to be analyzed in an appropriate volume of water, transfer to a dansyl tube, and dry *in*

vacuo to leave a film of peptide (1–5 nmol) in the bottom of the tube

2. Dissolve the dried peptide in sodium bicarbonate (0.2M, 10 μ L) and then add dansyl chloride solution (10 μ L) and mix
3. Seal the tube with parafilm and incubate at 37°C for 1 h, or at room temperature for 3 h
4. Dry the sample *in vacuo*. Because of the small volume of liquid present, this will only take about 5 min.
5. Add 6N HCl (50 μ L) to the sample, seal the tube in an oxygen flame, and place at 105°C overnight (18 h).
6. When the tube has cooled, open the top of the tube using a glass knife, and dry the sample *in vacuo*. If phosphorus pentoxide is present in the desiccator as a drying agent, and the desiccator is placed in a water bath at 50–60°C, drying should take about 30 min.
7. Dissolve the dried sample in 50% pyridine (10 μ L) and, using a microsyringe, load 1 μ L aliquots at the origin on *each side* of a polyamide plate. This is best done in a stream of warm air. Do not allow the diameter of the spot to exceed 3–4 mm
8. On the *reverse side only*, also load 0.5 μ L of the standard mixture at the origin.
9. When the loaded samples are completely dry, the plate is placed in the first chromatography solvent and allowed to develop until the solvent front is about 1 cm from the top of the plate. This takes about 10 min, but can vary depending on room temperature.
10. Dry both sides of the plate by placing it in a stream of warm air. This can take 5–10 min since one is evaporating an aqueous solvent
11. If the plate is now viewed under UV light, a blue fluorescent streak will be seen spreading up the plate from the origin, and also some green fluorescent spots may be seen within this streak. However, no interpretations can be made at this stage
12. The dansyl plate is now developed in the second solvent, at right angles to the direction of development in the first solvent. The plate is therefore placed in the chromatography solvent so that the blue 'streak' runs along the bottom edge of the plate

- 13 The plate is now developed in the second solvent until the solvent front is about 1 cm from the top of the plate. This takes 10–15 min.
- 14 The plate is then dried in a stream of warm air. This will only take 2–3 min since the solvent is essentially organic. However, since benzene is involved, drying *must* be done in a fume cupboard.
- 15 The side of the plate containing the sample only should now be viewed under UV light. Three major fluorescent areas should be identified. Dansyl hydroxide (produced by hydrolysis of dansyl chloride) is seen as a blue fluorescent area at the bottom of the plate. Dansyl amide (produced by side reactions of dansyl chloride) has a blue–green fluorescence and is about one-third of the way up the plate. These two spots will be seen on all dansyl plates and serve as useful internal markers. Occasionally other marker spots are seen and these are described in the Notes section below. The third spot, which normally fluoresces green, will correspond to the dansyl derivative of the *N*-terminal amino acid of the peptide or protein. However, if the peptide is not pure, further dansyl derivatives will of course be seen. The separation of dansyl derivatives after solvent 2 is shown in Fig. 2. Solvent 2 essentially causes separation of the dansyl derivatives of hydrophobic and some neutral amino acids, whereas derivatives of charged and other neutral amino acids remain at the lower end of the chromatogram.
16. A reasonable identification of any faster-moving dansyl derivatives can be made after solvent 2 by comparing their positions, relative to the internal marker spots, with the diagram shown in Fig. 2. Unambiguous identification is made by turning the plate over and comparing the position of the derivative on this side with the standard samples that were also loaded on this side.

N.B. Both sides of the plate are totally independent chromatograms. There is no suggestion that fluorescent spots can be seen *through* the plate from one side to the other.

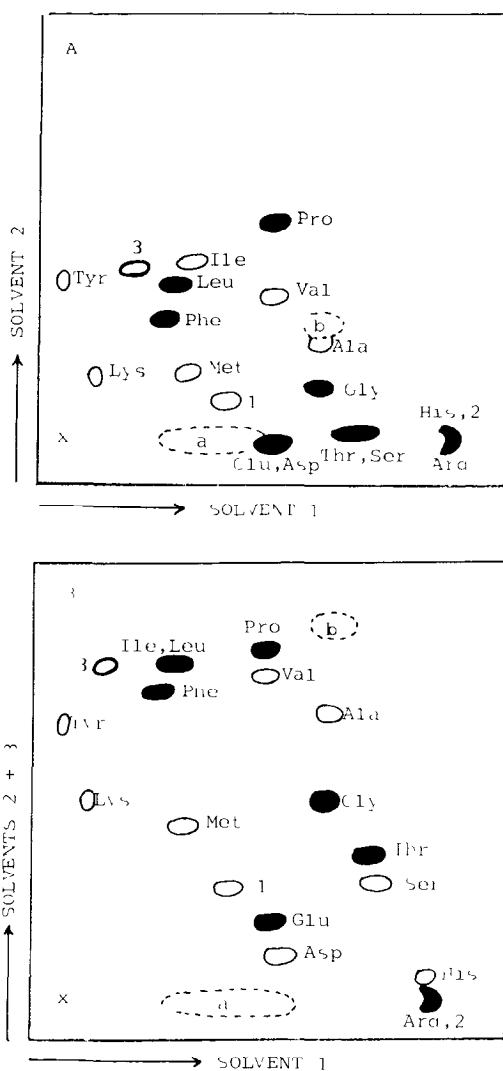


Fig 2 Diagrams showing the separation of dansyl amino acids on polyamide plates after two solvents (A), and after three solvents (B) a = dansyl hydroxide, b = dansyl amide, 1 = tyrosine (*o*-DNS-derivative), 2 = lysine (ϵ -DNS-derivative); 3 = histidine (bis-DNS-derivative) The standard dansyl amino acids that are used are indicated as black spots

17. Having recorded one's observations after the second solvent, the plate is now run in solvent 3 in the *same direction* as solvent 2. The plate is run until the solvent is 1 cm from the top, and this again takes 10–15 min.
18. After drying the plate in a stream of warm air (1–2 min), the plate is again viewed under UV light. The fast-running derivatives seen in solvent 2 have now run to the top of the plate and are generally indistinguishable (hence the need to record one's observations after solvent 2). However, the slow-moving derivatives in solvent 2 have now been separated by solvent 3 and can be identified if present. The separation obtained after solvent 3 is also shown in Fig. 2. The sum of the observations made after solvents 2 and 3 should identify the number and relative intensities of *N*-terminal amino acids present in the original sample.

Notes

1. It is important that the initial coupling reaction between dansyl chloride and the peptide occurs in the pH range 9.5–10.5. This pH provides a compromise between the unwanted effect of the aqueous hydrolysis of dansyl chloride and the necessity for the *N*-terminal amino group to be unprotonated for reaction with dansyl chloride. The condition used, 50% acetone in bicarbonate buffer, provides the necessary environment. The presence of buffer or salts in the peptide (or protein) sample should therefore be avoided to prevent altering the pH to a value outside the required range.
2. Because of the unpleasant and irritant nature of pyridine vapor, loading of samples onto the dansyl plates should preferably be carried out in a fume cupboard.
3. The viewing of dansyl plates under UV light should always be done wearing protective glasses or goggles. Failure to do so will result in a most painful (although temporary) conjunctivitis.
4. Most dansyl derivatives are recovered in high (> 90%) yield. However, some destruction of pro-

line, serine, and threonine residues occurs during acid hydrolysis, resulting in yields of approximately 25, 65, and 70%, respectively. When viewing these derivatives, therefore, their apparent intensities should be visually 'scaled-up' accordingly.

5. The sensitivity of the dansyl method is such that as little as 1–5 ng of a dansylated amino acid can be visualized on a chromatogram
6. The side chains of both tyrosine and lysine residues also react with dansyl chloride. When these residues are present in a peptide (or protein), the chromatogram will show the additional spots, *o*-DNS-Tyr and ϵ -DNS-Lys, which can be regarded as additional internal marker spots. The positions of these residues are shown in Fig. 2. These spots should not be confused with bis-DNS-Lys and bis-DNS-Tyr, which are produced when either lysine or tyrosine is the *N*-terminal amino acid
7. At the overnight hydrolysis step, dansyl derivatives of asparagine or glutamine are hydrolyzed to the corresponding aspartic or glutamic acid derivatives. Residues identified as DNS-Asp or DNS-Glu are therefore generally referred to as Asx or Glx, since the original nature of this residue (acid or amide) is not known. This is of little consequence if one is looking for a single *N*-terminal residue to confirm the purity of a peptide or protein. It does, however, cause difficulties in the dansyl-Edman method for peptide sequencing (see Chapter 24) where the residue has to be identified unambiguously
8. When the first two residues in the peptide or protein are hydrophobic residues, a complication can occur. The peptide bond between these two residues is particularly (although not totally) resistant to acid hydrolysis. Under normal conditions, therefore, some dansyl derivative of the first amino acid is produced, together with some dansyl derivative of the *N*-terminal dipeptide. Such dipeptide derivatives generally run on chromatograms in the region of phenylalanine and valine. However, their behavior in solvents 2 and 3, and their positions relative to the marker derivatives should prevent misidentification

as phenylalanine or valine. Such dipeptide spots are also produced when the first residue is hydrophobic and the second residue is proline, and these dipeptide derivatives run in the region of proline. However, since some of the *N*-terminal derivative is always produced, there is no problem in identifying the *N*-terminal residue when this situation arises. A comprehensive description of the chromatographic behavior of dansyl-dipeptide derivatives has been produced (2).

9. Three residues are difficult to identify in the three solvent system described in the methods section, DNS-Arg and DNS-His because they are masked by the ϵ -DNS-Lys spot, and DNS-Cys because it is masked by DNS-hydroxide. If these residues are suspected, a fourth solvent is used. For arginine and histidine, the solvent is

0.05M trisodium phosphate:ethanol (3.1, v/v)

For cysteine the solvent is

1M ammonia:ethanol (1 1, v/v)

Both solvents are run in the same direction as solvents 2 and 3, and the residues are identified by comparison with relevant standards loaded on the reverse side of the plate.

10. When working with small peptides it is often of use also to carry out the procedure known as "double dansylation." Having identified the *N*-terminal residue, the remaining material in the dansyl tube is dried down and the dansylation process (steps 2-4) repeated. The sample is then redissolved in 50% pyridine (10 μ L) and a 1 μ L aliquot is examined chromatographically. The chromatogram will now reveal the dansyl derivative of *each* amino acid present in the peptide. Therefore for relatively small peptides (< 10 residues) a quantitative estimation of the amino acid composition of the peptide can be obtained. This method is not suitable for larger peptides or proteins since most residues will be present more than once in this case, and it is not possible to

- quantitatively differentiate spots of differing intensity.
- 11 Although the side chain DNS-derivative will be formed during dansylation if histidine is present in the peptide or protein sequence, this derivative is unstable to acid and is not seen during *N*-terminal analysis. Consequently, *N*-terminal histidine yields only the α -DNS derivative and not the bis-DNS compound as might be expected. The bis-DNS derivative is observed, however, if the mixture of free amino acids formed by acid hydrolysis of a histidine-containing peptide is dansylated and subsequently analyzed chromatographically (i.e., during "double dansylation").
 12. The dansylation method described here was developed for use with peptides. However, this method can also be applied quite successfully to proteins, although some difficulties arise. These are caused mainly by insolubility problems, which can limit the amount of reaction between the dansyl chloride and protein thus resulting in a lower yield of dansyl derivative, and the presence of large amounts of *o*-DNS-Tyr and ϵ -DNS-Lys on the chromatogram that can mask DNS-Asp and DNS-Glu. A modification of the basic procedure described here for the dansylation of proteins has been described (3).

References

- 1 Gray, W R, and Hartley, B S (1963) A fluorescent end group reagent for peptides and proteins *Biochem J* **89**, 59P
- 2 Sutton, M R, and Bradshaw, R A (1978) Identification of dansyl dipeptides. *Anal Biochem* **88**, 344-346
- 3 Gray, W R (1967) in *Methods in Enzymology* Vol XI (ed Hirs, C H W.) p 149 Academic Press, New York

Chapter 24

The Dansyl-Edman Method for Peptide Sequencing

John M. Walker

*School of Biological and Environmental
Sciences, The Hatfield Polytechnic,
Hatfield, Hertfordshire, England*

Introduction

The dansyl-Edman method for peptide sequencing uses the Edman degradation (*see* Chapter 26) to sequentially remove amino acids from the *N*-terminus of a peptide. Following the cleavage step of the Edman degradation, the thiazolinone derivative is extracted with an organic solvent *and discarded*. This contrasts with the direct Edman degradation method (Chapter 26), where the thiazolinone is collected, converted to the more stable PTH derivative, and identified. Instead, a small fraction (~5%) of the remaining peptide is taken and the newly liberated *N*-terminal amino acid determined in this sample by the dansyl method (*see* Chapter 23). Although the dansyl-Edman method results in successively less peptide

being present at each cycle of the Edman degradation, this loss of material is more than compensated for by the fact that the dansyl method for identifying *N*-terminal amino acids is about one hundred times more sensitive than methods for identifying PTH amino acids. The dansyl-Edman method described here was originally introduced by Hartley (1).

Materials

1. Ground glass stoppered test tubes (approx 65×10 mm, e.g., Quickfit MF 24/0). All reactions are carried out in this 'sequencing' tube.
2. 50% Pyridine (aqueous, made with AR pyridine) Store under nitrogen at 4°C in the dark. Some discoloration will occur with time, but this will not affect results.
3. Phenylisothiocyanate (5% v/v) in pyridine (AR) Store under nitrogen at 4°C in the dark. Some discoloration will occur with time, but this will not affect results. The phenylisothiocyanate should be of high purity and is best purchased as "sequenator grade." Make up fresh about once a month.
4. Water-saturated *n*-butyl acetate. Store at room temperature.
5. Anhydrous trifluoroacetic acid (TFA) Store at room temperature under nitrogen.

Method

1. Dissolve the peptide to be sequenced in an appropriate volume of water, transfer to a sequencing tube, and dry *in vacuo* to leave a film of peptide in the bottom of the tube (see Note 1).
2. Dissolve the peptide in 50% pyridine (200 μ L) and remove an aliquot (5 μ L) for *N*-terminal analysis by the dansyl method (see Chapter 23).
3. Add 5% phenylisothiocyanate (100 μ L) to the sequencing tube, mix gently, flush with nitrogen and incubate the stoppered tube at 50°C for 45 min.

4. Following this incubation, unstopper the tube and place it *in vacuo* for 30–40 min. The desiccator should contain a beaker of phosphorus pentoxide to act as a drying agent, and if possible the desiccator should be placed in a water bath at 50–60°C. When dry, a white 'crust' will be seen in the bottom of the tube. This completes the coupling reaction.
5. Add TFA (200 μ L) to the test-tube, flush with nitrogen and incubate the stoppered tube at 50°C for 15 min.
6. Following incubation, place the test tube *in vacuo* for 5 min. TFA is a very volatile acid and evaporates rapidly. This completes the cleavage reaction.
7. 'Dissolve' the contents of the tube in water (200 μ L). Do not worry if the material in the tube does not all appear to dissolve. Many of the side-products produced in the previous reactions will not in fact be soluble.
8. Add *n*-butyl acetate (1.5 mL) to the tube, mix vigorously for 10 s, and then centrifuge in a bench centrifuge for 3 min.
9. Taking care not to disturb the lower aqueous layer, carefully remove the upper organic layer and discard.
10. Repeat this butyl acetate extraction procedure once more and then place the test-tube containing the aqueous layer *in vacuo* (with the desiccator standing in a 60°C water bath if possible) until dry (30–40 min).
11. Redissolve the dried material in the test tube in 50% pyridine (200 μ L) and remove an aliquot (5 μ L) to determine the newly liberated *N*-terminal amino acid (the second one in the peptide sequence) by the dansyl method.
12. A further cycle of the Edman degradation can now be carried out by returning to step 3. Proceed in this manner until the peptide has been completely sequenced.

Notes

1. Manual sequencing is normally carried out on peptides between 2 and 30 residues in length and requires 1–5 nmol of peptide.

- 2 Since the manipulative procedures are relatively simple it is quite normal to carry out the sequencing procedure on 8 or 12 peptides at one time
- 3 A repetitive yield of 90–95% is generally obtained for the dansyl-Edman degradation. Such repetitive yields usually allow the determination of sequences up to 15 residues in length, but in favorable circumstances somewhat longer sequences can be determined.
4. The amino acid sequence of the peptide is easily determined by identifying the new *N*-terminal amino acid produced after each cycle of the Edman degradation. However, because the Edman degradation does not result in 100% cleavage at each step, a background of *N*-terminal amino acids builds up as the number of cycles increases. Also, as sequencing proceeds, some fluorescent spots reflecting an accumulation of side products can be seen towards the top of the plates. For longer runs (10–20 cycles) this can cause some difficulty in identifying the newly liberated *N*-terminal amino acid. This problem is best overcome by placing the dansyl plates from consecutive cycles adjacent to one another and viewing them at the same time. By comparison with the previous plate, the increase of the new residue at each cycle, over and above the background spots, should be apparent.
- 5 A single cycle takes approximately 2.5 h to complete. When this method is being used routinely, it is quite easy to carry out three or four cycles on eight or more peptides in a normal day's work. During the incubation and drying steps the dansyl samples from the previous days sequencing can be identified.
- 6 As sequencing proceeds, it will generally be necessary to increase the amount of aliquot taken for dansylation at the beginning of each cycle, since the amount of peptide being sequenced is reduced at each cycle by this method. The amount to be taken should be determined by examination of the intensity of spots being seen on the dansyl plates.
7. It is most important that the sample is completely dry following the coupling reaction. Any traces of water present at the cleavage reaction step will introduce

hydrolytic conditions that will cause internal cleavages in the peptide and a corresponding increase in the background of *N*-terminal amino acids.

- 8 Very occasionally it will prove difficult to completely dry the peptide following the coupling reaction and the peptide appears oily. If this happens, add ethanol (100 μ L) to the sample, mix, and place under vacuum. This should result in a dry sample.
- 9 When removing butyl acetate at the organic extraction step, take great care not to remove *any* of the aqueous layer as this will considerably reduce the amount of peptide available for sequencing. Leave a small layer of butyl acetate above the aqueous phase. This will quickly evaporate at the drying step.
- 10 Vacuum pumps used for this work should be protected by cold traps. Considerable quantities of volatile organic compounds and acids will be drawn into the pump if suitable precautions are not taken.
- 11 The identification of *N*-terminal amino acids by the dansyl method is essentially as described in Chapter 23. However, certain observations peculiar to the dansyl-Edman method are described below.
12. Tryptophan cannot be identified by the dansyl method as it is destroyed at the acid hydrolysis step. However, where there is tryptophan present in the sequence, an intense purple color is seen at the cleavage (TFA) step involving the tryptophan residue that unambiguously identifies the tryptophan residue.
13. If there is a lysine residue present in the peptide, a strong ϵ -DNS-lysine spot will be seen when the dansyl derivative of the *N*-terminal amino acid is studied. However, when later residues are investigated the ϵ -DNS-lysine will be dramatically reduced in intensity or absent. This is because the amino groups on the lysine side chains are progressively blocked by reaction with phenylisothiocyanate at each coupling step of the Edman degradation.
14. The reactions of the lysine side chains with phenylisothiocyanate causes some confusion when identifying lysine residues. With a lysine residue as the *N*-terminal residue of the peptide it will be

identified as bis-DNS-lysine. However, lysine residues further down the chain will be identified as the α -DNS- ϵ -phenylthiocarbamyl derivative because of the side chain reaction with phenylisothiocyanate. This derivative runs in the same position as DNS-phenylalanine in the second solvent, but moves to between DNS-leucine and DNS-isoleucine in the third solvent. Care must therefore be taken not to misidentify a lysine residue as a phenylalanine residue.

15. When glutamine is exposed as the new *N*-terminal amino acid during the Edman degradation, this residue will sometimes cyclize to form the pyroglutamyl derivative. This does not have a free amino group, and therefore effectively blocks the Edman degradation. If this happens, a weak DNS-glutamic acid residue is usually seen at this step, and then no other residues are detected on further cycles. There is little one can do to overcome this problem once it has occurred, although an enzyme that cleaves off pyroglutamyl derivatives has been reported (2)
16. The main disadvantage of the dansyl-Edman method compared to the direct Edman method is the fact that the dansyl method cannot differentiate acid and amide residues. Sequences determined by the dansyl-Edman method therefore usually include residues identified as Asx and Glx. This is most unsatisfactory since it means the residue has not been unambiguously identified, but often the acid or amide nature of an Asx or Glx residue can be deduced from the electrophoretic mobility of the peptide (3)
17. Having identified any given residue it can prove particularly useful to carry out the procedure referred to as "double dansylation" on this sample (*see* Chapter 23). This double-dansylated sample will identify the amino acids remaining beyond this residue. Double dansylation at each step should reveal a progressive decrease in the residues remaining in the peptide, and give an excellent indication of the amount of residues remaining to be sequenced at any given cycle

References

1. Hartley, B. S. (1970) Strategy and tactics in protein chemistry *Biochem J* **119**, 805–822
2. Ulliana, J. A., and Doolittle, R. F. (1969) *Arch Biochem Biophys* **131**, 561
3. Offord, R. E. (1966) Electrophoretic mobilities of peptides on paper and their use in the determination of amide groups *Nature* **211**, 591–593

Chapter 25

Microsequencing of Peptides and Proteins with 4-*N,N*- Dimethylamino- azobenzene- 4'-isothiocyanate

***Brigitte Wittmann-Liebold
and Makoto Kimura***

*Max-Planck-Institut für Molekulare Genetik,
Abteilung Wittmann, D-1000 Berlin 33
(Dahlem), West Germany*

Introduction

Manual methods for the stepwise *N*-terminal degradation of polypeptides have been widely applied in protein chemistry. The technique most commonly used up to now, has been the Edman degradation, carried out either

manually (1) or automatically (2–4). Although nowadays the reaction can be performed with high efficiency and automatically in a sequencer, the manual methods are still of value. The reasons for this are: (i) The manual methods can be easily applied in any laboratory, even by an inexperienced researcher, with a minimum of equipment at low cost, (ii) It is possible to simultaneously screen many peptides for purity, and to gain information in a reasonable time about the *N*-terminal sequences of sets of peptides generated by proteolytic or chemical cleavage, (iii) The selection of fragments of high purity for the sequencer or those peptides that need to be sequenced in a machine because of length, hydrophobicity, or difficult sequence stretches is made easier.

For many years, the manual Edman degradation has been performed either directly (see Chapter 26) or by the dansyl-Edman technique (see Chapter 24). The latter technique has been applied to thousands of peptides and has proved especially valuable when proteins available in submicromol amounts must be sequenced. However, this method has several drawbacks, such as the unavoidable deamidation of glutamine and asparagine during acid hydrolysis of the dansyl-peptide (see Chapter 24 for a detailed discussion of this method).

In 1976, Chang et al. proposed a new chromophoric reagent for protein sequence analysis, 4-*N,N*-dimethylaminoazobenzene-4'-isothiocyanate (DABITC), which allows the identification of the released amino acids as red-colored derivatives (5,6). Several other modified Edman-type reagents have been proposed for sequential degradation of peptides, however, it was shown that isothiocyanate derivatives with bulky side chains couple less efficiently than PITC itself. Also, in the case of the DABITC reagent, coupling efficiency was low, about 60%, for a reasonable reaction time and temperature, e.g., 40 min and 50°C. Therefore it was necessary to complete the coupling with PITC prior to the cleavage reaction (7) in order to avoid severe carryover during degradation. This DABITC/PITC double coupling technique was adequate for practical sequencing in the low nanomol range. In the past few years (8,9), these improvements have enabled the complete analysis of the primary structures of all 53

proteins derived from the *Escherichia coli* ribosome. In this chapter we describe the DABITC/PITC coupling method as it is currently being used for microsequencing. Further, we describe the use of this coupling method in connection with manually performed solid-phase degradations.

Materials

Since the use of high purity reagents is necessary for optimal results when sequencing proteins and peptides, purification procedures for most of the reagents used are included in this section.

- 1 The degradations are carried out in thick-walled glass-stoppered centrifugation tubes (borosilicate glass) and the peptides to be degraded remain in these tubes during the degradation process. The tubes are narrowed at the bottom and have the following sizes: for 5–10 nmol of peptide length, 45 mm and i.d., 0.8 mm, for 300 pmol to 2 nmol, 35 mm $\times$ 0.3 mm. In the latter case, micro-injection tubes, suitable for 5–100 μ L volumes, and fitted with silicon stoppers are used. All tubes must fit snugly into thermostatically heated blocks maintained at 55 °C.
- 2 For all centrifugation steps use a bench centrifuge (e.g., Labofuge I supplied by Heraeus/Christ) equipped with adaptors capable of holding the stoppered tubes.
- 3 Two vacuum pumps that provide a vacuum of 2 Pa measured at the pump inlet. One pump is used for the removal of alkaline solvents after the coupling and the other for the removal of the acid after cleavage. Each vacuum line has a glass distributor for at least six desiccators, the acid vapors are trapped by KOH, and both pumps are protected by a cryostat placed in the vacuum lines.
- 4 For all experiments, nitrogen having 99.999 vol %, and containing less than 2 vpm = 2×10^{-4} vol % oxygen should be used.

5. Polyamide thin-layer sheets (DC-Mikropolyamide Foils, F1700, 15 × 15 cm, purchased from Schleicher & Schull) or silica gel thin-layer sheets (HPTLC Kieselgel, 60 F254, 10 × 10 cm, from Merck) are used for the identification of the DABTH-amino acid derivatives. A paper cutter is used to cut the 5 × 5 cm, or 2.5 × 2.5 cm-sized sheets.
6. *p*-Phenylisothiocyanate (PITC) of the fourth highest purity grade is redistilled under reduced pressure (oil pump) and nitrogen, while keeping the distillate at 4°C. Small amounts for the manual degradations are kept in sealed glass ampules under nitrogen at -20°C, larger amounts are stored as stock in 3–4 mL portions also in ampules, under identical conditions.
7. 4-*N,N*-dimethylaminoazobenzene-4'-isothiocyanate (DABITC) is recrystallized from acetone and 1 g is then dissolved in 70 mL of boiling acetone, filtered through a paper filter, and allowed to cool slowly. This procedure yields about 0.7 g of brown flakes with a mp of 169–170°C. The reagent is kept as stock in solid form under nitrogen. For manual sequencing, it is dissolved in redistilled acetone and portions of 0.6 mg are transferred to 1-mL plastic tubes, dried *in vacuo*, and stored at -20°C.
8. *p*-Phenylene diisothiocyanate (DITC) is recrystallized from acetone. A 2 g quantity of DITC is dissolved in 10 mL of boiling acetone and allowed to cool slowly. This yields 1.55 g of needles, mp 129–131°C, that give a clear, colorless solution in DMF.
9. 1-Ethyl-3-(3-dimethylaminopropyl)-carbodiimide/HCl (EDC), mp 112–115°C, is used without further purification.
10. CPG-10, controlled pore glass (Serva) 40–80 μm diameter (mesh-size 200–400) with a nominal pore diameter of 75 Å.
11. 3-Aminopropyl triethoxysilane (APTS) (Pierce) is used without further purification.
12. Aminopropyl glass beads (APG). 4 g of CPG beads are degassed *in vacuo* for 2 h at 180°C (water pump), then cooled *in vacuo*. A 30 mL quantity of dry toluene and 3 mL of APTS are added. After degassing, the reaction is performed at 75°C for 24 h in a stoppered flask un-

der nitrogen, with gentle stirring. The resin is filtered on a sintered glass filter and washed alternately with toluene, acetone, and methanol (2×50 mL, each). The APG is dried *in vacuo* over P_2O_5 at room temperature and stored at 4°C under nitrogen.

13. *p*-Phenylene diisothiocyanate activated glass (DITC-glass). 1 g of DITC is dissolved in 13 mL of DMF, and 2 g of APG are added in portions, with gentle stirring over a 1-h period. The mixture is kept at room temperature for 2 h and the beads are washed with DMF and methanol (2×10 mL, each) on a sintered glass filter. The resin is dried *in vacuo*. Storage is as described for APG.
14. Acetone, pro analysis grade (Merck) A 2.5 L quantity of acetone is passed through a column (id of 3.5 cm) of silica (Silica 100–200 active; filled to 8 cm height) and Al_2O_3 (alumina, grade neutral, activity I, height 10 cm), and then redistilled over charcoal (pro analysis grade) in a 30 cm glass column filled with glass rings, bp 56°C . The solvent is kept over molecular sieve type-3 Å pellets, 2 mm (Merck).
15. Acetic acid, pro analysis grade (Merck) is used without further purification.
16. *n*-Butyl acetate, pro analysis grade (Merck) is used without further purification.
17. Chloroform, pro analysis grade (Merck), is used without further purification.
18. Dimethylformamide (DMF), pro analysis grade (Merck), is purified as follows. Add to 263 mL of DMF, 34 mL of benzene and 12 mL of H_2O . Redistill and discard the first 60 mL of distillation product, up to 150°C . The product above this temperature is collected and redistilled over P_2O_5 under reduced pressure (water pump). DMF is stored over molecular sieve pellets (see acetone) and under nitrogen.
19. Ethanolamine (EA), synthesis grade (Merck), is redistilled. Its bp at 15 mm Hg is 77°C . Store in ampules under nitrogen at -20°C .
20. Ethyl acetate, pro analysis grade (Merck), is purified and stored as given for acetone (bp 77°C).
21. Formic acid, pro analysis grade (Merck), is used without further purification.

22. *n*-Heptane, Uvasol grade (Merck), is used without further purification.
23. *n*-Hexane, pro analysis grade (Merck), is used without further purification.
24. Methanol, Uvasol grade (Merck) is purified and stored as given for acetone (bp 64°C).
25. *N*-methyl morpholine (NMM), synthesis grade (Merck), is distilled over ninhydrin and redistilled over a 30 cm column filled with glass rings. It is stored in ampules under nitrogen at -20°C.
26. Pyridine (PYR), pro analysis grade (Merck), is thrice distilled, once over KOH, then over ninhydrin, and finally over KOH (bp 114–116°C). It is stored in 100-mL glass stoppered flasks (sealed with parafilm) under nitrogen at -20°C.
27. Toluene, pro analysis grade (Merck), is purified as described for acetone, and stored over molecular sieve.
28. Triethylamine (TEA), synthesis grade from Merck, is distilled from phthalic anhydride (bp 89°C). It is stored in ampules under nitrogen at -20°C.
29. Trifluoroacetic acid (TFA), purum grade (Fluka), is distilled over BaO, redistilled from $\text{CaSO}_4 \times 0.5 \text{ H}_2\text{O}$ (dried at 500°C immediately before use) over a 30 cm column filled with glass rings (bp 72–73°C).
30. Attachment buffer 1: *N*-methyl morpholine/TFA buffer, pH 9.5: 5 mL of NMM and 5 mL of H_2O are mixed under nitrogen (resultant pH 11.4). About four drops of TFA are added to adjust pH to 9.5. Use freshly prepared buffer and store it under nitrogen.
31. Attachment buffer 2. pyridine/HCl buffer, pH 5.0. 8 mL of pyridine is added to 80 mL of H_2O , they are mixed under nitrogen, and then 5.3 mL of 32% HCl and H_2O is used to adjust to a final volume of 100 mL at pH 5.0. Keep the buffer under nitrogen.
32. Attachment buffer 3 anhydrous buffer, DMF/TEA 2 mL volume of triethylamine is added to 200 mL of DMF. Alternatively, attachment of peptide under anhydrous conditions is carried out in DMF alone.
33. Attachment buffer 4. bicarbonate buffer, pH 9.0: 0.2M NaHCO_3 is adjusted to pH 9.0 with 1M NaOH.
34. High performance liquid chromatography (HPLC) of the DABTH-amino acid derivatives is performed on a

250 × 4 mm column filled with Shandon MOS-Hypersil, 5 μ m, at 45°C (see Fig. 2.)

Methods

Manual DABITC/PITC Double Coupling Method

The following procedure is for peptide amounts of 2–5 nmol. With amounts below 2 nmol it is recommended that volumes about one-quarter of those given here be used and that the reaction be performed in restricted tubes (see Materials).

1. To the dried peptide add 80 μ L of 50% pyridine in water, 40 μ L of DABITC solution (2.3 mg/mL in pyridine), flush with nitrogen, and incubate at 55°C for 30 min.
2. Add 5 μ L of PITC and incubate at 55°C for a further 30 min. This addition is necessary to avoid any carryover from one degradation step to the next. However, if the N-terminal residue of a peptide alone is wanted (end-group determination), this second coupling can be eliminated (see ref. 22).
3. This solution is then extracted with 4 × 400 μ L of *n*-heptane/ethyl acetate (2:1, v/v) under nitrogen with stirring. Centrifuge and remove the upper organic layer with care. Great care should be taken to avoid withdrawal of the interphase layer, which often carries peptide precipitate when removing the upper phase. This is the reason why we prefer to make four extractions and to leave about 50 μ L of the top layer at each centrifugation step.
4. Following extraction, dry the lower aqueous phase *in vacuo* (~30 min). The complete dryness of the lower phase is crucial since any residual reagent or pyridine will lead to progressive salt accumulation at the cleavage step. An efficient vacuum must therefore be maintained to guarantee short, but complete drying stages

This is especially important if longer polypeptides are being degraded

- 5 To the dried sample add 50 μL of anhydrous TFA, flush with nitrogen, and incubate at 55°C for 10 min. TFA is then removed *in vacuo* (~ 10 min). This completes the cleavage reaction. Repetitions of the cleavage (2–4 times) are recommended in the case of hydrophobic peptide bonds, mainly, if Pro-Pro, Pro-Ile, Pro-Val, Pro-Phe, Ile-Ile, Val-Val, Ile-Val, and Val-Ile are involved.
6. To extract the thiazolinone, add 30 μL of water, flush with nitrogen, add 50 μL of *n*-butyl acetate. Vortex and centrifuge. Remove the upper organic layer, repeat the extraction once, combine the extracts, and dry *in vacuo*. The aqueous phase is also dried *in vacuo* and is subjected to further cycles.
- 7 Conversion of the dried thiazolinone to the respective DABTH-amino acid derivative is carried out by adding 40 μL of 50% TFA, flushing with nitrogen, and incubating at 55°C for 20 min. The sample is then dried *in vacuo*, redissolved in 2–5 μL of ethanol and aliquots taken for analysis (see the section on Identification of the DABTH–Amino Acid Derivatives).

Attachment Procedures to Glass Supports

Attachment of Lysine Peptides via Epsilon-Amino Groups to DITC–Glass

1. Transfer 5–10 nmol of salt-free peptide to a solid-phase tube and dry *in vacuo*.
2. Redissolve the sample in 400 μL of 0.2M $\text{NaHCO}_3/\text{NaOH}$ buffer, pH 9.0 (alternatively use attachment buffer 1 or 2). Check the solubility of the peptide by withdrawal of an aliquot for picomole amino acid analysis.
3. Add 10 mg of DITC–glass (prewashed in methanol), flush with nitrogen, degas *in vacuo*, and stir at 50°C for 60 min

4. To saturate remaining sites, add 20 μL of ethanolamine, flush with nitrogen, and incubate at 50°C for 15 min
5. Wash the beads twice with water (2 mL) and methanol (2 mL), and then dry *in vacuo* (but see note at end of the section on attachment via a Homoserine Lactone Residue)

Attachment of Peptides via C-Terminal Carboxyl Groups

1. Transfer 5–10 nmol of peptide to a solid-phase tube and dry *in vacuo* over P_2O_5 .
2. Add 100 μL of anhydrous TFA, flush with nitrogen, and incubate for 15 min at room temperature.
3. Dry the sample *in vacuo* (20–30 min), add 2 mg of EDC in 200 μL of DMF, 10 mg of APG (pre-washed in methanol and DMF), flush with nitrogen, degas, and keep at 40°C for 60 min with gentle stirring.
4. Centrifuge and wash the beads with water and methanol ($2 \times 2 \text{ mL}$)
5. To saturate the remaining sites, add 20 μL of PITC in 100 μL of DMF and 50 μL of pyridine in 200 μL of DMF. Flush with nitrogen, degas, and incubate at 50°C for 20 min.
6. Centrifuge, wash the beads with DMF and methanol ($2 \times 2 \text{ mL}$), and dry under vacuum (but see also note at end of the section on Attachment via a Homoserine Lactone Residue).

Attachment via a Homoserine Lactone Residue

1. Transfer 5–10 nmol of homoserine-containing peptide to a solid-phase tube and dry *in vacuo* over P_2O_5
2. Add 300 μL of anhydrous TFA, flush with nitrogen, and incubate at room temperature for 1 h. This will convert all homoserine residues to the lactone. Dry the sample *in vacuo*.
3. To achieve attachment, dissolve the peptide lactone in 300 μL of DMF, add 10 mg of APG (pre-washed in DMF), and 50 μL of TEA, flush with nitrogen, degas *in vacuo*, then incubate at 45°C for 2 h. Centrifuge and dry the sample *in vacuo*.

The final drying of the glass beads after attachment may be omitted if the covalently bound peptide is subjected to degradation immediately. Saturation of the residual amino groups of the APG after peptide attachment should be made with PITC. This can be done with the first coupling of the first degradation cycle if DABITC is replaced by PITC (therefore, no DABTH-amino acid can be obtained for the first residue of the peptide)

The Manual Solid-Phase DABITC/PITC Sequencing Method

The method described here is an improved version of one described previously (10,11). The reaction is carried out in larger Edman Tubes (1 × 7 cm) with gentle stirring by means of a stirring bar. A magnetic stirrer is placed below the heating block. Special care has to be taken to prevent physical losses of glass-attached peptide during vacuum drying. This can be a problem when removing the TFA after cleavage. 'Bumping out' of the glass beads *in vacuo* is prevented by means of a glass adaptor fitted with a G-2 sinter glass filter that replaces the stopper during the drying stages and restricts the vacuum. We further recommend that drying should be performed as follows: first, dry for 5 min in a Speed Vac Concentrator, then for 20 min in a desiccator

1. To the glass-attached peptide add 400 μL of 50% pyridine and 200 μL of DABITC solution (2.3 mg/mL in pyridine). Flush with nitrogen and incubate at 55°C for 30 min.
2. Add 20 μL of PITC, flush with nitrogen, and incubate at 55°C for a further 30 min. (At the first cycle, omit DABITC solution, replace with 50 μL of PITC). This completes the coupling reaction.
3. Centrifuge and remove the supernatant. Wash the beads twice with 500 μL of pyridine and then twice with 500 μL of methanol. Dry *in vacuo* (see above).
4. Add 200 μL of anhydrous TFA, flush with nitrogen, and incubate at 55°C for 8 min. Dry *in vacuo*. This completes the cleavage reaction.

- 5 The thiazolinone is extracted with 400 μL of methanol, and then 200 μL of methanol. The residual glass-attached peptide is dried *in vacuo* and then subjected to further cycles.
6. The combined methanol extracts are dried *in vacuo*, and the thiazolinone is converted by the addition of 80 μL of 50% TFA. The tube is flushed with nitrogen and incubated at 55°C for 20 min. The converted sample is then dried *in vacuo* and identified as described below.

Identification of the DABTH–Amino Acid Derivatives

Two-Dimensional Thin-Layer Chromatography (23)

1. Standard markers for chromatography are prepared as follows: add 500 μL of pyridine to a test tube and 30 μL each of diethylamine and ethanolamine. Add 250 μL of DABITC solution (2.3 mg/mL in pyridine) and incubate at 55°C for 1 h. Dry *in vacuo*, dissolve in 1 mL of ethanol, and apply a small spot together with each sample.
- 2 Dissolve the DABTH extract in 2–5 μL of ethanol (depending on the initial amount of peptide) and load 0.5 μL aliquots onto polyamide sheets (2.5 $\times$ 2.5 cm) together with a standard marker sample. It may be necessary to spot more with increasing number of degradation cycles because of extraction losses of peptide.
- 3 The chromatogram is then developed in two dimensions (1 min each). The solvents are:

1st dimension 33% acetic acid in water
2nd dimension toluene : *n*-hexane : acetic acid, 2:1:1, v/v/v
- 4 All DABTH-derivatives except DABTH-Ile/Leu can be separated with these solvent mixtures as illustrated in Fig. 1. They are visible as red spots after exposure to acid vapors (hold over a flask containing 12M HCl) whereas byproducts of the degradation are not visible or form blue-colored spots. The positions of the released DABTH-derivatives are correlated to the posi-

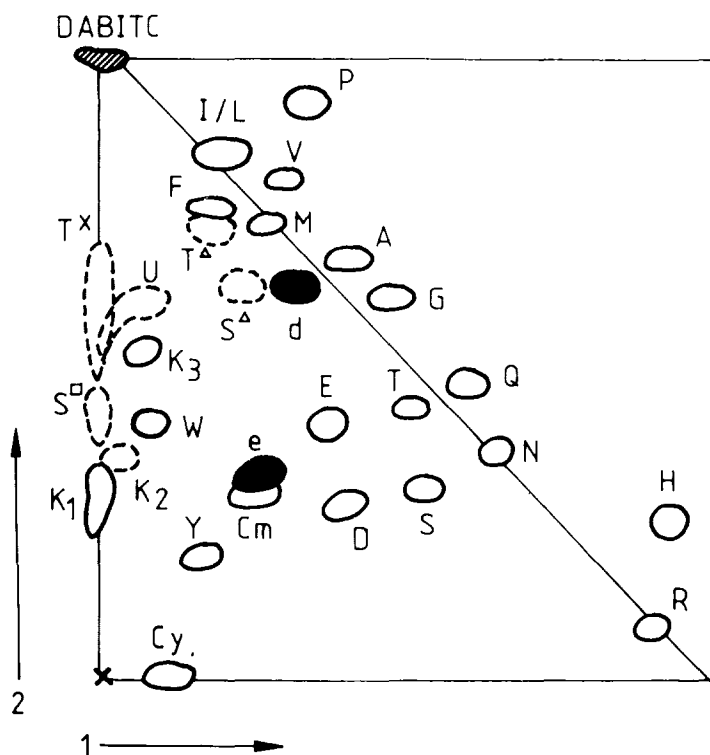


Fig. 1 Two-dimensional thin-layer chromatography of DABTH-amino acid derivatives. About 20–50 pmol of standard DABTH-amino acid derivatives are applied to 2.5×2.5 cm polyamide sheets (Schleicher and Schull). Details of the chromatography systems and the marked spots are given in the text. The DABTH-amino acid derivatives are denoted by the single-letter code, T^{Δ} , dehydrated DABTH-threonine, T^x , product formed after β -elimination of DABTH-Thr, S^{Δ} and $S^{\square}$, corresponding products formed by DABTH-Ser, U, thiourea derivative; K_1 , α -DABTH-epsilon-DABTC-Lys, K_2 , α -PTC-epsilon-DABTC-Lys, K_3 , α -DABTH-epsilon-PTC-Lys, spots "e" and "d" are blue-colored reference markers.

tions of two markers, DABITC-reacted diethylamine (spot "d" of Fig. 1) and ethanolamine (spot "e" of Fig. 1).

5. Thin-layer separation of DABTH-Ile and DABTH-Leu can be performed by one-dimensional chromatography on silica sheets (HPTLC, see materials section) using

formic acid:water:ethanol, 1.10:9, v/v/v, as the solvent mixture (12). As this type of thin-layer needs more material only isoleucine and leucine derivatives released in amounts from more than 5 nmol starting peptide material can be identified by this means. Alternatively, HPLC separations, as listed below, may be employed.

HPLC Separation of DABTH-Amino Acid Derivatives

DABTH-derivatives can be separated by isocratic elution in about 30 min (13). The self-packed column contains C₈ reversed-phase material, MOS-Hypersil, pore size 5 μ m (Shandon) and has a dimension of 250 $\times$ 4.6 mm. The solvent system used is 50% 12 mM Na acetate, pH 5.0/50% acetonitrile/0.5% 1,2-dichloroethane. The flow rate is 1.2 mL/min at 45°C. The separation of the DABTH-amino acid derivatives is presented in Fig. 2. DABTH-tryptophan elutes between valine and proline, and the derivative of histidine after alanine. DABTH-arginine is retained on the column under these conditions, but can be determined in a separate injection. It is eluted from the column, separated from all other derivatives, with 20% 12 mM Na acetate, pH 5.0/80% acetonitrile/0.5% 1,2-dichloroethane. This buffer may also be used to clean the column.

Notes

Manual Liquid-Phase DABITC/ PITC Method

- 1 The general reaction scheme (for the manual method) is presented in Fig. 3. A characteristic color change occurs between the reagent (purple), the thiocarbamoylpeptide (blue), and the released amino acid derivative (red) because of the differences in the resonance structures of the dimethylaminoazobenzene ring after exposure to the acid.
- 2 The method described can be applied to peptides and proteins. Care has to be taken to avoid salt contamination, as this prolongs all the drying stages, and can prevent the sample from being dried at all, in a reasonable time. After enzymatic cleavages peptides are

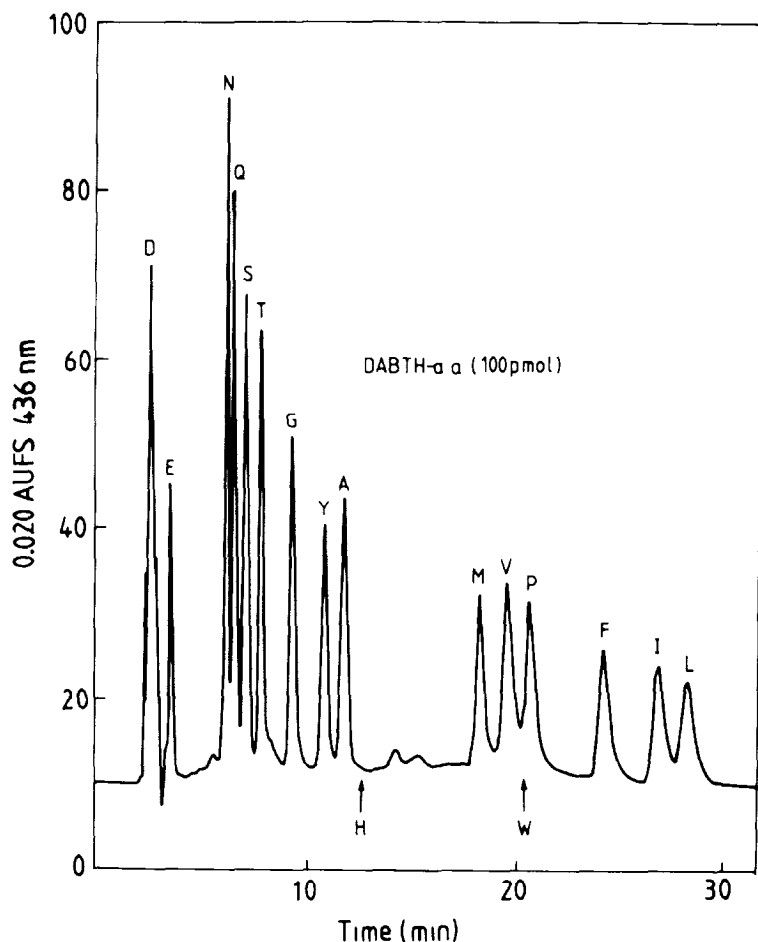


Fig. 2 Isocratic HPLC-separation of standard DABTH-amino acid derivatives. Separation was made on Shandon support, Hypersil MOS (C_8), $5\ \mu\text{m}$, column size was $250 \times 4.6\ \text{mm}$, that of the pre-column (filled with the same support) was $40 \times 4.6\ \text{mm}$, flow rate was $1.2\ \text{mL/min}$. Elution at 45°C was made with 50% $12\ \text{mM}$ Na acetate buffer, pH 5.0/50% acetonitrile/0.5% 1,2-dichloroethane. Injected were $100\ \text{pmol}$ of a standard amino acid-DABTH mixture; measurements were made at $436\ \text{nm}$ and $0.02\ \text{AUFS}$. All derivatives can be separated under these conditions, see text. However, DABTH-Arg does not elute from the column with this solvent mixture, but migrates separately from all other derivatives by eluting with 20% $12\ \text{mM}$ Na acetate buffer, pH 5.0/80% acetonitrile/0.5% 1,2-dichloroethane.

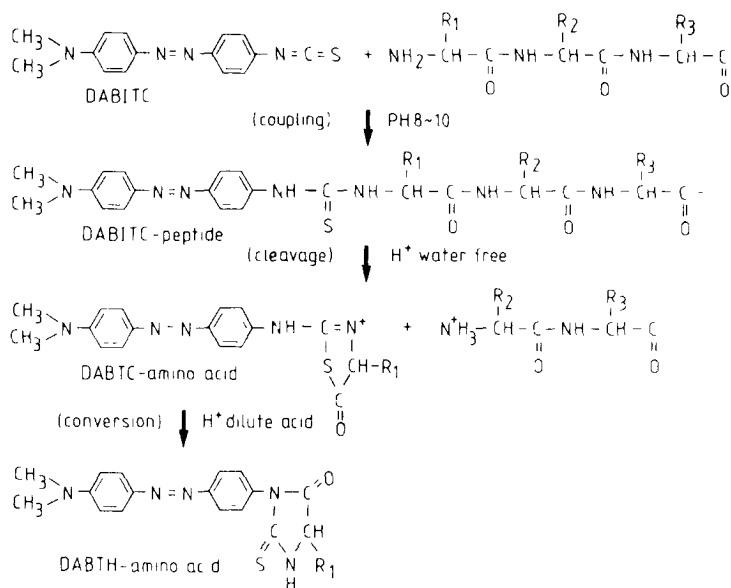


Fig. 3. Reaction scheme of the Edman-type degradation of polypeptides employing 4-*N,N*-dimethylaminoazobenzene-4'-isothiocyanate. The degradation consists of three parts: the coupling with the reagent in alkaline medium, the cleavage under acidic, water-free conditions, and the isomerization of the thiazolinone to the hydantoin amino acid derivative.

normally isolated by standard thin-layer techniques (*see* Chapter 4), and their elution is carried out preferably in dilute acids, 0.07% ammonia or 20% pyridine in water (for details, *see* ref. 14). Peptides deriving from chemical cleavages are desalted on Sephadex columns prior to microsequencing. Alternatively, modern HPLC techniques may be employed for peptide purification (*see* Chapter 5).

Similarly, proteins should be isolated without recourse to ammonium sulfate precipitations, SDS or any method using high amounts of salt. Most suitable are proteins isolated by gel filtration on Sephadex, or by HPLC-methods employing buffers of low salt concentrations (see Chapter 2). Examples where proteins isolated from HPLC columns have been directly used for microsequencing are presented elsewhere (15,16).

3. The length of a polypeptide does not limit the application of the method, whereas with the dansyl-Edman technique, difficulties arise with larger sized peptides or proteins. The formation of a red-colored product with DABITC, which is easily visible by eye on thin-layer sheets, down to 20 pmol is a considerable advantage. On the other hand, byproducts of the reaction, which are visible in UV light, are not seen. All polypeptides containing asparagine or glutamine are preferably degraded with this method. Here, the choice of the conversion medium is important. TFA in water (30–50%) produces high yields of the amide derivatives (DABTH-Asn and -Gln), free of the acid derivatives, provided the time for the reaction and the temperature of the block are optimized with standard peptides. Further, arginine and histidine residues are recovered in good yields. On the other hand, serine containing peptides give higher yields of DABTH-Ser if the conversion is performed in 1M HCl, or acetic acid saturated with HCl gas.
4. Serine residues give several spots on the thin-layer chromatogram and lysine yields a number of products because of the use of two coupling reagents, DABITC and PITC, and the presence of α - and epsilon-amino groups during the reaction (*see* Fig. 1). When sequencing less than 1 nmol of peptide or protein, those amino acids that produce more than one spot (lysine, serine, cysteine, and threonine) are difficult to identify or not seen at all. Cysteine can be identified after oxidation or alkylation of the residue in the peptide.
5. Because of the extractions, hydrophobic peptides or those that have the hydrophilic residues in the first part of the sequence are washed out during extraction. In such cases it is better to attach the peptide covalently to a support prior to manually performing the DABITC/PITC degradation.

Manual Solid-Phase DABITC/ PITC Method

6. Peptide attachment to solid supports is essentially as described for the use in solid-phase sequencers (17)

In principle, all the supports described for solid-phase sequencing, such as polystyrene or glass, may be used. However, the cleanest results are obtainable for peptides that are linked to glass supports (18). Therefore, we have described only these attachment procedures

- 7 All types of peptides including the hydrophobic ones are suitable for the solid-phase DABITC method provided they contain a group for the attachment. Best peptide attachment has to be selected according to the solubility and the chemical nature of the peptide; all peptides containing a C-terminal lysine are preferably attached via the epsilon-amino group to DITC-activated glass (17) (with yields of 80% or better), all small peptides with free carboxyl-termini but lacking a C-terminal lysine can be attached to APG via carboxyl-activation with EDC (19,20), with sufficient yield (approximately 50–80%); peptides containing homoserine are bound to APG (17) in yields above 80%. Proteins are best attached via their side chain lysine residues. If several lysines are present, a quantitative reaction of all lysines is not necessary for a tight binding of the protein to the glass. Such bound proteins may be satisfactorily sequenced. It is frequently observed that the sequencing ability of the bound protein decreases with increased reaction of all lysines. Polypeptides eluted from gels, even in the presence of SDS, can be subjected to solid-phase sequencing by the manual DABITC method
- 8 All peptides that are attached to the support in good yields can easily be sequenced to their C-terminal end by the manual method. Only in cases of steric hindrance, because of hydrophobic sequences involving proline, isoleucine or valine, is a degradation with repeated cleavages necessary to avoid severe carryover. A peptide or protein may be attached to glass in the presence of SDS (about 0.1%), which is often necessary for hydrophobic peptides. The easiest attachment is under waterfree conditions in DMF, with or without the addition of a base, after pretreatment with TFA (*see Methods*).
- 9 Limits to the solid-phase method arise from the low solubility of some of the polypeptides during the at-

tachment Sometimes, insoluble polypeptides behave better with the liquid-phase DABITC technique Even if they are not soluble in 50% pyridine, they react with the DABITC/PITC reagents to a certain degree, and an efficient degradation can be obtained if the coupling (and extraction thereafter) is repeated several times Polypeptides that form a separate layer between the phases or at the bottom of the tube can still become degraded

- 10 All peptides lacking a lysine (or aminoethylated cysteine) in the C-terminal region, a C-terminal homoserine or a free C-terminal carboxyl-group (carrying a C-terminal amide) cannot be attached to a solid support by the methods described. Covalent binding to a support may then be achieved via tyrosine, tryptophan, or the cysteine-side chain Carboxyl-attachment is limited by lower yields. Further, peptides with C-terminal glutamine and proline cannot be linked via their C-terminal carboxyl-group to APG. However, the presence of several side chain carboxyl-groups neither disturbs this attachment nor sequencing of the peptide significantly Under mild attachment conditions (attachment at room temperature for 30 min) at least 30–50% of glutamic acid and aspartic acid residues can be positively identified upon degradation. Attachment via the carboxyl terminus of a protein is also possible, but although good attachment yields have been observed, the sequencing results are poor
- 11 The glass beads used for coupling are stable except in the presence of strong bases, especially at high temperature. Therefore, attachment should be carried out under the conditions described in the Methods section, and the coupling buffers should be kept below pH 10. The glass supports decrease in stability in TFA, as they are sensitive to hydrogen fluorides. Therefore, all cleavage times should be kept as short as possible and the acid removed quickly

Identification of the DABTH-Amino Acid Derivatives

- 12 Steel holders for drying 10 sheets after chromatography (by means of a cold fan) are helpful Plexiglass

holders have the advantage that they are acid resistant and may be placed in a desiccator over HCl vapors for color development of the amino acid derivatives on the sheets. The best solvent chambers for the small sheets are flat-bottomed 50-mL beakers. The rim of the beaker is cut away and the cut surface polished, so that a polished glass plate can be used to close the chamber.

- 13 The color difference between DABITC, DABTC-peptides, and DABTH-derivatives greatly facilitates the identification of released amino acids. On thin-layers, 10 pmol of the DABTH-amino acid derivatives is the lower detection limit, quantitative determinations by HPLC can be made with at least 20 pmol.
- 14 Serine and threonine give additional spots, namely the dehydrated derivatives, and an additional blue colored spot, T^x, near thiourea (marked with U), lysine forms as main product α -DABTH-epsilon-PTC-lysine, a red spot which is marked as K3 in Fig. 1, homoserine moves to an almost identical position as DABTH-Thr, but can be differentiated because of the missing blue-colored extra spot described for serine and threonine above. Carboxymethylated cysteine can be identified as a spot that moves into the position of the ethanolamine marker, as shown in Fig. 1.
15. Problems with the identification of the DABTH-amino acid derivatives can be caused by.
 - (a) Low purity of the DABITC reagent, which is indicated by the formation of an extra spot in the left upper corner of the chromatogram.
 - (b) Inefficient extraction after the coupling leads to a similar spot, which disturbs the migration of the DABTH-derivatives.
 - (c) The occurrence of double spots for each of the hydrophobic residues, because of incomplete conversion, or destruction (after too long an exposure to the dilute acid).
 - (d) Fast migration of the spots into the front of the second dimension, because the solvent mixture quickly changes its composition (second dimension solvent mixture should be stored at 4°C and used for less than 1 h in the small beaker).
 - (e) Low resolution of spots, mainly in the area

DABTH-His/Arg, caused by salt contamination. This can result from increasing salt accumulation during the degradation (incomplete or inefficient drying) or from salt contamination of the original DABTH-amino acid extract. In both cases, it is helpful to dry the DABTH-amino acid extract, to reextract it with water/butyl acetate, and to repeat the identification.

Application to Ribosomal Proteins

16. The DABITC methods described here have been applied to a large number of proteins and peptides derived from different sources over the last few years. Most of the peptides from the *E. coli* ribosomal proteins, whose sequences were finished in 1979–1982, were sequenced using these techniques. Especially difficult sequence areas can be determined in this manner. Further, purity controls of many ribosomal proteins isolated from other organisms were carried out using the manual method, in order to identify which proteins were pure enough to merit their sequencing in a sequencer.

More recently, the DABITC techniques have enabled the complete sequence analysis of about 16 ribosomal proteins isolated from *Bacillus stearothermophilus* in a rather short period of time and using only a few milligrams of material (21). The sequence determination of protein S5 from *B. stearothermophilus* may serve as an example (24). This protein has been manually sequenced exclusively by the liquid and solid-phase methods. Long sequence stretches of up to 30 residues were sequenced after attachment of the peptides via the C-terminal carboxyl-groups to aminopropyl glass. The combined manual techniques enabled to complete the sequence of this protein in a few months using only 3–4 mg. In summary, all the methods described here are applicable to any type of protein, and the inexpensive instrumentation makes microsequencing possible in almost all laboratories.

Acknowledgment

We are grateful to Mr. Keith Ashman for carefully reading the English version of the manuscript.

References

- 1 Edman P and Henschen A. (1975) Sequence Determination, in *Protein Sequence Determination*, Needleman S B, ed, Springer Verlag, Berlin, pp. 232–279.
- 2 Edman P and Begg G (1967) A protein sequenator *Eur J Biochem* **1**, 80–91.
- 3 Laursen R A (1971) Solid-phase Edman degradation, an automatic peptide sequencer. *Eur J Biochem* **20**, 89–102
- 4 Hewick R, Hunkapiller M, Hood L E, and Dreyer W J (1981) A gas-liquid solid phase peptide and protein sequenator *J Biol Chem* **256**, 7990–7997
- 5 Chang J Y, Creaser E H, and Bentley K W (1976) 4-*N,N*-Dimethylaminoazobenzene-4'-isothiocyanate, A new chromophoric reagent for protein sequence analysis *Biochem J* **153**, 607–611
- 6 Chang J Y and Creaser E H (1976) A novel manual method for protein sequence analysis *Biochem J* **157**, 77–85.
- 7 Chang J. Y, Brauer D, and Wittmann-Liebold B (1978) Micro-sequence analysis of peptides and proteins using 4-*N,N*-dimethylaminoazobenzene-4'-isothiocyanate/phenylisothiocyanate double coupling method, *FEBS Lett* **93**, 205–214
- 8 Wittmann H G, Littlechild J, and Wittmann-Liebold B (1980) Components of Bacterial Ribosomes in *Ribosomes*, Chambliss G, Craven G R, Davies J, Davis K., Kahan L, and Nomura M, eds, University Park Press, Baltimore, Maryland, pp 51–88
- 9 Giri L, Hill W E, Wittmann H G, and Wittmann-Liebold B. (1984) Ribosomal proteins their structure and spatial arrangement in prokaryotic ribosomes *Adv Protein Chem* **36**, 2–4648
- 10 Chang J Y (1979) Manual solid phase sequence analysis of polypeptides using 4-*N,N*-dimethylaminoazobenzene-4'-isoth *BBA* **578**, 188–195
- 11 Wittmann-Liebold B (1981) Micro-sequencing by manual and automated methods as applied to ribosomal proteins, in *Chemical Synthesis and Sequencing of Peptides and Proteins*

- Liu T , Schechter A , Henrikson R , and Condliffe P eds , Elsevier/North Holland, New York, pp 75–110
- 12 Yang C Y (1979) Die Trennung der 4,4-dimethylamino-phenyl-azophenylthiohydantoin-Derivate des Leucins und Isoleucins über Polyamid-dünnschichtplatten im Picomolbereich *Hoppe-Seyler's Z Physiol Chem* **360** 1673–1675
 - 13 Lehmann A and Wittmann-Liebold B (1984) Complete separation and quantitative determination of DABTH-amino acid derivatives by isocratic reversed phase high performance liquid chromatography *FEBS Lett* **196**, 360–364
 - 14 Wittmann-Liebold B and Lehmann A (1980) New Approaches to Sequencing by Micro- and Automatic Solid Phase Technique, in *Methods in Peptide and Protein Sequence Analysis*, Birr Ch ed , Elsevier/North Holland, Amsterdam, pp 49–72
 - 15 Kamp R M , Yao Z J , Bosserhoff A , and Wittmann-Liebold B , (1983) Purification of *Escherichia coli* 30S ribosomal proteins by high performance liquid chromatography *Hoppe-Seyler's Z Physiol Chem* **364**, 1777–1793
 - 16 Kamp R M and Wittmann-Liebold B (1984) Purification of *Escherichia coli* 50S ribosomal proteins by high performance liquid chromatography *FEBS Lett* **167**, 59–63
 - 17 Laursen R A (1977) Coupling Techniques in Solid Phase Sequencing, in *Meth Enzymol* **47**, Hirs C H W , and Timasheff S N , eds , pp 277–288
 - 18 Machleidt, W , Wachter, E , Scheulen, M , and Otto, J (1973), Coupling proteins to aminopropyl glass *FEBS Lett* **37**, 217–220
 - 19 Wittmann-Liebold B and Lehmann A (1975) Comparison of various Techniques Applied to the Amino Acid Sequence Determination of Ribosomal Proteins, in *Solid Phase Methods in Protein Sequence Analysis*, Laursen R A , ed , Pierce Chem Corp , Rockford, Ill , pp 81–90
 - 20 Salnikow J , Lehmann A and Wittmann-Liebold B (1981) Improved automated solid phase micro-sequencing of peptides using DABITC *Anal Biochem* **117**, 433–442
 - 21 Kimura M and Chow C K (1984) Complete amino acid sequences of ribosomal proteins L17, L27 and S9 from *Bacillus stearothermophilus* *Eur J Biochem* **139**, 225–234
 - 22 Chang J Y (1980) Amino-terminal analysis of polypeptide using dimethylaminoazobenzene isothiocyanate *Anal Biochem* **102**, 384–392
 - 23 Chang J. Y and Creaser E H (1977) Improved chromatographic identification of coloured amino acid thiohydantoin *J Chrom* **132**, 303–307
 - 24 Kimura, M (1984) Proteins of the *Bacillus stearothermophilus* Ribosome The Amino Acid Sequence of Proteins S5 and L30 *J Biol Chem* , **259**, 1051–1055

Chapter 26

Manual Edman Degradation of Proteins and Peptides

Per Klemm

*Department of Microbiology, The Technical
University of Denmark, Lyngby,
Denmark*

Introduction

The Edman or phenylisothiocyanate degradation (1) has been employed for the determination of the primary structures of peptides and proteins for approximately three decades. The relative simplicity of the method and its high efficiency in the sequential removal of amino acid residues from the amino terminus of a peptide or protein has resulted in a widespread popularity and usage. In spite of the full automatization of the procedure by Edman and Begg in the nineteen sixties (2), the manual version of the sequential amino acid degradation still remains a very realistic and efficient alternative.

The major advantages of manual Edman degradation of proteins or peptides reside in the cheapness and ease with which it can be established in any ordinary laboratory. Indeed in its most simple form all that is needed, apart from the necessary chemicals is a desiccator, a good vacuum pump, and a setup for thin-layer chromatography for identification purposes. Furthermore, since all operations are so easy to perform, little or no previous training is required. The manual method cannot compete with the automated versions either in repetitive yield, i.e., the amount of amino acid derivative recovered per cycle, nor in the number of cycles performed per day (unless you want to work overnight) on a specific peptide or protein. Consequently the size of an amino sequence that can be established from the *N*-terminal region of a peptide or protein falls in the range of 10–30 depending on the amount of starting material and the particular sequence. With automatic equipment this result is normally surpassed by a factor of 2. However, several samples, i.e., 6–12 can adequately be degraded simultaneously with little extra work in the case of the manual procedure—a capability automatic sequencing equipment is not endowed with.

Manual Edman degradation can favorably be applied to the following kind of problems

- 1 Characterization of *N*-terminal regions for identification purposes
- 2 Elucidation of short *N*-terminal primary structures for use in connection with DNA-sequence information for the establishment of reading frames and starting points
- 3 Limited sequencing projects, e.g., a peptide hormone
- 4 Sequencing of large number of peptides of up to 30–50 residues in connection with elucidation of complete protein sequences

The reactions involved in the isothiocyanate degradation are depicted in Fig. 1. The entire cycle can conveniently be divided into the following steps. coupling, wash, cyclization, and extraction (Fig. 2).

1. *The coupling reaction.* In this initial reaction, the free α -amino group of the peptide chain reacts with

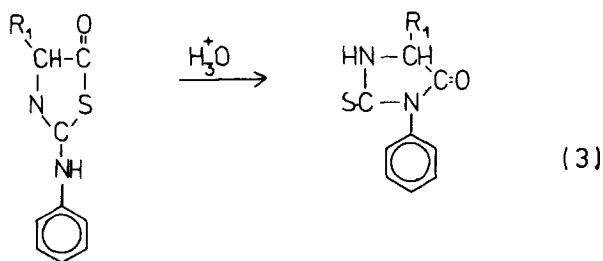
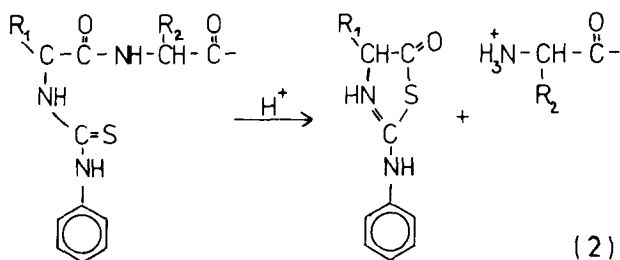
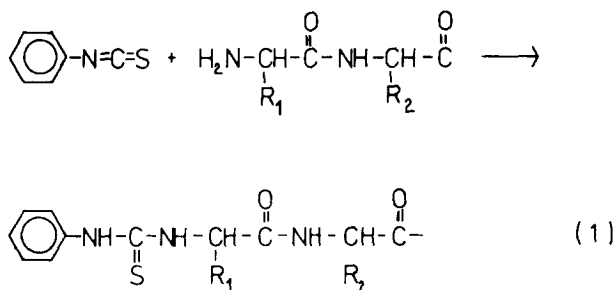


Fig. 1. The Edman degradation reactions and conversion step (1) the coupling reaction, (2) the cleavage reaction, (3) the conversion step

phenylisothiocyanate to form a phenylthiocarbamyl derivative (Fig. 1 1) The coupling reaction, which requires a charged amino group ($pK_a \approx 9.5$) is consequently performed in an alkaline milieu, e.g., aqueous pyridine. Furthermore, in order to avoid oxidation, the coupling is profitably carried out in a nitrogen atmos-

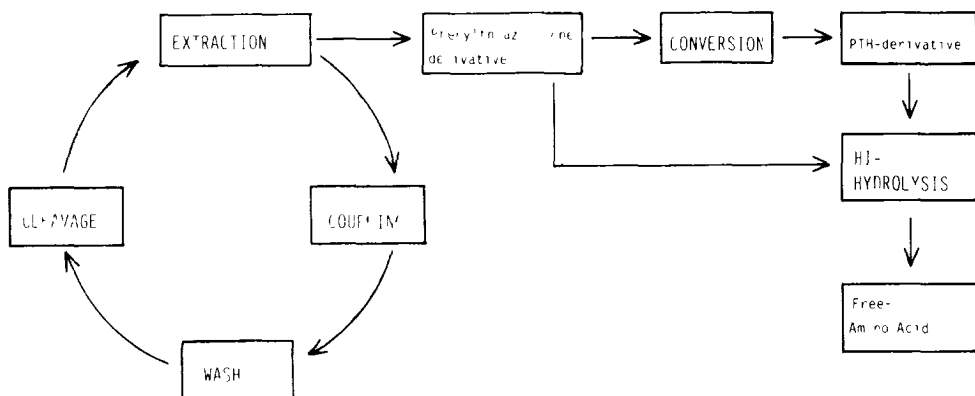


Fig. 2 Diagram showing the steps involved in the manual Edman degradation

phere. The reaction is generally completed in 10–20 min at 50°C in the case of peptides, but usually takes longer with proteins, in which the α -amino group can be less accessible.

2. *Washing* In order to remove excess reagent and byproducts from the coupling mixture, the latter is washed with organic solvents, ideally without removing any protein or peptide. In the case of proteins this is normally not a problem. However, with highly hydrophobic peptides the risk of losses resulting from extraction in the organic phase is normally present. If identification by means of the parent amino acid is employed (see later) the washing step can be dispensed of
3. *Cleavage reaction.* The next reaction (Fig. 1.2) involves cyclization of the phenylthiocarbamyl-peptide and concerted cleavage of the peptide bond. This results in the release of the phenylthiazoline derivative of the first amino acid and demasking of the α -amino group of the second residue. The cleavage reaction requires a strongly acidic milieu in order to proceed. At the same time the presence of water should be avoided to prevent hydrolysis of the peptide chain. The solvent of choice for this reaction is trifluoroacetic acid or other perfluorated carbon acids, which apart from providing an adequate low pH are very good solvents and like-

wise volatile. The cyclization proceeds rapidly at 50°C and is completed within a few minutes, although proline residues are liberated at a slower rate. The conditions used in the cleavage step are unfortunately conducive for dehydration of serine and threonine residues. Special precautions to diminish this have been described extensively elsewhere (3).

4. *Extraction* After evaporation of the cyclization medium, the residual polypeptide chain is separated from the phenylthiazolinone derivative of the terminal amino acid by liquid partition. This is best performed by extraction of a neutral aqueous solution, e.g., 5% pyridine, that is a good solvent for peptides with a moderate hydrophobic organic solvent like *n*-butyl acetate. The extraction is best carried out on an ice bath in order to diminish side reactions and to provide better separation.

The cycle is then completed and after drying a new series of reactions can be started that bring the Edman chemistry to play on the new *N*-terminal amino acid now constituted by the original second residue.

The phenylthiazolinone derivatives could in principle be used for identification of the respective amino acid residues. However, they are extremely unstable and in practice are therefore not useful for this purpose. Instead, the phenylthiazolinones are converted into the corresponding phenylthiohydantoin (PTH) derivatives, which are quite stable and can be separated and identified by various methods. The conversion reaction (Fig. 1) takes place by heating in dilute acid, e.g., 20% trifluoroacetic acid.

Identification of PTH-derivatives is normally performed by thin-layer chromatography or high performance liquid chromatography (HPLC). Both methods provide excellent resolution and reproducibility.

Thin-layer chromatography of PTH-derivatives is best performed on silica sheets precoated with a fluorescent indicator. Ascending chromatography is carried out with one or two mixtures of organic solvents and takes less than 20 min to do. A reliable system is described in the Method section.

High performance liquid chromatography of PTH-derivatives in reverse phase systems consisting of a hy-

drocarbon resin eluted with increasing amounts of organic solvent in water provides a fast, reliable and extremely sensitive means for identification, a description of which is outside the scope of this article, but has been treated extensively elsewhere (4).

Alternatively, or to supplement the results obtained by analysis of the PTH-derivatives, these can be converted into the corresponding parent amino acids. The regeneration is performed by hydrolysis with hydroiodic acid. If analysis of the parent amino acids is used as a unique means of identification, the thiazolinone derivatives may be hydrolyzed with hydroiodic acid without prior conversion into the PTH-derivatives.

Materials

1. Phenylisothiocyanate is redistilled and stored in ampules of 0.5 mL in the dark at -20°C
2. Pyridine is likewise redistilled before use and stored at -20°C
3. Solutions of 2.5% phenylisothiocyanate in 50% pyridine for peptide sequencing are freshly prepared and kept under N_2 at -20°C as working solution for up to 2 wk with no adverse effects
4. *n*-Butyl acetate-saturated with water, for extraction of phenylthiazolinones, is kept at 0°C
5. 0.5M NaHCO_3 , adjust to pH 9.8 with 1M NaOH.
6. 10% SDS is kept at room temperature
7. 0.1% ninhydrin, 5% collidine in ethanol for spraying of thin-layer sheets is freshly made prior to use

Methods

Procedure for Proteins

1. The protein (5–30 nmol) is freeze-dried from a volatile buffer system, e.g., ammonium hydrogen carbonate. The presence of non-amine containing detergents does not interfere with the degradation reactions. A 10×75

mm pyrex tube can conveniently be used and all subsequent reactions can be carried out in this

2. Add 150 μL of 0.5M NaHCO_3 and 20 μL of 10% sodium dodecyl sulfate. Flush with nitrogen for approx. 20 s, add 10 μL of phenylisothiocyanate, flush again with nitrogen for approx. 20 s, and seal the tube (parafilm or rubber cork). Mix vigorously and place the tube in a water bath at 50°C for 45 min and shake thoroughly at intervals of 5 min.
3. Freeze the tube and add 1 mL of ice-cold acetone (stored in freezer). Shake thoroughly and collect the flocculent precipitate by centrifugation, withdraw and discard the supernatant, and repeat the washing twice. Evaporate the last acetone by flushing with nitrogen while vortexing the tube to distribute the precipitate over the lower part of the tube. Place the tube at an approx. 45° inclination in a desiccator and dry under vacuum for 10 min at 60°.
4. When dry, the sample is ready for the cleavage reaction. Cool to room temperature and add 150 μL of trifluoroacetic acid. Flush with nitrogen for 10 s, seal the tube and incubate for 5 min at 50°C. Evaporate the trifluoroacetic acid by flushing with nitrogen in the hood (use gloves). Dry the precipitate under vacuum in a desiccator for 10 min at 60°C.
5. Place the tube in an ice bath, add 100 μL of ice cold 5% pyridine and extract three times with 300 μL of ice cold *n*-butyl acetate saturated with water (shake and centrifugate). During extraction take great care not to suck up any of the intervening layer separating the two layers where the protein tends to accumulate. Dry the water-phase in the desiccator under vacuum for 20 min. The protein is now ready for the next cycle of Edman degradation.
6. The *n*-butyl acetate containing the thiazolinone derivative is now dried down under vacuum in a desiccator at 60°C (takes approx. 10 min), (Do not leave the tube in the desiccator longer than necessary to avoid side-reactions). Add 75 μL of 20% trifluoroacetic acid, seal the tube, and leave at 60°C for 15 min (use the desiccator). Dry under vacuum at 60°C for approx. 10 min. The phenylthiohydantoin is now ready for analysis, and will keep well in a freezer for weeks.

Procedure for Peptides

1. Lyophilize the peptide (2–30 nmol) from a volatile buffer in a test tube. Good results have been obtained with drawn-out Pasteur pipets, which provides large inner surface for the amounts of solvent used.
2. For the coupling reaction add 50 μL of 2.5% phenylisothiocyanate in 50% pyridine, flush with nitrogen for 10 s, and seal the tube (small rubber corks are excellent). Incubate for 20 min in a 50°C waterbath.
3. Extract the solution once with 300 μL heptane:ethyl acetate (10:1) and once with heptane:ethyl acetate (1:2). Centrifuge and discard the organic layer. Dry under vacuum at 60°C in a desiccator for 10 min.
4. For the cleavage step add 50 μL of trifluoroacetic acid and leave the tube open in a 45°C desiccator (evacuated for 3 s) for 10 min. The trifluoroacetic acid is removed by applying vacuum to the desiccator for 10 min.
5. To extract the released thiazolinone place the tube in an ice bath, add 50 μL of pyridine and extract three times with 250 μL ice-cold butyl acetate saturated with water. The remaining solution is then dried under vacuum in a 60°C desiccator for 15 min and the peptide is now ready for the next degradation cycle.
6. The thiazolinone present in the *n*-butyl acetate extract is then converted to the PTH derivative as described for proteins (see above).

Identification of PTH-Derivatives

Thin-Layer Chromatography

1. Thin-layer chromatography of PTH-amino derivatives is carried out on silica-gel thin-layer sheets containing fluorescence indicator. The chromatogram is developed in two solvents.
2. Approx. 2 μL aliquots of derivative taken up in methanol are applied by capillary tubes. Ascending chromatography is performed in chloroform/methanol (9:1, v/v) as solvent one and in toluene/methanol (8:1, v/v) as solvent two. Mixtures of standard PTH amino acids are also applied to the chromatogram in lanes adjacent to the unknown sample.

3. The plates are dried in a stream of cold air after each solvent and immediately inspected and photographically recorded under 254 nm light (see Fig 3)
4. Further information can be obtained by spraying the sheets with a solution of ninhydrin/collidine in ethanol, which is then treated in an oven at 110°C for about 5 min before inspection. Using this method, many PTH derivatives develop specific colors, which aids in their identification (see Fig 3). Plates thus stained should be read within 30 min since they tend to turn completely red

Conversion with Hydriodic Acid

1. Thiazolinone or thiohydantoin derivatives are converted into free amino acids by adding 75 μ L of 56% HI to the dried derivative contained in a 7 $\times$ 50 mm pyrex tube

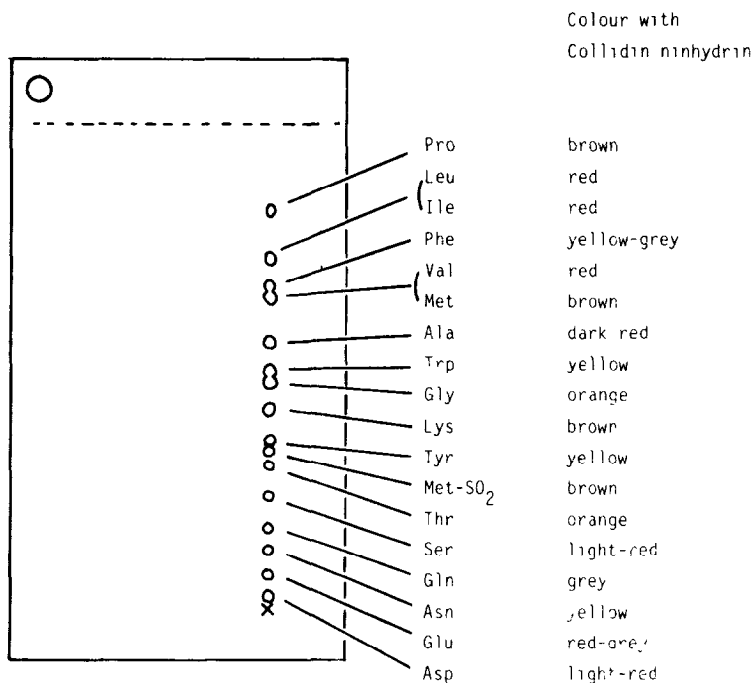


Fig 3. Separation of PTH-amino acids on silica gel thin layers, showing the colors produced with the ninhydrin/collidine stain

2. After evacuation the tube is drawn out as an ampule, closed and incubated at 130°C for 10–15 h
3. The solution is then dried down in a 60°C desiccator under vacuum, and the content submitted to amino acid analysis

Notes

1. Cycling times are 2 and 1.5 h for the protein and peptide versions, respectively. However, when a large number of samples are processed simultaneously, extra time for handling must be included
2. The repetitive yields for the presented methods are around 90%, differing slightly from protein to protein
3. For the best results great care should be taken to use very clean glassware and pure chemicals
4. Several of the chemicals involved in the Edman chemistry are toxic, notably phenylisothiocyanate and trifluoroacetic acid, and great care should be taken to minimize exposure to these. The extensive use of a hood and gloves is highly recommended.
5. The peptide procedure works well for most peptides up to 50 residues and have even been used for peptides up to 80 residues. Amounts of 2–30 nmol of peptide can be used with the indicated amounts of reagents and solvents. More than 30 residues have been sequenced from a particular peptide with this method
6. Under the conversion with 20% TFA side chains of Asn- and Gln-derivatives are partially hydrolyzed and the presence of either of these residues will inevitably result in detection of the corresponding acid as well as the amide in roughly equimolar quantities in later analysis. Furthermore, threonine and serine residues, already largely dehydrated during the cleavage reaction, are prone to further degradation in the conversion step. Subsequently, PTH-threonine is normally identified in the form of its dehydrated PTH-derivative. Serine often gives rise to multiple products because of polymerization, which can be seen as multiple tell-tale spots/peaks during later identification

- 7 On thin-layer sheets, the PTH-derivatives are visualized under UV-illumination as dark spots, but tend to disappear on prolonged illumination. Consequently, their positions are best registered by photography (green filter) for permanent record. To complement the information provided by the one-dimensional pattern (see Fig. 3), the thin-layer sheet can be stained. Several of the PTH-derivatives give quite specific color reactions with the ninhydrin/collidine stain (Fig. 3). As little as 0.1 nmol of a PTH-derivative can be detected by thin-layer chromatography.
8. It is quite normal to load up to a dozen samples on adjacent tracks on a chromatogram, with standard mixtures of PTH amino acids flanking these samples. Do not be alarmed if the chromatogram at first appears complex. Samples will all contain considerable amounts of side-products, particularly phenylthiourea and diphenylthiourea, which will often be present in greater amounts than the PTH-derivative being detected. However, background spots should appear fairly constant in each sample, whereas spots corresponding to the PTH-amino acids are detected by their appearance/disappearance in adjacent samples. Much of the side-product material can in fact be removed by heating the plate at 150°C for 10 min.
9. The hydriodic acid hydrolysis method gives reliable results, but information about amides, methionine, and tryptophan residues is lost and serine and threonine are recovered as alanine and α -amino butyric acid, respectively. Identification of the amino acids can be performed with thin-layer chromatography or preferably with an automatic analyzer.

References

- 1 Edman, P. (1950) Method for the determination of the amino acid sequence in peptides. *Acta Chem Scand* **4**, 283–293.
- 2 Edman, P., and Begg, G. (1967) A protein sequenator. *Eur J Biochem* **1**, 80–91.

- 3 Tarr, G E (1977) Improved manual sequencing methods
Meth Enzymol **47**, 335–357
- 4 Downing, M R., and Mann, K G (1976) High pressure liquid chromatographic analysis of amino acid phenylthiohydantoins *Anal Biochem* **74**, 298–319

Chapter 27

Carboxy-Terminal Sequence Determination of Proteins and Peptides with Carboxypeptidase Y

Per Klemm

*Department of Microbiology, The Technical
University of Denmark, Lyngby,
Denmark*

Introduction

No useful chemical method (similar to that of the Edman degradation) exists that allows a sequence investigation from the carboxy-terminal end of proteins and peptides. Efforts have therefore been centered around the exploitation of enzymatic methods, notably the use of carboxypeptidases for that purpose. Carboxypeptidases are enzymes that remove amino acids one at a time from the carboxy-terminus of a peptide chain

Several types of carboxypeptidases have been used in protein chemical investigations, such as pancreatic carboxypeptidases A and B, but their use is limited since they have rather narrow specificities (1). However, carboxypeptidase Y (CPY) from bakers yeast has properties that makes it the enzyme of choice for C-terminal investigations. In contrast to most other commercially available carboxypeptidases, CPY has a broad specificity and is able to release all amino acids, including proline, stepwise from the C-terminus of a peptide chain (2). Furthermore, since the enzyme retains its activity for extended periods in 6M urea or even in 1% sodium dodecyl sulfate solutions, it is excellently suited for the study of proteins that have inaccessible and difficult accessible C-termini under normal assay conditions (3,4).

The C-terminal sequence of a given protein or peptide is investigated by studying the kinetics of amino acid release from the peptide chain after addition of CPY. This is done by withdrawing aliquots of the incubation mixture at different time intervals, stopping the reaction (by acidification), and determination of the amounts of free amino acids present in the samples (Fig. 1).

In contrast to the Edman degradation an investigation with a carboxypeptidase seldom results in "linear" sequence data. However, in favorable cases (depending on the protein or peptide), the rate of appearance of the released amino acids during the digestion will give sufficient evidence to establish the C-terminal sequence. The method presented here has allowed establishment of C-terminal sequences of up to 12 residues flawlessly.

Materials

1. Carboxypeptidase Y (EC 3.4.17.4). The enzyme is stored at -20°C , and solutions of CPY in pyridine acetate buffer are made freshly just before use in each case.
2. 0.1M pyridine acetate buffer, pH 5.6, can be stored at 0°C for long periods.
3. Digestion buffer; 0.1M pyridine acetate, pH 5.6, containing 1% SDS and being 0.1 mM in norleucine.

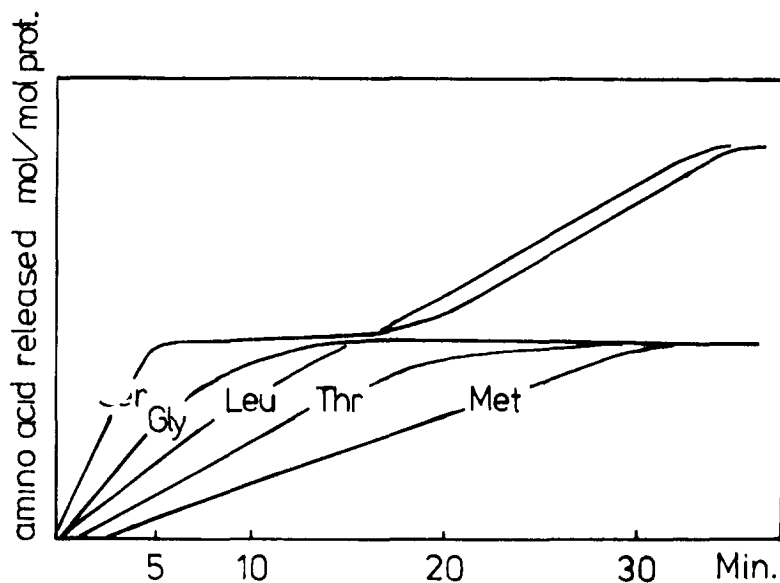


Fig. 1 Rate of release of amino acids from the C-terminal of the CFA/1-protein Ser - Leu - Val - Met - Thr - Leu - Gly - Ser (4)

Methods

Procedure for Proteins

A prerequisite for the described procedure is access to an automatic amino acid analyzer. The amino acid norleucine (not naturally occurring) is used as internal standard.

1. 20 nmol of protein is lyophilized from a salt-free solution. The protein is dissolved in 200 μL of digestion buffer and the preparation then incubated at 60°C for 20 min in order to denature the protein.
2. After cooling to room temperature a 25 μL aliquot is withdrawn as $T = 0$ background.
3. Incubation takes place at room temperature. 0.2 nmol carboxypeptidase Y (MW of CPY is 61,000) in 5–10 μL

of 0.1 M pyridine-acetate, pH 5.6, is added, the solution is thoroughly mixed, and 25 μ L aliquots are withdrawn at $T = 1, 2, 5, 10, 20, 30,$ and 60 min. In order to stop the reaction, 5 μ L of glacial acid is immediately added to a sample. Thereafter, it should be frozen and lyophilized. The latter can conveniently be done when all samples have been collected.

- 4 After freeze drying the samples can be submitted to amino acid analysis, without further preparation. The low amount of sodium dodecyl sulfate applied to the amino acid analyzer does not interfere with the elution profiles, nor with the well-being of the analyzer.

Procedure for Peptides

In the case of C-terminal analysis of peptides, the procedure given for proteins can be applied with the following modifications

- 1 Sodium dodecyl sulfate can be omitted from the incubation buffer since the C-terminal of peptides should be readily accessible
2. The time intervals for sample withdrawal should be shortened since the rate of appearance of free amino acids tends to be faster than in the case of proteins. Use instead $T = 0, \frac{1}{2}, 1, 2, 5, 10, 15,$ and 20 min

Notes

- 1 Repeated freezing and thawing of solutions of carboxypeptidase Y can lead to denaturing of the enzyme. Likewise, prolonged exposure to room temperature should be avoided as autodigestion of the enzyme can occur. The use of very clean glassware is a prerequisite for good results.

References

- 1 Ambler, R. P. (1972) Enzymatic hydrolysis with carboxypeptidases *Meth. Enzymol.* **25**, 143–154

- 2 Hayashi, R , Moore, S , and Stein, W H (1973) Carboxypeptidase from yeast *J Biol Chem* **248**, 2296–2302.
- 3 Gaastra, W , Klemm, P , Walker, J M , and de Graaf, F K (1979) K88 fimbrial proteins Amino- and carboxyl terminal sequences of intact proteins and cyanogen bromide fragments. *FEMS Microbiol Lett* **6**, 15–18
- 4 Klemm, P (1982) Primary structure of the CFA/1 fimbrial protein from human enterotoxigenic *Escherichia coli* strains *Eur J Biochem* **124**, 339–348

Chapter 28

Immunization and Fusion Protocols for Hybridoma Production

J. N. Wood

*Department of Experimental Immunobiology,
Wellcome Research Laboratories,
Beckenham, Kent*

Introduction

Kohler and Milsteins' (1) technique of monoclonal antibody production is now being exploited in most areas of biology. The essence of the method is to immortalize and then select for clones of plasma cells secreting antibody against a desired antigen. Individual plasma cells secrete antibody of a single antigenic specificity and monoclonal antibodies are thus obtainable from a cell line derived from a single plasma cell. Once established, clonal lines of hybridomas (hybrids composed of plasma cells fused with immortal myeloma cells) provide an infinite supply of antibodies with reproducible properties. Monoclonal anti-

bodies may be produced without extensive purification of the immunogen, and without the tedious cross-adsorption steps often necessary for production of specific antisera. So far, only mouse and rat hybridomas are being produced routinely, while human monoclonal antibodies of potential therapeutic value are technically more difficult to obtain.

Three prerequisites must be satisfied for hybridoma production. It must be possible to induce antibody-secreting plasma cells by immunization *in vivo* or *in vitro*. Secondly, a suitable method of immortalization either by fusion with a transformed cell line or direct viral transformation must be available. Thirdly, a quick simple and reliable assay for detecting desired antibodies is necessary for selecting hybridoma clones.

In this chapter the simplest method of immunization and plasma cell immortalization for production of rodent antibodies is outlined. The basic scheme is shown in Fig. 1. Should the antigen of interest be small, say less than 5000 daltons, conjugation of this material to a high molecular weight carrier may be necessary to elicit an adequate immune response. Bovine thyroglobulin (mol. wt. 650,000) is a cheap and effective carrier (4). The method of coupling depends on the availability of reactive groups in the desired immunogen. Here the use of a water-soluble carbodiimide (EDAC) which crosslinks amino and carboxyl groups, and glutaraldehyde, which predominantly couples amino groups, is described. As an example, we have chosen the pentapeptide Met-enkephalin.

To elicit a strong immune response, adjuvants are usually employed during immunization. Freund's complete adjuvant contains heat-killed bacteria to stimulate the animal's immune system, with an oil base in which the immunogen is emulsified to ensure slow release. We describe a protocol for immunizing BALB/c mice.

Where very small amounts of immunogen are available, or human monoclonals are desired, *in vitro* immunization of cultured plasma cells may be necessary. Some most impressive examples of this technique are now appearing, but it is difficult and beyond the scope of this book (2). While B cell immortalization is usually carried out by polyethylene glycol-induced fusion with a mye-

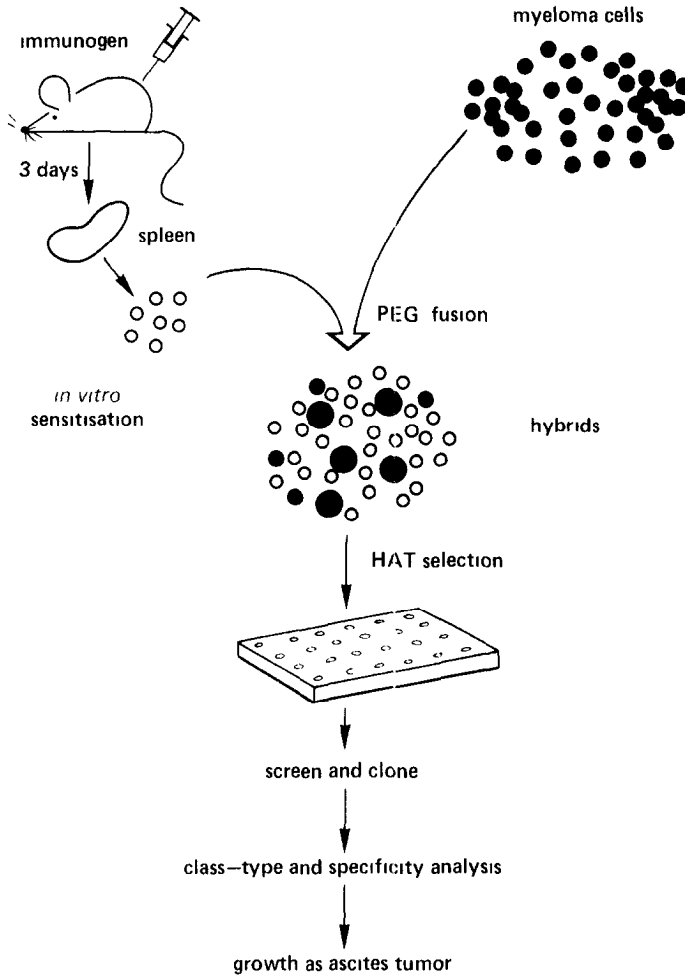
Production of monoclonal antibodies

Fig 1 Diagram showing the steps involved in producing monoclonal antibodies.

loma line, recent developments with high efficiency electric fusion may make this the future method of choice (3).

After polyethylene glycol-induced fusion of immunized splenocytes and myeloma cells, the immortalized hybridomas are outnumbered by residual immortal myeloma cells. In order to kill the myelomas and select

uniquely for hybrid growth, the HAT selection system is used. Aminopterin (A) inhibits the *de novo* synthesis of purines by impairing tetrahydrofolate metabolism. In normal cells a second scavenging pathway of purine synthesis, using the enzyme HGPRT (hypoxanthine guanine phosphoribosyl transferase) is present. Hybridoma cells derived from splenocytes can thus still grow in the presence of aminopterin when the scavenging pathway is boosted by exogenous additions of hypoxanthine and thymidine (H and T). However, the myeloma cells used are deficient in the enzyme HGPRT, and they eventually die in the presence of HAT medium. After cell fusion, cells are therefore grown in medium containing HAT for 2 wk, and then transferred to HT medium for a further week (in case residual aminopterin is still blocking the *de novo* synthesis of purines). The HGPRT⁻ mutant myeloma cell line is obtained by passage in azaguanine (20 µg/mL). Cells with normal levels of HGPRT take up the 8-azaguanine and incorporate it into DNA, leading to cell death.

Simple procedures for the conjugation of haptens, production of hybridomas, their cloning, and storage are presented below.

Materials

Conjugation and Immunization

1. Thyroglobulin (bovine). Store frozen (−20°C or better −70°C) 10 mg/mL in PBS.
2. Glutaraldehyde 5% solution in PBS made fresh from 25% stock stored at 4°C in the dark.
3. EDAC [1-ethyl-3-(3-dimethyl aminopropyl) carbodiimide hydrochloride].
4. Hydroxylamine: 1M solution made fresh in water
5. PBS (phosphate-buffered saline). KH₂PO₄, 1.5 mM (0.2 g/L), Na₂HPO₄, 8.1 mM (1.15 g/L), KCl 2.7 mM (0.2 g/L), NaCl 140 mM (8.0 g/L).

Cell Fusion

1. Polyethylene glycol (1500–4000 form) (PEG) 50% solution in PBS. Store at 4°C in the dark after autoclaving.

- 2 Growth medium DMEM or RPMI 1640 supplemented with antibiotics (penicillin 100 U/mL and streptomycin 100 μ g/mL), glutamine 2 mM and 10% fetal calf serum. Store at 4°C preferably for not more than 1 month.
- 3 HT medium Dissolve 272 mg hypoxanthine (H) in 70 mL of 0.1M NaOH and make up to 150 mL with water. Dissolve 77.6 mg thymidine (T) in 20 mL H₂O, add to hypoxanthine stock and make up to 200 mL. Filter, sterilize, and store at -20°C in aliquots. Dilute in growth medium 100-fold to give a final concentration of H 13.6 μ g/mL and T 3.88 μ g/mL in HT medium.
- 4 HAT medium dissolve 3.82 mg aminopterin in 2 mL NaOH (0.1N) make up to 200 mL, filter, sterilize, and store in aliquots at -20°C. Dilute 100-fold into HT medium to give a final concentration of 0.19 μ g/mL as HAT medium.

Cell Storage and Ascites Production

- 1 10 percent analar DMSO (dimethyl sulfoxide) in growth medium, freshly prepared
- 2 Pristane, store at room temperature

Method

Conjugation of Low Molecular Weight Haptens

Carbodiimide Crosslinking

- 1 Dissolve the antigen in PBS at a molar ratio of 100 to 1 with thyroglobulin (e.g., 1 mg enkephalin, 10 mg thyroglobulin in 1 mL PBS). Add a 200 molar excess of EDAC (0.5 mg) to the mixture, to crosslink amino and carboxyl groups.
- 2 Mix well and leave at room temperature overnight.
- 3 Add an equal volume of 1M hydroxylamine, and incubate for 4 more hours. This is to reverse a possible modification induced in the tyrosine aromatic ring by the carbodiimide.

4. Dialyze the solution exhaustively against PBS at 4°C, i.e., 100 vol of PBS changed five times.
5. Freeze aliquots of the solution at -20°C, or lyophilize them.

Glutaraldehyde Crosslinking

1. Dissolve the antigen in a 100:1 ratio with thyroglobulin, e.g., 1 mg enkephalin with 10 mg thyroglobulin in PBS
2. Add 50 μ L of 5% glutaraldehyde in PBS and agitate at room temperature for 30 min.
3. Dialyze against PBS at 4°C (five changes of 100 vol of PBS)
4. Store frozen aliquots at -20°C or lyophilize.

In order to estimate the level of binding, it may be useful to add an aliquot of radiolabeled antigen and calculate percentage uptake into nondialyzable material.

Immunization

See also Chapter 32

1. Emulsify the antigen in Freund's complete adjuvant using an emulsifier, a chopping blade device, ultrasonication, or agitation between two syringes connected by narrow rubber tubing. Use an equal volume of adjuvant to immunogen. A satisfactory emulsion should not separate into phases after prolonged storage at 4°C (weeks).
2. Inject from 1 to 100 μ g/animal in about 0.15 mL volume. Intraperitoneal injections are probably the easiest. Immunize several animals if enough material is available
3. Repeat immunizations with antigen emulsified in incomplete Freund's adjuvant (no bacteria) after about 2 wk
4. After a further boost without adjuvant, tail bleed the mice for an antibody test, a few hundred microliters should be sufficient
5. Allow the blood to clot in a microfuge tube containing a glass capillary. After 1 h at room temperature remove the clot with the capillary, and microfuge the sera.

6. Test the sera at various dilutions compared with a normal mouse serum control (*see* Chapters 30 and 31). Ideally a well-immunized mouse should be giving a titer at a dilution of one in a thousand of 50% maximum binding. The mice may be identified by numbering with an ear punch or, less satisfactorily, painting rings on the tails with colored marker pens.
7. When satisfactorily immunized, a final injection without adjuvant should be given 3–4 d before fusion. Intraperitoneal or intravenous injection (tail vein) are satisfactory.

Fusion Protocol

1. Grow HAT-sensitive myeloma cells (commercially available) that do not produce endogenous immunoglobulin (NSO, AG8 653, SP2₀ for BALB/c mice, Y123 or YO for Lou rats) in growth medium at a density of about 10^5 cells/mL to give enough cells for a fusion (2×10^6 for a mouse, 1×10^8 for a rat)
2. Kill the immunized animal by cervical dislocation and aseptically remove its spleen 3 or 4 d after a final immunization of antigen without adjuvant.
3. Dissociate the spleen in medium without serum in a laminar flow hood, by homogenizing with a loose potter, teasing with blunt forceps, or passage through a sterile sieve.
4. Wash the splenocytes and the myelomas separately in about 50 mL serum-free growth medium by bench centrifugation.
5. Resuspend both splenocytes and myeloma cells separately in about 25 mL serum-free growth medium and count them with a hemocytometer. A mouse spleen contains between 50 and 200×10^6 lymphocytes, whereas a rat spleen has about 2×10^8 lymphocytes.
6. Mix the cells in a ratio of 1 myeloma to 10 splenocytes for a mouse fusion, or 1 to 2 for a rat fusion. Spin the mixed cells down with a bench centrifuge and aspirate all the medium.
7. Flick the cells around the bottom of the centrifuge tube. Add 1 mL of warm PEG₁₅₀₀ solution dropwise over 30 s. Gently agitate the cells for 1 min.

- 8 Add 20 mL of serum-free medium dropwise over 5 min to the cell pellet, slowly diluting out the PEG.
- 9 Bench centrifuge out the cells and resuspend in 30–50 mL of warm HAT medium.
- 10 Plate out the cells at $1\text{--}2 \times 10^5$ cells/well in 96-well microtiter plates. If a complicated screening assay is being used, plate into 24 well-plates. We generally use four plates per BALB/c fusion.
11. After 5 d feed the cells with HAT medium
- 12 At 10–14 d (visible colonies) screen the tissue culture supernatants
13. Expand positive clones in HT medium, from 96 to 24 well-plates, thence to 25 mL flasks. The cells should now be cloned, and some frozen as soon as possible.

Cloning

Positive hybridomas must be grown from a single cell to obtain monoclonal rather than a mixture of antibodies. This cloning process is most easily achieved by growing up hybrids from very low cell densities. The presence of supporting feeder cells is necessary to support growth of the hybrids, which usually die at the low cell densities (~ 1 cell/well) necessary for cloning. Here we describe a simple and effective cloning method using mouse splenocytes as feeder cells.

- 1 Tease out splenocytes from stock mouse spleens as described in the fusion protocol. Plate out 4×96 well plates/spleen in growth medium containing HT ($100 \mu\text{L}/\text{well}$)
2. The next day, very carefully count a suspension of positive hybridomas. Serially dilute the cells to give concentrations of 0.1, 0.3, 1, 3, 10, and 30 cells/ $100 \mu\text{L}$ (about 5 mL of each concentration). Plate out cells at each density in half a 96 well plate ($100 \mu\text{L}/\text{well}$) containing feeders
- 3 When visible colonies appear (10 d), screen (Chapters 29,30) and grow several positive wells from the lowest dilutions

4. Freeze positive cells and repeat the cloning. When a cell line is truly cloned, all wells positive for growth will also be positive by screening.

Freezing Cells

Cells can be stored in liquid nitrogen or, less satisfactorily, in -70°C freezers when suspended at high concentration in 10% DMSO.

1. Make sure the cells are in exponential growth. Bench centrifuge and resuspend in 10% DMSO/growth medium at a density of 0.5×10^6 cells/mL.
2. Freeze down in ampules at $1^{\circ}\text{C}/\text{min}$ by placing the cells in a polystyrene container in a -70°C freezer, or an igloo containing cardice, or the neck of a liquid nitrogen container.
3. When frozen, place in a liquid nitrogen container, be sure the ampules are indelibly marked.
4. Unfreeze cells when required by warming to 37°C quickly. Resuspend then in a large volume of growth medium, bench centrifuge, resuspend, and plate out doubling dilutions in a 24-well plate.

Ascitic Tumor Production

By growing a hybridoma line as an ascitic tumor in an isogenic animal, high concentrations of monoclonal antibody can be rapidly obtained. To diminish the possibility of solid tumor formation, the animals are primed with pristane, a mineral oil.

1. Inject pristane intraperitoneally using a glass syringe 0.2 mL/mouse, 1 mL/rat.
2. At 10 d to 1 month later, inject $2-4 \times 10^6$ cells/animal ip.
3. When the tumor is clearly visible as a large swelling (1-3 wk), tap the ascitic fluid into a heparinized tube using a 21 gage needle, spin out the cells, and freeze the ascitic fluid.
4. Continue to tap the tumors. If the animals are distressed, kill them and aspirate the ascitic fluid with a Pasteur pipet.

Notes

1. It is imperative that animal work comply with legislative regulations and be supervised by a skilled technician. Do not undertake the techniques described here without training by an experienced worker.
2. Hybridomas are easy to make if one is competent at tissue culture. Minor contamination can be knocked out by adding concentrated NaOH to infected wells. Always use the same tissue culture plastic type through the experiments, since changes in plastic type may kill the cells. It is wise to gain experience in a laboratory in which hybridomas are being made.
3. If too many positive wells are obtained hybrids can be frozen down at very low density (see ref. 5).

References

- 1 Kohler, G., and Milstein, C. (1975) Continuous cultures of fused cells secreting antibody of predefined specificity. *Nature* **256**, 495–497.
- 2 Reading, C. L. (1982) Theory and methods for immunization in culture and monoclonal antibody production. *J Immunol Methods* **53**, 261–291.
- 3 Zimmerman, U. (1982) Electric field-mediated fusion and related electrical phenomena. *Biochim Biophys Acta* **694**, 227–277.
- 4 Skowsky, W. R., and Fisher, D. A. (1972) The use of thyroglobulin to induce antigenicity to small molecules. *J Lab Clin Med* **80**, 134–144.
- 5 Wells, D. E., and Price, P. J. (1983) Simple rapid methods for freezing hybridomas in 96 well microculture plates. *J Immunol Methods* **59**, 49–52.

Chapter 29

Immunofluorescence and Immunoperoxidase Screening of Hybridomas

J. N. Wood

*Department of Experimental Immunobiology,
Wellcome Research Laboratories,
Beckenham, Kent*

Introduction

As mentioned in the previous chapter, the provision of a rapid, simple assay for screening monoclonal antibodies is essential for their production. Although solid-phase binding assays are the method of choice (Chapter 30) a number of applications require visualization of antibody binding to cultured cells or tissue sections.

To give some examples, if a cell type can be defined by immunodetection of a specific surface antigen, then purification and characterization of this particular cell type

may be possible. Such an approach has been successfully used in defining lymphocyte subpopulations, and determining the origin of neuronal cell types in culture. The ability to visualize a viral antigen may elucidate the role of virus infection and expression in cellular transformation. Spatial or temporal variation in an antigen expression during cell growth or differentiation may also provide useful markers to define different sorts of cellular organization, while disease-state-specific antigen expression may enable the production of diagnostic monoclonal antibodies, even though the nature of the antigen is unknown. This chapter is concerned with the simplest method of fixing cells and tissue sections, and the visualization of antibody binding by fluorescent- or peroxidase-linked second antibodies.

Fluorescent antispecies second antibody, produced by coupling a fluorochrome to the affinity-purified antisera, is used to visualize monoclonal antibody binding to cells or tissue sections by means of fluorescence microscopy.

Indirect immunoperoxidase staining gives a permanent record of antibody binding, and may thus be preferable to immunofluorescence screening, which can fade. In essence, the peroxidase attached to the second anti-species antibody is used to produce an insoluble stain at the site of antibody binding. Excellent commercial enzyme-linked anti-species antibodies are now available. The sensitivity of the technique may be much enhanced by the use of Stenberger's peroxidase-antiperoxidase method (2), but for routine screening this should be unnecessary.

There are no hard and fast rules for predicting the best fixation method for an antigen. Acid alcohol, paraformaldehyde, and picric acid are commonly used, as are unfixed frozen sections. A very useful method has been described by Lane (1), where cultured cells are fixed and dried on tissue culture plates using acetone/methanol, which does not render the plastic opaque. This enables many samples to be screened on a single petri dish.

As well as describing methods of visualizing antibody staining, a protocol for coupling fluorescent dyes to antibodies is described. Using fluorescein or rhodamine isothiocyanate, the dyes may be directly conjugated to an-

antibodies at alkaline pH without any coupling agent. Peroxidase conjugation is covered in the next chapter.

Materials

Fixation of Cultured Cells

1. 90% Methanol/10% acetic acid. Store at -20°C .
2. 50% Acetone/50% methanol. Store at room temperature.
3. 40% Formaldehyde. Store at room temperature. Dilute 1:10 in phosphate-buffered saline (PBS) for working solution (*see* Chapter 28).

Tissue Sections

1. 40% Formaldehyde (*see* above).
2. 4% Paraformaldehyde. Make fresh by dissolving 8 g paraformaldehyde in 50 mL H_2O at 60°C , adding NaOH (1N) dropwise, in a fume hood. Add 50 mL double strength PBS (*see* Chapter 28) when the paraformaldehyde is dissolved.
3. 5% Sucrose in PBS (*see* Chapter 28)
4. Glycerol.

Immunofluorescence

1. Normal goat serum. Store at 4°C with 0.02% sodium azide.
2. PBS (*see* Chapter 28).
3. Goat anti-mouse heavy and light chains (FITC- or TRITC-conjugated) Store frozen aliquots in the dark. Dilute 1:50 in PBS as a working solution and microfuge before use.
4. Mounting medium: 70% glycerol, 0.02% azide, 0.5% NaCl, 0.02% NaOH, 0.4% glycine

Immunoperoxidase

1. 70% Ethanol
2. 100% Methanol, 0.1% H_2O_2 (make fresh from 30% H_2O_2 stock)

3. Peroxidase-conjugated goat anti-mouse heavy and light chains. Store frozen aliquots, dilute 1:50 in PBS and microfuge before use
4. 4-Chloro-1-naphthol. 3 mg/mL in methanol. (Store in the dark at 4°C) Dilute 1·5 v/v in PBS with 0.01% H₂O₂ immediately before use
5. Mounting medium, *see* above.

Conjugating Fluorochromes

- 1 Sodium borate (50 mM, pH 9.3) containing 0.4M NaCl. Store at 4°C.
- 2 Dimethyl sulfoxide (DMSO).
- 3 TRITC (Tetramethyl rhodamine isothiocyanate). (Store dry, -20°C in the dark).
4. FITC (fluorescein isothiocyanate). (Store dry, -20°C in the dark).
5. Purified antibody.

Methods

Fixation of Cultured Cells

1. Wash the cell cultures twice in PBS to remove serum.
- 2 Fix cells in
 - (a) 90% methanol-10% acetic acid (-20°C) for 2 min or
 - (b) 4% formaldehyde in PBS for 30 min or
 - (c) 4% paraformaldehyde in PBS for 30 min
3. Wash cells in PBS.
- 4 Proceed directly to antibody staining (2.3) if few samples are being tested. Otherwise continue with steps 5 and 6
5. Add acetone/methanol (1:1 v/v) for 2 min
- 6 Aspirate the acetone/methanol and air-dry.

Tissue Sections

For best results, it may be necessary to perfuse fresh tissue with fixation medium. However, soaking small

chunks of tissue in fixative prior to sectioning, or using frozen unfixed tissue sections may be adequate

- 1 Immerse fresh tissue in 4% paraformaldehyde overnight at 4°C, using 0.5 cm cube chunks of tissue
- 2 Rinse in PBS, and soak in 5% sucrose PBS for several hours.
3. Cut cryostat sections and lift them off the knife onto coverslips smeared with a thin layer of glycerol (a thumbprint)
4. Store sections at -20°C, but try to use them as quickly as possible.

Indirect Immunofluorescence

Cells

- 1 Spot 5 μ L samples of tissue culture fluid in a known order onto a petri dish containing fixed air-dried cells, previously marked with ~ 0.5 cm square grids
2. Incubate in a moist box for 1 h at room temperature, being very careful not to allow the samples to run into each other
- 3 Wash the plate four times in PBS, with a final longer wash (5 min) in PBS 1% normal goat serum [this is to block any nonspecific second (goat) antibody-binding sites, if you are using rabbit anti-mouse then use 1% normal rabbit serum].
4. Add $1/40$ dilution of microfuged goat anti-mouse globulins (FITC or TRITC labeled) Cut down the volume required by covering the petri dish with coverslips when the second antibody has been applied.
5. After 1 h, wash the petri dish five times in PBS, floating away the coverslips If samples are too near the rim of the dish to examine microscopically, tear off the sides with pliers.
- 6 Coverslip the samples after addition of mounting medium (seal them with nail varnish), and examine by epifluorescence microscopy.

If you are careful, it may be possible to quickly screen the samples at low power (obviously without using oil or

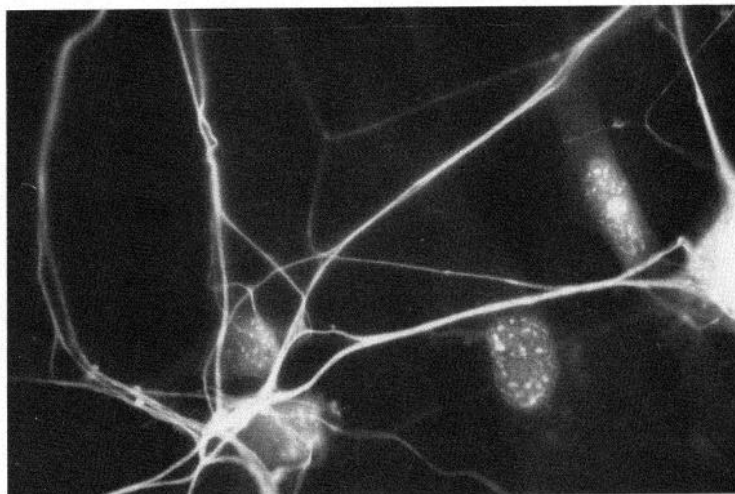


Fig. 1. Indirect immunofluorescence of an anti-neurofilament monoclonal antibody staining rat dorsal root ganglia cultures [$\times 710$].

water immersion lenses) directly after washing, and without bothering to mount or coverslip the cells.

Coverslip Mounted Sections or Cells

1. Add 100 μL of tissue culture fluid to the section or fixed cells on the coverslip, and keep them in a wet box for 1 h at room temperature. Number the coverslips with a diamond pen.
2. Using coverslip racks, or Coplin jars, wash the coverslips four times in PBS, and once for 5 min in 1% normal goat serum (see above).
3. Replace the coverslips the right way up (check the numbering) in the wet box, and add 50 μL 1:50 FITC or TRITC goat anti-mouse globulins.
4. After 1 h, wash five times in PBS, mount in mounting medium on slides, seal with nail varnish, and examine.

Immunoperoxidase Screening

1. Using fixed cells in petri dishes, or sections on coverslips, immerse in:

- (a) 70% ethanol, 10 min
- (b) 100% methanol, 0.1% H_2O_2 , 10 min
- (c) 70% ethanol, 10 min

This inactivates endogenous peroxidases.

2. Incubate in hybridoma supernatant and wash as described in the immunofluorescence protocol.
3. Add 1/50 dilution of peroxidase anti-species antibody for 1 h at room temperature
4. Wash four times in PBS
5. Add 4-chloro-1-naphthol working solution for 10–20 min.
6. Wash in PBS to stop the reaction, and mount in glycerol medium. The reaction should be visible to the naked eye even before microscopic examination

Conjugation of Fluorochromes to Proteins

With the commercial availability of fluorescent antispecies antisera, indirect immunofluorescence is the usual method of visualizing monoclonal antibody binding to cells or tissue sections. However, direct conjugation of a fluorochrome to a monoclonal antibody may occasionally be useful, for example, in double labeling experiments with monoclonals of similar isotype. Direct conjugation of an enzyme, for example peroxidase, may be useful for a similar reason, and this is covered in Chapter 30.

1. Dissolve purified antibody (2 mg/mL) in 50 mM borate, pH 9.3, containing 0.4M NaCl.
2. Dissolve 3 mg FITC in 100 mL 50 mM borate buffer, pH 9.3. For TRITC, dissolve 3 mg in 0.2 mL DMSO, then make up to 100 mL of 50 mM borate, pH 9.3.
3. Dialyze the antibody (1–5 mL) against the dissolved fluorochrome overnight at 4°C in the dark.
4. Dialyze the conjugated antibody against several changes of PBS at 4°C in the dark for 3 more days.
5. Store the antibody in frozen aliquots (–20°C) in the dark.

Notes

1. Cutting sections requires a sharp knife, careful reading of the cryostat instruction manual, and some perseverance. It is easy enough to cut thick (30 μm) sections, which are quite adequate for preliminary screening.
2. To examine surface staining of live cells, it may be necessary to inhibit possible internalization or shedding of the antigen-antibody complex. For this reason, the cells are stained at 4°C in the presence of 0.02% sodium azide. Follow the previous protocol, using chilled reagents, unfixed cells, and 4°C incubations. For suspension cells, incubations and washings can be carried out using small plastic tubes (LP3) and a bench centrifuge. At the end of the staining protocol, small volumes (10–20 μL) can be mounted and screened as usual.
3. One potential problem is the presence of endogenous peroxidase in some tissues and cell types. For this reason β -D-galactosidase-coupled second antibodies together with a suitable substrate (e.g., *o*-nitrophenyl β -galactopyronoxide) may sometimes be useful. In the protocol described, an inactivation step for endogenous peroxidase is included although this step may be unnecessary for the cell type that you are investigating.

References

1. Lane, D. P., and Lane, E. B. (1981) A rapid antibody assay system for screening hybridoma cultures, *J. Immunol. Methods* **47**, 303.
2. Sternberger, L. A. (1974) *Immunocytochemistry*, Prentice-Hall, Englewood Cliffs, NJ.

Chapter 30

Solid-Phase Screening of Monoclonal Antibodies

J. N. Wood

*Department of Experimental Immunobiology,
Wellcome Research Laboratories,
Beckenham, Kent*

Introduction

By far the most convenient and simple assays for screening monoclonal antibody-producing hybridomas are those utilizing solid-phase binding assays, where radiolabelled or enzyme linked antispecies antibodies are used to detect binding of the hybridoma supernatants to insolubilized antigen. Such procedures require an adequate supply of antigen, preferably purified, which can be bound to a solid support, usually through noncovalent binding to PVC microtiter plates or nitrocellulose sheets.

The advantage of such methods lies in their simplicity and speed. Hundreds of samples can be easily screened in a day. After negative clones have been discarded, considerable information about the affinity of the antibody, and

species and tissue distribution of the antigen can be rapidly obtained with minor modifications of the assay. This allows a further level of selection to be exercised on the antibody before exhaustive characterization.

In this chapter, simple solid-phase binding assays, iodination techniques, and conjugation of enzymes to antibodies are described.

The two solid-phase assays, which differ in the incubation conditions for attachment of the antigen to plastic (PVC) plates should be compared for their sensitivity with the particular antigen being investigated. Should antigen be in short supply, nitrocellulose has advantages as a solid support, in that very small amounts of antigen may be used, and a fine discrimination between real and background binding can readily be made, either by radioautography or enzyme-linked assays (2).

Although commercial iodinated antispecies antibodies are available, they are expensive, and it may be necessary to prepare one's own. The protocols described below may be used to iodinate either antigens or antibodies via tyrosine aromatic rings to a high specific activity (*see Chapter 31*).

An alternative approach, using enzyme-linked second antibody to visualize antibody binding is equally good. If labelled antigen is available, the assay may be carried out more conveniently by "dotting" the hybridoma supernatants on a grid, and soaking the whole sheet in labelled antigen. Positive wells will be detectable by a dark spot on autoradiography.

Materials

Iodination

Chloramine T

1. Chloramine T, fresh solution. 5 mg/mL in water
2. Dowex 1 X-8 resin. 2 mL in a plastic syringe freshly equilibrated with PBS (phosphate buffered saline) (Chapter 28) containing 0.1% bovine hemoglobin.
3. Tyrosine in PBS, fresh saturated solution (1 mL)

4. 25% Trichloroacetic acid, store at room temperature (stable)
5. Na ^{125}I high-specific activity (carrier-free) in NaOH
Always use a fume hood when diluting this into PBS for iodinations.

Iodogen

Iodogen (1,3,4,6-chloro-3a,6a-diphenyl-glycoluril).

Make up at 100 $\mu\text{g}/\text{mL}$ in dichloromethane and add 25 $\mu\text{L}/\text{microfuge tube}$. Air dry the open microfuge tubes by rotating them briskly, and when dry, store them at -20°C (stable for weeks)

Peroxidase Conjugation

1. Horse radish peroxidase (store dry at -20°C).
2. Immunoglobulin (preferably affinity-purified).
3. Glutaraldehyde: 0.02% made fresh in PBS.
4. Sephadex-G200 column equilibrated with PBS.

Plate Binding Assay

Method A

1. PBS containing 5% bovine hemoglobin and 0.02% sodium azide, store at 4°C .
2. ^{125}I antispecies antibody [e.g., rabbit anti-mouse, preferably F(ab')_2 minimum specific activity 10 $\mu\text{Ci}/\mu\text{g}$]. Stored at 4°C with 0.02% sodium azide.

Method B

1. Adsorption buffer. 50 mM bicarbonate, pH 9.6 (1.59 g Na_2CO_3 , 2.93 g NaHCO_3 , 0.2 g NaN_3 in 1 L). Store at 4°C
2. Wash buffer. Tween 40 (0.05%) in PBS containing 1% hemoglobin and 0.02% NaN_3 (store at 4°C)
3. ^{125}I second antibody, as above

Enzyme-Linked Assay

1. Peroxidase-conjugated antispecies antibody [preferably F(ab')_2] store frozen aliquots after reconstitution, and dilute to 1/60 in PBS and microfuge before use.

2. 4-Chloro-1-naphthol, 3 mg/mL in methanol stored at 4°C in the dark (weeks). Working solution 1:5 v/v in PBS with 0.01% hydrogen peroxide.
3. Hydrogen peroxide 30%. Store at 4°C

Nitrocellulose Binding Assay

1. 5% Bovine hemoglobin in PBS plus 0.02% sodium azide (4°C).
2. 1:5 dilution in PBS of above.
3. ^{125}I -antispecies antibody [e.g., rabbit anti-mouse, preferably F(ab')_2 minimum specific activity 10 $\mu\text{Ci}/\mu\text{g}$]
or
- 3a. Alternatively, enzyme-linked antibody reagents marked under ELISA.

Methods

Iodination

Chloramine T Iodination

This fast oxidation may reduce the antigenicity of some molecules, but is usually satisfactory for iodinating immunoglobulins.

1. In a fume hood, mix the antibody or protein (200 μg in 250 μL PBS) with ^{125}I (sodium salt, 250 μCi in PBS) and 25 μL Chloramine T.
2. After 1 min at room temperature add 50 μL tyrosine in PBS.
3. Chromatograph the solution on a Dowex 1 X-8 column (2 mL in a disposable syringe) equilibrated with PBS containing 0.1% hemoglobin
4. Collect the column eluate (first 3 mL) and discard the column to which free ^{125}I is bound.
5. Measure incorporation by mixing 10 μL aliquot with 100 μL of 1% hemoglobin in PBS. Add 1 mL 25% TCA, leave for 1 hr at 4°C then microfuge (10,000g, 5 min) and measure both the precipitated and soluble counts. The percentage of counts incorporated (precipitate) should be < 80%.

Iodination with Iodogen

This slightly milder process uses iodogen-coated tubes and may be preferable for labelling monoclonal antibodies or labile antigens.

1. Add antibody or antigen (100 μg in 100 μL PBS) to a coated iodogen tube.
2. Add 250 μCi ^{125}I (sodium salt diluted in PBS) and mix.
3. After 20 min at room temperature, chromatograph on a Dowex column as described above.

Conjugation of Peroxidase to Immunoglobulin

1. Mix 1 mg of immunoglobulin with 5 mg of horseradish peroxidase in 1 mL PBS.
2. Add 1 mL of glutaraldehyde and mix for 30 min at room temperature.
3. Chromatograph on a column of G200 Sephadex equilibrated with PBS (column size about 25×2 cm) and retain the first peak of conjugated antibody. Store frozen aliquots that should be stable indefinitely.

Solid-Phase Binding Assays on PVC Plates

Two protocols are described, one or the other of which has proved satisfactory with protein antigens ranging from small peptides to whole brain homogenates (1)

Binding Assay A

1. Pipet 50 μL of antigen (1–100 $\mu\text{g/mL}$) in PBS into each well of a 96-well PVC plate and incubate overnight at room temperature, to allow antigen to bind to the well surface
2. Remove the surplus antigen and wash three times with 5% (w/v) hemoglobin in PBS (wash bottle).
3. Fill the wells with Hb solution and incubate for 30 min at room temperature, to block further binding of protein to the surfaces of the wells, then wash twice more with the same solution.

4. Pipet dilutions of antiserum, or samples of hybridoma supernatants into the wells (45 μL /well) and incubate for 1 h at room temperature. Remove the antiserum by aspiration, or more simply by flicking the contents of the plate into a sink, and wash five times with 1% (w/v) Hb in PBS
5. Pipet the second antiserum (^{125}I -labeled rabbit anti-mouse globulins, 45 μL /well about 20,000 cpm/well) into the wells and incubate 1 h at room temperature. Remove the labeled antiserum and wash six times with 1% Hb in PBS, followed by one wash with PBS. Allow the plate to dry, then cut the wells and count samples with a gamma counter

Binding Assay B

- 1 Incubate the antigen (1–100 $\mu\text{g/mL}$) in adsorption buffer overnight on 96-well plates (50 μL /well) at 4°C
- 2 Sink the plate in wash buffer for 2 h to block remaining sites. Rinse the plate briefly in water, and drain. The plates may be stored at -20°C until required, or used immediately
- 3 Add dilutions of antisera, or tissue culture supernatants (50 μL /well) overnight at 4°C
- 4 Wash four times in wash buffer
- 5 Incubate ^{125}I -antispecies antibody (40 μL /well) containing 20,000 cpm for 1 h at room temperature
- 6 Wash four times in wash buffer, dry, and count

Enzyme-Linked Immunoassay (ELISA)

Both of the assays above can be adapted by using peroxidase-conjugated second antibody.

At step 5, substitute 1/60 diluted peroxidase conjugated antibody for a 1 h incubation

- 6 Wash four times in PBS
- 7 Add working solution of fresh 4-chloro-1-naphthol (50 μL /well)

8. When dark blue spots have appeared, stop the reaction by washing the plate

Nitrocellulose Assay

- 1 Draw a 0.5 cm square grid on the nitrocellulose sheet using an ordinary biro, and "dot" the antigen solution centrally with a micropipet (1 μ L of 1 mg/mL in PBS). Air dry
2. Store the sheet desiccated at -20°C until required (months)
- 3 Number the nitrocellulose squares with a biro, and then soak the grid in 5% Hb for 1 h
- 4 Cut out the numbered squares from the grid and incubate them overnight at 4°C in neat tissue culture supernatant or dilutions of antisera (serial tenfold) Use 5 mL bijoux containing about 0.25 mL sample volume
5. Wash the squares four times in PBS, then add $\sim 20,000$ cpm/square of ^{125}I antispecies antibody (10 $\mu\text{Ci}/\mu\text{g}$) in 1% Hb and incubate for one more hour.
- 6 Wash the squares five times in PBS, dry, and autoradiograph

Notes

1. Neither PVC plates nor nitrocellulose blots should ever be allowed to dry out during assays
2. Always include a blank plate as a control in these assays, as nonspecific binding is a common problem
- 3 Always compare normal serum with immunized mouse serum at various dilutions during an assay to give low and high control points.
4. When starting to use these assays, it may be necessary to optimize antigen binding by varying the incubation conditions (pH, ionic strength, time) For example, anionic carbohydrates may require poly-L-lysine coating of the plate (100 $\mu\text{g}/\text{mL}$ overnight) to facilitate antigen binding

References

- 1 Johnstone, A., and Thorpe, R (1982) *Immunochemistry in Practice* Blackwell, Oxford
- 2 Hawkes, R , Niday, E , and Gordon, J (1982) A dot immunobinding assay for monoclonal and other antibodies *Anal Biochem.* **119**, 142–147

Chapter 31

Subclass Analysis and Purification of Monoclonal Antibodies

J. N. Wood

*Department of Experimental Immunobiology,
Wellcome Research Laboratories,
Beckenham, Kent*

Introduction

Once hybridoma lines have been established that secrete monoclonal antibodies of the desired specificity, it is useful to characterize the antibody subclass before mass-producing the antibody (1,2) IgG isotypes may be separated by chromatography on protein A-sepharose, and such affinity columns are useful for concentrating as well as purifying antibody from tissue culture supernatants. When the isotype has been determined, using commercial isotype-specific antisera conjugated to different fluorochromes and monoclonal antibodies of different isotype and specificity, double indirect immunofluorescence

can allow similarities in antigen distribution to be assessed. Ideally, antibody should be purified by affinity chromatography using insolubilized antigen columns (see Chapter 3), although such a technique obviously requires adequate supplies of purified antigen. For many practical purposes, a crude globulin fraction from ascitic fluid is quite adequate, however.

This chapter describes a rapid simple subclass determination, concentration and purification of antibodies, and the preparation of antibody fragments, for use where possible nonspecific binding may create problems of interpretation.

The method of subclass determination described is an obvious development from the nitrocellulose dot-blot technique described in the previous chapter. By "dotting" dilute samples of isotype specific antisera on nitrocellulose sheets, and then immersing them in hybridoma supernatants, the subclass of the antibody can be detected by addition of a third radiolabelled antispecies antibody. Figure 1 shows a typical analysis of six different hybridoma supernatants. By incorporating an antiglobulin antibody on the sheets, one can also be sure that the hybridoma is still producing antibody. The method is far superior to the conventional double agar diffusion test in sensitivity, economy of reagents, simplicity, and speed.

Sodium sulfate may be used to concentrate and purify the globulin fraction from ascitic fluid. If the antibody is of the G1, G2a, or G2b subclass, a more effective purification may be obtained using protein A affinity chromatography, however.

Staphylococcus aureus (Cowan strain) has a surface protein (protein A) that binds the Fc portion of IgG. The different domain structure of the IgG subclasses is reflected in different affinities for protein A, which can be insolubilized on a sepharose matrix. By sequential elution of such an affinity column at different pHs, protein A-bound IgG can thus be purified into different subclasses. The protocol described is suitable for mouse immunoglobulins.

In order to ascribe binding to the variable region of antibody, and discount potential artifacts caused by nonspecific binding, it may be useful to produce a $F(ab')_2$

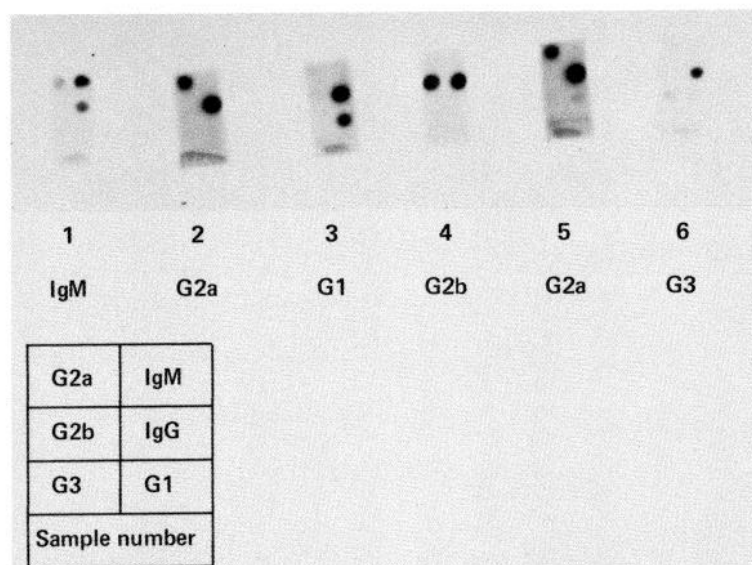


Fig. 1. Subclass determination: 1 mL samples of hybridoma supernatants were analyzed. The dark spots correspond to the antibody subclass marked on the grid. The samples shown were autoradiographed for 3 d, but a clear answer could be obtained after overnight autoradiography or the use of enzyme linked reagents (*see* Chapter 30).

fraction. This immunoglobulin fragment is prepared by pepsin hydrolysis, which cleaves the molecule distal to heavy chain disulfide bonds, so that a bivalent antibody fragment— $F(ab')_2$ —is left while the constant (Fc) region of the antibody is degraded.

Materials

Subclass Determination

1. Nitrocellulose sheets.
2. Commercial antispecies anti-isotype antisera and antiglobulin antisera: e.g., rabbit anti-mouse anti G1, G2a, G2b, G3, M, and A. Store at 4°C in 0.02% azide.
3. ^{125}I -antispecies antibody (specific activity > 10 $\mu\text{Ci}/\mu\text{g}$) preferably $F(ab')_2$ (*see* Chapter 30).

Globulin Concentration

1. 27% Na₂SO₄. Store at room temperature in water
2. Phosphate-buffered saline (PBS) (*see* Chapter 29)

Protein-A Purification

1. Protein-A-sepharose column, 5 mL commercial column
2. 0.1M Phosphate buffer, pH 8.0 (store at 4°C)
3. Sodium citrate buffers, 20 mL each of
 - 100 mM pH 6.0
 - 100 mM pH 4.5
 - 100 mM pH 3.5
 - 100 mM Citric acid

Store frozen

4. 2M Trizma base in water. Store frozen.

Preparation of F(ab')₂ Fragments

1. 0.1M sodium acetate, pH 4.5.
2. Pepsin, made up fresh from 10 mg/mL in 0.14M sodium acetate, pH 4.5.
3. 2M Trizma base in water
4. Sephadex G-200 column (20 × 1 cm) equilibrated in PBS (*see* Chapter 28).

Purification of IgM

1. Sepharose, 4B column (50 × 2.5 cm) equilibrated in PBS.

Methods

Subclass Determination

1. Dot 1/100 dilutions of commercial anti-isotype antisera in a marked pattern on nitrocellulose sheets and air dry.

2. The sheets may be stored dry at -20°C indefinitely.
3. When samples are available, block the grids in 1 mL 5% hemoglobin in PBS in a bijou on a spiramix.
4. Quickly transfer the grids to Universals containing 1 mL tissue culture fluid and rotate for 1 h at room temperature
5. Wash the grids three times in large volumes of PBS (5 min/wash)
6. Add 10^5 cpm F(ab')_2 ^{125}I -antispecies antibody ($10 \mu\text{Ci}/\mu\text{g}$) in 1% hemoglobin in PBS for 1 h at room temperature
7. Wash four times in PBS
8. Dry, autoradiograph overnight at -70°C , and develop

Globulin Concentration

1. Add two volumes of 27% sodium sulfate to 1 vol of ascitic fluid at room temperature with stirring.
2. Leave the solution for 1 h at 37°C
3. Centrifuge at $10,000g$, 10 min.
4. Resuspend the pellet in a volume of PBS equivalent to the original volume of ascites.
5. Dialyze exhaustively against PBS
6. Store frozen or freeze dried.

Protein-A Affinity Chromatography

1. Pack a 1 mL protein-A Sepharose column for every 10 mg of mouse IgG in your sample
2. Equilibrate the column with 0.1M phosphate buffer, pH 8.0, at 4°C
3. Apply sample in the same buffer at 1 mL/min or less.
4. The run-through will contain IgG3, as well as IgA, most IgM, and other proteins.
5. Sequentially elute the column with 10 mL of
 - a. pH 6.0 citrate buffer for IgG1
 - b. pH 4.5 citrate buffer for IgG2a
 - c. pH 3.5 citrate buffer for IgG2b
 - d. citric acid to clear the column

6. Neutralize both the acidified sample eluates and the column with Trizma base as quickly as possible.

The column should be kept at 4°C in azide containing (0.05%) PBS and can be re-used indefinitely. Tissue culture monoclonal antibody from fetal calf serum containing medium should be essentially pure, whereas ascitic fluid-containing antibody may contain some contaminating host IgG.

Production of F(ab')₂ Fragments

1. Dissolve 10 mg IgG in 1 mL acetate buffer.
2. Add 20 µL of pepsin (10 mg/mL in acetate buffer) and incubate overnight at 37°C
- 3 Neutralize the mixture with Tris to inhibit further proteolysis.
4. Separate the products of digestion by gel exclusion chromatography on a Sephadex-G200 column equilibrated with PBS.
- 5 The first major peak is the F(ab')₂ material. A shoulder of undigested IgG may run ahead of the main peak if digestion was incomplete, and this material should be discarded

Purification of IgM

IgM monoclonal antibodies may best be purified from concentrated globulin extracts by gel exclusion chromatography on Sepharose-4B columns. The size (~900,000 daltons) of IgM is reflected in its elution position as the first major peak

- 1 Concentrate the antibody by sodium sulfate precipitation.
2. Resuspend the precipitate in a minimal volume of PBS, and apply to a Sepharose-4B column.
3. Collect the first peak of OD₂₈₀-absorbing material.

Notes

Although antisera are commonly stored at 4°C, in our experience a number of monoclonal antibodies stored in

tissue culture medium at this temperature have proved to be unstable. It is safer to store the antibodies aliquoted at -20°C , and discard unfrozen material after a single use. A number of textbooks claim that IgM antibodies are more stable at 4°C than -20°C , but this is not true for many of the monoclonal IgMs we have studied.

References

- 1 Johnstone, A P , and Thorpe, R (1982) *Immunochemistry in Practice*, Blackwell, Oxford
- 2 Ey, P L , Prowse, S J , and Jenkin, C P (1978) Isolation of pure IgG1, IgG2b immunoglobulins from mouse serum using protein-A sepharose *Immunochemistry* **15**, 429–438

Chapter 32

The Production of Antisera

G. S. Bailey

*Department of Chemistry, University of Essex,
Colchester, Essex, England*

Introduction

Suitable antisera are essential for use in all immunochemical procedures. Three important properties of an antiserum are avidity, specificity, and titer. The avidity of an antiserum is a measure of the strength of the interactions of its antibodies with an antigen. The specificity of an antiserum is a measure of the ability of its antibodies to distinguish the immunogen from related antigens. The titer of an antiserum is the final (optimal) dilution at which it is employed in the procedure; it depends on the concentrations of the antibodies present and on their affinities for the antigen. The values of those parameters required for a particular antiserum very much depend on the usage to which the antiserum will be put. For example, for use in radioimmunoassay, it is best to have a monospecific antiserum of high avidity, whereas for use in immunoaffinity chromatography the monospecific antiserum should not

possess too high an avidity otherwise it may prove impossible to elute the desired antigen without extensive denaturation.

A substance that, when injected into a suitable animal, gives rise to an immune response is called an immunogen. The immunogenicity of a substance is dependent on many factors, such as its size, shape, chemical composition, and structural difference from any related molecular species indigenous to the injected animal. Normally, cellular (particulate) materials are very immunogenic and induce a rapid immune response. However, the resultant antisera do not usually possess a high degree of specificity and do not store well (1).

Soluble immunogens differ widely in their ability to produce an immune response. In general, polypeptides and proteins of molecular weight above 5000 and certain large polysaccharides can be effective immunogens. Smaller molecules, such as peptides, oligosaccharides, and steroids, can often be rendered immunogenic by chemical coupling to a protein that by itself will produce an immune response (2). For most situations it is best to use the most highly purified sample available for injecting into the animal. Furthermore, it is usual to inject a mixture of the potential immunogen and an adjuvant that will stimulate antibody production.

Classically, the immune response is described as occurring in two phases. Initial administration of the immunogen induces the primary phase (response) during which only small amounts of antibody molecules are produced as the antibody producing system is primed. Further administration of the immunogen results in the secondary phase (response) during which large amounts of antibody molecules are produced by the large number of specifically programmed lymphocytes. In practice, the time-scale and nature of the immunization process does not lead to a clear recognition of two distinct phases.

All of the factors that influence antibody production have not yet been elucidated. Thus the raising of an antiserum is, to some extent, a hit or miss affair. Individual animals can respond quite differently to the same process of immunization and thus it is best to use a number of animals. The species of animal chosen for immunization will

depend on particular circumstances, but, in general, rabbits are often used

Many different methods of producing antisera have been described, varying in amount of immunogen required (often milligram quantities), route of injection, and frequency of injection (3). This chapter will describe a method of antiserum production (4) that has been successfully utilized in the author's laboratory using small doses (microgram quantities) of soluble protein as immunogen (5).

Materials

1. Six rabbits each of 2 kg body weight. Various types can be used, e g , New Zealand whites, Dutch, and so on.
2. Solution of the purified immunogen in an appropriate buffer to maintain its stability
3. Complete and incomplete Freund's adjuvant, available from various commercial sources.
4. Heat lamp.

Method

The following procedure is carried out for each rabbit in turn

1. Thoroughly mix one volume of the immunogen solution with three volumes of complete Freund's adjuvant with the aid of a glass pestle and mortar. Initially the mixture is very viscous, but after about 5 min the viscosity becomes less. The mixture can then be transferred to a syringe. The syringe is emptied and refilled with the mixture a number of times resulting in the formation of a stable emulsion that can be injected intradermally into the prepared rabbit. The emulsion should be used within 1 h of preparation.
2. The rabbit is prepared by cutting away the long hair along the center of its back. The short hair is removed by shaving. The emulsion of immunogen and complete Freund's adjuvant is injected via a 1 mL syringe

plus 21 gage needle into two rows of five sites equidistantly spaced along the rabbit's back, each row being about 2 cm from the backbone, such that each site receives 0.1 mL of emulsion containing 1–10 μg immunogen, i.e., a total dose of 10–100 μg immunogen/rabbit

3. After 8–10 wk, a test bleeding is carried out on the rabbit. The fur on the back of one ear is removed by shaving. The eyes of the rabbit are protected while the shaven ear is heated for less than 1 min with a heat lamp to expand the vein. The expanded vein just below the surface of the back of the ear is nicked with a scalpel blade. Blood is collected in a glass vessel (up to 20 mL can be collected in 10 min) removing the clot from the puncture wound by rubbing the ear with cotton wool from time to time. The blood is allowed to clot standing at room temperature for a few hours. The serum is separated from the clot by centrifugation and can be stored at 4°C in the presence of 0.1% sodium azide as antibacterial agent until tested.
4. If on testing the serum shows the characteristics required for its particular usage, then further bleeding can be carried out. Up to three bleedings can be made on successive days. After that it is best to allow the rabbit to rest for about a month before further bleeding.
5. If the original antiserum is unsatisfactory or if the quality of the bleedings taken over a period of several weeks or months starts to become unsatisfactory, then the rabbit can be boosted, i.e., receive a second injection of immunogen.
6. For the booster injection, half the original dose of immunogen is administered in incomplete Freund's adjuvant. The emulsion is prepared as detailed before, but it is injected *subcutaneously* into the rabbit (for example, into the fold of skin of the neck).
7. After 10 d a test bleeding is obtained from the rabbit and the serum so produced is analyzed.
8. Further bleedings can be carried out over a period of time if the boosted antiserum is satisfactory.
9. If after boosting, the antiserum is not of the desired quality it is best to disregard that rabbit. Hopefully,

one or more of the other rabbits in the group will have produced good antisera either directly or after boosting. However, in some cases, particularly with weak immunogens, it may be necessary to repeat the process of immunization with a new group of rabbits or other animals.

10. Each sample of antiserum can now be tested for its ability to form an immune precipitate with the immunogen by carrying out immunodiffusion and immunoelectrophoresis. For example, the specificity of the antibodies can be determined by running the antiserum against the immunogen and related antigens in Ouchterlony double diffusion (*see* Chapter 33). The titer and a measure of the avidity of the antiserum can be obtained by radioimmunoassay (*see* Chapter 37)

Notes

- 1 Many antisera can be satisfactorily stored at 4°C in the presence of an antibacterial agent for many months. After some time, the antiserum solution may become turbid and even contain a precipitate (mostly of denatured lipoprotein). Even so, there should be no significant reduction in the quality of the antiserum. If necessary, the solution can be clarified by membrane filtration. For prolonged storage, the antiserum can be kept at -20°C in small quantities so as to avoid repeated thawing and freezing.
- 2 Animals other than rabbits can be used for immunization, e.g., mice, rats, guinea pigs, sheep, goat, and so on. The rabbit is often a good initial choice, but if results are unsatisfactory, other species can be tried. Obviously only small volumes of antisera can be generated in the small species, whereas large volumes can be obtained from the larger species. The latter though do require more immunogen and are more expensive to maintain.
3. A clean sample of antiserum should be straw-colored. Pink coloration is due to partial hemolysis, but should not affect the properties of the antiserum.

4. The dose of immunogen employed can be of crucial importance in many procedures for antibody production. A state of tolerance can be induced in the animal with little or no production of antibody if too much or too little immunogen is repeatedly given over a relatively short period of time. The method described in this chapter should not suffer from that effect since there is a gap of at least 10 wk between the initial and booster injections. In general, the lower the dose of antigen the greater is the avidity of the antiserum.
5. Since the antisera produced by conventional methods consist of mixtures of different antibody molecules, it is to be expected that the properties of the antisera collected during the prolonged period of immunization may change. Thus each bleeding should be tested for specificity, titer, and avidity.

References

- 1 Hurn, B. A. L., and Chantler, S. M. (1980) Production of Reagent Antibodies in *Methods in Enzymology* **70** (eds Van Vunakis, H., and Langone, J. J.) pp 104–141. Academic Press, New York.
- 2 Erlanger, B. F. (1980) The Preparation of Antigenic Hapten-Carrier Conjugates. A survey, in *Methods in Enzymology* **70** (eds Van Vunakis, H., and Langone, J. J.), pp 85–104. Academic Press, New York.
- 3 Weir, D. M. (1978) *Handbook of Experimental Immunology* **3**, third edition. Blackwell, Oxford.
- 4 Vaitukaitis, J. L. (1981) Production of Antisera with Small Doses of Immunogen. Multiple Intradermal Injections in *Methods in Enzymology* **73** (eds. Langone, J. J., and Van Vunakis, H.) pp 46–57. Academic Press, New York.
- 5 Hussain, M., and Bailey, G. S. (1982) An improved method of isolation of rat pancreatic prokallikrein. *Biochim Biophys Acta* **719**, 40–46.

Chapter 33

Immunodiffusion in Gels

G. S. Bailey

*Department of Chemistry, University of Essex,
Colchester, Essex, England*

Introduction

Immunodiffusion in gels encompasses a variety of techniques that are useful for the analysis of antigens and antibodies (1). The fundamental immunochemical principles behind their use are exactly the same as those that apply to antigen–antibody interactions in the liquid state. Thus an antigen will rapidly react its specific antibody to form a complex, the composition of which will depend on the nature, concentrations, and proportions of the initial reactants. As increasing amounts of a multivalent antigen are allowed to react with a fixed amount of antibody, precipitation occurs, in part because of extensive crosslinking between the reactant molecules. Initially the antibody is in excess and all of the added antigen is present in the form of an insoluble antigen–antibody aggregate. Addition of more antigen leads to the formation of more immune precipitate. However, a point is reached beyond which further addition of antigen produces an excess of antigen and leads to a reduction in the amount of the precipitate (*see*

Fig. 1) because of the formation of soluble antigen-antibody complexes. The analysis of such interactions occurring in gels is of much higher sensitivity and resolution than that for the liquid state, thus explaining the extensive use of immunochemical gel techniques.

The methods employing immunodiffusion in gels are often classified as simple (single) diffusion or double diffusion. In simple diffusion, one of the reactants (often the antigen) is allowed to diffuse from solution into a gel containing the corresponding reactant, whereas in double diffusion both antigen and antibody diffuse into the gel. A variety of information, both qualitative and quantitative, can be obtained from the numerous techniques (2). This chapter will be concerned with two methods that are very widely employed for immunochemical analysis.

Ouchterlony Double Immunodiffusion

This technique is often used in qualitative analysis of antigens and antisera (2).

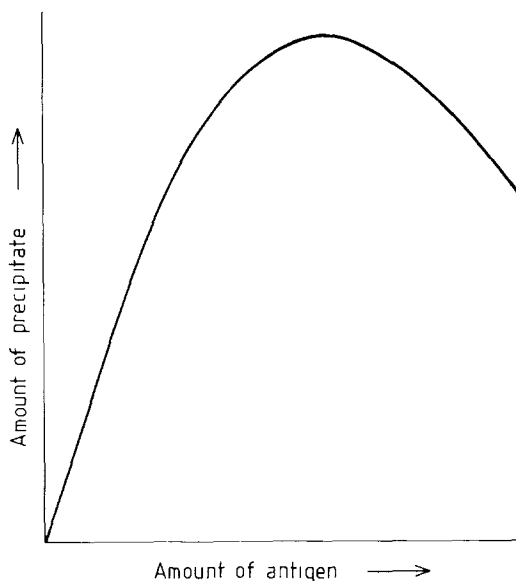


Fig 1 Variation in the amount of immune precipitate on addition of increasing amounts of antigen to a fixed amount of antibody

Single Radial Immunodiffusion (SRID)

This method is used for the quantitative analysis of antigens (3).

Materials

1. Agarose.
2. 0.07M barbitone buffer, pH 8.6, containing 0.01% thimerosal as antibacterial agent. The buffer is prepared by dissolving sodium barbitone (14.5 g), disodium hydrogen phosphate decahydrate (7.16 g), boric acid (6.2 g), and sodium ethyl mercurithiosalicylate (0.1 g) in distilled water and making the final volume up to 1 L.
3. Solutions of antigen and antiserum
4. Plastic or glass Petri dishes or rectangular plates.
5. Gel punch and template. Suitable gel punches of various sizes and templates of various designs can be obtained from commercial sources.
6. Flat level surface.
7. Humidity chamber at constant temperature.
8. 0.1M sodium chloride in distilled water.
9. Staining solution. The solution is prepared by mixing ethanol (90 mL), glacial acetic acid (20 mL), and distilled water (90 mL), and then adding Coomassie Brilliant Blue R-250 (1 g). The solution can be re-used several times.
10. Destaining solution. The composition is the same as that of the staining solution but without the dye.

Method

Ouchterlony Double Immunodiffusion

1. Agarose (1 g) is dissolved in the barbitone buffer (100 mL) by heating to 90°C on a water bath with constant stirring.

2. The agarose solution is poured to a depth of 1–2 mm into the Petri dishes or onto the rectangular plates that had previously been set on a horizontal level surface. The gels are allowed to form on cooling and when set (5–10 min) can be stored at 4°C in a moist atmosphere for at least 1 wk
3. A template of the desired pattern, according to the number of samples to be analyzed, is positioned on top of the gel. Commonly used patterns are shown in Fig. 2. The gel punch, which is connected to a water vacuum pump, is inserted into the gel in turn through each hole of the template so that the wells are cleanly formed as the resultant agarose plugs are sucked out.
4. The wells are filled with the solutions of antigen and antisera until the meniscus just disappears. The concentrations of antigen solutions and the dilution of the antiserum to be used have to be established largely by trial and error by running pilot experiments with solutions of different dilutions. However, as a rough guide, for the analysis of antigens, the following concentrations of antigen solutions can be run against undiluted antiserum: for a pure antigen, 1 mg/mL; for a partially pure antigen, 50 mg/mL, for a very impure antigen, 500 mg/mL, using 5 μ L samples of antigen solutions and antiserum.
5. The gel plate is then left in a moist atmosphere, e.g., in a humidity chamber at a constant temperature of 20°C for 24 h

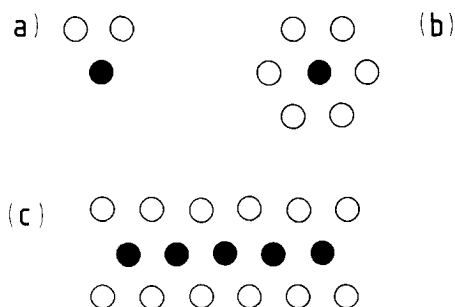


Fig. 2. Patterns of wells often used in double immunodiffusion ○, well contains antigen, ●, well contains antiserum

6. The precipitin lines can then be recorded *either* directly by photography with dark-field illumination or by drawing the naked eye observation on suitable oblique illumination of the plate against a dark background *or* after staining. Prior to staining, excess moisture is removed from the gel plate by application of a 1 kg weight (e.g., a liter beaker full of water) to a wad of filter paper placed on top of the gel for 15 min. Soluble protein is removed by washing the gel (3×15 min) in 0.1M sodium chloride solution followed by further pressing. The gel is dried using cold air from a hair dryer and is then placed in the staining solution for 5 min. The plate is finally washed with distilled water and placed in the destaining solution for about 10 min.

Single Radial Immunodiffusion (SRID)

1. Agarose (2 g) is dissolved in the barbitone buffer (100 mL) by heating to 90°C on a water bath with constant stirring.
2. The solution is allowed to cool to 55°C and is mixed with an equal volume of a suitable dilution of the monospecific antiserum also at 55°C. The optimal dilution of the antiserum has to be found from pilot experiments using different dilutions of the antiserum. The dilution chosen depends in part on the range of antigen concentrations that is required to be measured.
3. The agarose solution containing the antiserum is poured onto the rectangular plates that had previously been set on a horizontal level surface. The gels are allowed to form on cooling and the gel plates are normally used within 24 h, but can be stored for longer periods at 4°C provided dessication is avoided.
4. Using a gel punch and a template, a row of the desired number of wells of diameter 2 mm is cut into the gel
5. At least three of the wells are filled with standard solutions of antigen of known concentration. The other wells are filled with the solutions containing the antigen at unknown concentrations. For the most accurate work it is necessary to add all samples via a microsyringe.

- 6 The gel plate is then left in a humidity chamber for an appropriate period of time. As the antigen diffuses into the gel containing the antiserum a disc of immune precipitate is formed. The final, maximum area of that disc is directly proportional to the initial concentration of the antigen in the well (and inversely proportional to the antibody concentration in the gel). The time required to achieve maximum area of precipitation depends on the velocity of diffusion of the antigen. That, in turn, is dependent on temperature and molecular size of the antigen. In general, the diffusion at room temperature must be allowed to continue for a few days, taking measurements of the areas of the discs every 24 h until no further increase takes place.
- 7 The area of each disc is measured in terms of its diameter, which can be measured directly with the aid of a magnifying glass on suitable oblique illumination of the plate positioned over a dark background. For increased resolution of the discs, the plate can be placed in 1% tannic acid for 3 min prior to viewing.
8. At the end of the diffusion process the plate can be stained. Soluble protein is removed from the gel by washing for 2 d with several changes of 0.1 M sodium chloride solution. The plate is then pressed, dried, stained and destained as detailed for Ouchterlony double immunodiffusion (*see above*).
9. A standard graph is constructed by plotting the diameters of the discs against the logarithm of the antigen solutions of known concentration. The concentrations of the antigen in the test samples can then be determined by simple interpolation (*see Fig. 3*).

Notes

1. Ouchterlony double diffusion is frequently used for comparing different antigen preparations.

If the antigen solution contains several different antigens that can react with the antibodies of the antiserum, then multiple lines of precipitation will be produced. The relative position of each line is determined by the local concentration of each antigen and antibody.

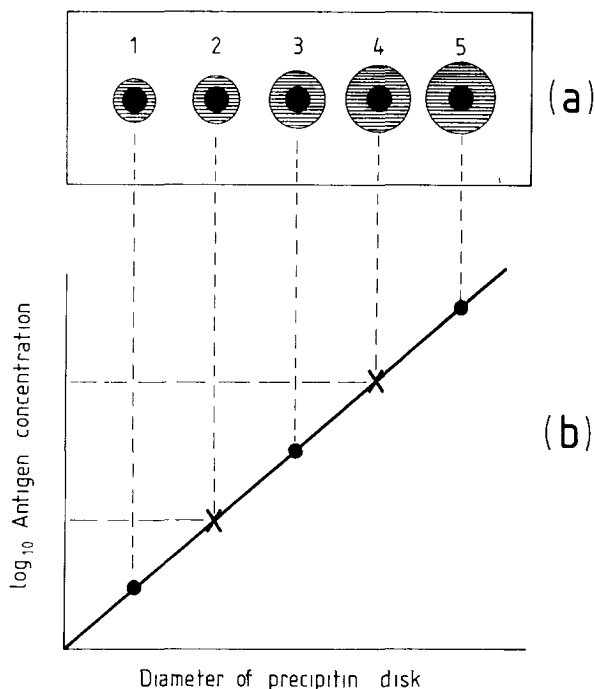


Fig 3 Single radial immunodiffusion (a) Precipitin discs formed at termination of diffusion of antigen into gel containing monospecific antiserum Wells 1, 3, and 5 contained standard antigen solutions of increasing concentration Wells 2 and 4 contained samples of the antigen at unknown concentrations Shaded area, immunoprecipitate (b) Semilog plot of diameter of the precipitin discs of standard antigen solutions (●) against concentration Measurement of the diameters of the precipitin discs of the unknown solutions (X) allows an estimation of the antigen concentration to be made by simple interpolation

in the gel In turn, those local concentrations depend not only on the initial concentrations of the reactants in the wells, but on the rates at which they diffuse through the gel and hence are also dependent on molecular size.

If different antigen preparations, each containing a single antigenic species capable of reacting with the antiserum used, are allowed to diffuse from separate wells, then the degree of similarity of the antigens can be assessed by observation of the geometrical pattern

produced. For example, Figure 4 shows the four basic precipitin patterns that can be produced in a balanced system by two related antigens interacting with an antiserum that contains antibodies that can recognise both antigens.

Pattern (a) is called the "pattern of identity" or "pattern of coalescence" and indicates that the antibodies in the antiserum employed cannot distinguish the two antigens, i.e., the two antigens are immunologically identical as far as that antiserum is concerned.

Pattern (b) is called the "pattern of non-identity" or "pattern of absence of coalescence" and indicates that none of the antibodies in the antiserum employed react with antigenic determinants that may be present in both antigens, i.e., the two antigens are unrelated as far as that antiserum is concerned.

Pattern (c) is called a "pattern of partial identity" or "pattern of partial coalescence of one antigen" and indicates that more of the antibodies in the antiserum employed react with one antigen (that diffusing from the left-hand well in Fig. 4c) than the other antigen. The "spur", extending beyond the point of partial coalescence, is thought to result from the determinants present in one antigen but lacking in the other antigen

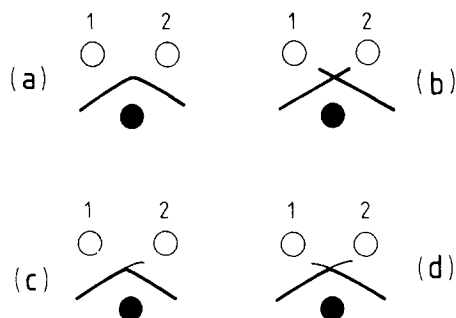


Fig. 4 Patterns of precipitin lines (a) pattern of coalescence, (b) pattern of absence of coalescence, (c) pattern of partial coalescence of one antigen, (d) pattern of partial coalescence of two antigens Well 1 contains antigen 1, well 2 contains antigen 2, well ● contains antiserum,—precipitin line.

Pattern (d) is called "the pattern of partial coalescence of two antigens" and is another type of "pattern of partial identity." It indicates that some of the antibodies in the antiserum employed react with both antigens, whereas other antibodies only react with one or other of the antigens. Thus the two antigens have at least one antigenic determinant in common, but also differ in other antigenic determinants.

2. The following comments relate to Ouchterlony Double Immunodiffusion

- (a) If glass petri dishes or rectangular plates are used for this technique, it normally is necessary for them to be precoated with a 0.5% agarose gel prior to formation of the main gel to prevent the main gel from becoming detached during the washing and staining procedures.
- (b) Buffers of different composition and pH from the described buffer can be used if more convenient, e.g., 0.05M phosphate buffer, pH 7.2, containing 0.1M sodium chloride.
- (c) The agarose gel solution, once made, can be separated into test tubes and stored at 4°C for several weeks. Approximately 12 mL of 1% agarose gel solution is required for an area of 75 mL. The contents of each test tube can be simply melted as required.
- (d) Diffusion of antigens and antibodies is more rapid at higher temperatures. Thus double immunodiffusion is often run for 48 h at 4°C, or 24 h at room temperature or for as little as 3 h at 37°C. It is important to maintain a constant temperature otherwise artifacts may be produced.
- (e) Other artifacts can be produced by refilling of the wells or by denaturation of the antigen during diffusion. If the latter is a possibility, it is best to run the immunodiffusion at 4°C.
- (f) The sensitivity of the Ouchterlony technique is largely determined by the relative concentrations of the antigens and antibodies, and by the separation of the wells. In the latter case, the closer the wells the greater is the sensitivity. It should be

possible to detect protein solutions of 10 $\mu\text{g/mL}$ concentration.

3. The following comments relate to Single Radial Immunodiffusion.
 - (a) If samples in neighboring wells are too concentrated, their precipitin discs will overlap and in that case the samples have to be run again at higher dilutions.
 - (b) If the antiserum employed is not monospecific for the required antigen in an impure sample a number of precipitin discs will be produced by the different antigens interacting with their corresponding antibodies. The problem then is to identify the disc produced by the antigen under study. One solution is to rerun each sample in duplicate with one well containing just the sample and the duplicate well containing the sample plus a fixed amount of standard antigen. The disc caused by the antigen under study can then be identified by its larger area around the duplicate well.
 - (c) For maximum sensitivity the antiserum used in the gel should be diluted with non-immune serum, i.e., serum from a non-immunized animal. Furthermore, the dimensions of the wells and volumes of samples and standards applied can be increased. It should be possible to measure protein solutions of 1 $\mu\text{g/mL}$ concentration.

References

1. Oudin, J. (1980) Immunochemical analysis by antigen-antibody precipitation in gels in *Methods in Enzymology* 70 (eds Van Vunakis, H., and Langone, J. J.) pp. 166–198. Academic Press, New York.
2. Ouchterlony, O. (1968) *Handbook of Immunodiffusion and Immuno-electrophoresis*. Ann Arbor Science Publications, Michigan.
3. Vaerman, J. P. (1981) Single radial immunodiffusion, in *Methods in Enzymology* 73 (eds Langone, J. J., and Van Vunakis, H.) pp. 291–305. Academic Press, New York.

Chapter 34

Crossed Immunoelectrophoresis

Graham B. Divall

*Metropolitan Police Forensic Science
Laboratory, London, England*

Introduction

Crossed immunoelectrophoresis is a simple, quick, and sensitive technique for the qualitative detection of a wide range of protein antigens

The method consists of making two small wells, approximately 5 mm apart, in a thin layer of agar buffered at pH 8.6. One of the wells (sample well) is filled with a solution thought to contain the protein in question. The second (antiserum) well is filled with a monospecific antiserum against that protein. The whole gel is then subjected to electrophoresis for about 15 min. At pH 8.6, the vast majority of proteins are negatively charged and migrate towards the anode. The antibodies, contained in the γ -

globulin fraction of the antiserum, have a relatively low negative charge, but their net movement is towards the cathode as a result of electroendosmosis. Protein antigen and corresponding antibody are therefore caused to migrate towards each other in the gel and form lines of precipitation between the wells.

The phenomenon of electroendosmosis is caused by the presence of negatively charged groups in the agar gel and the surface adsorption of positive ions from the buffer. The agar gel as the supporting medium is, of course, fixed, but the adsorbed positive ions migrate towards the cathode. This accelerates the migration of cations towards the cathode and retards or even reverses the migration of anions towards the anode. In the case of crossed immunoelectrophoresis, the net effect of electrophoresis and electroendosmosis is for the γ -globulins to migrate towards the cathode.

The method is similar to simple gel immunodiffusion. However, it is quicker and more sensitive because all the antigen and antibody are driven towards each other under the influence of the electric field rather than being allowed to diffuse radially.

The technique is suitable for the detection of any protein with an electrophoretic mobility at pH 8.6 greater than γ -globulin and for which a monospecific precipitating antiserum is available. It is widely used to determine the species origin of bloodstains in forensic science and in many clinical screening procedures, for example, the detection of Australia Antigen.

Materials

1. Agar gel buffer, pH 8.6, consisting of: 7.0 g sodium barbiturate; 1.1 g diethyl barbituric acid, 1.02 g calcium lactate dissolved in 1 L of distilled water.
2. Electrophoresis tank buffer, pH 8.6, consisting of: 17.52 g sodium barbiturate; 2.76 g diethyl barbituric acid; and 0.77 g calcium lactate dissolved in 2 L of distilled water.
3. Staining solution, consisting of: 0.4 g naphthalene black 10 B, 200 mL methyl alcohol; 40 mL glacial acetic acid, and 200 mL distilled water.

4. Wash/destain solution, consisting of 500 mL methyl alcohol; 100 mL glacial acetic acid; and 500 mL distilled water.
5. 2% w/v stock agar in distilled water: Prepared by mixing 10 g dry agar powder with 500 mL distilled water in a 2 L glass beaker. Heat the suspension over a low Bunsen flame and stir continuously with a glass rod. Bring to the boil and continue boiling for a few minutes to ensure that the agar is completely dissolved. Tip the hot agar solution into a storage container (plastic sandwich boxes are ideal) and allow to cool and gel. Fit a lid and store 2% stock agar gel at 4°C.
6. Glass plates (microscope slides) 7.5 × 5 cm, approximately 1.5 mm thick.

Method

1. Take 50 g stock 2% agar and dissolve in 50 mL gel buffer by gentle heating in a boiling water bath or over a low Bunsen flame.
2. Take 12 glass plates. Scribe an identification mark in one corner of each plate with a diamond pencil. Clean the surfaces with a tissue moistened with methyl alcohol and place the slides on a horizontal surface.
3. Carefully transfer 7 mL of the hot agar solution to the surface of each glass plate. This is best achieved by using a large plastic hypodermic syringe. Leave the plates for 10 min to allow the agar to cool and gel, then place them in a moist chamber (plastic box with a tight fitting lid and lined with damp filter paper) and store at 4°C.
4. Make a series of paired wells in the gel. They should be approximately 1–1.5 mm in diameter and 5–8 mm apart. The wells are made and plugs of agar removed by means of a glass Pasteur pipet or a piece of metal tubing attached to a vacuum pump.
5. Place the gel plate on a dark surface with the identification mark in the top right-hand corner. For each pair of wells, fill the one on the left with a solution of the sample and the one on the right with antiserum. This can be difficult and is best done using finely

drawn Pasteur pipets. Be careful not to overfill the wells. For maximum reproducibility, use a microsyringe and fill the wells with 2–3 μL of sample solution or antiserum

6. Invert the gel plate (the solutions will be retained in their wells) and place it across the bridge of an electrophoresis tank with the cathode on the left and the anode on the right. The edges of the gel surface should rest on filter paper wicks to form a bridge gap of about 6.5 cm. The identification mark should now be visible in the bottom right-hand corner. The sample well of each pair will be nearest the cathode and the antiserum well nearest to the anode
7. Turn on the power pack and subject the gel plate to electrophoresis at 130 V (20 V/cm) for 15 min.
8. After electrophoresis, remove the plate from the tank and view the gel with oblique lighting. A positive result is indicated by a clear sharp white precipitin line between the two wells. Very weak results cannot be seen at this stage. Furthermore, for some samples such as those encountered in forensic analysis, a nonspecific protein precipitate forms between the wells and can be confused with a true immunoprecipitate. For these reasons, the gel is best washed and stained
9. Place the gel plate in a 1M saline bath for at least 24 h. During this time all protein will be washed out of the gel except for immunoprecipitates that have become enmeshed in the gel matrix. Transfer the gel plate to a distilled water bath for 2 h to remove the salt. The washing procedures are more efficient if the saline/water baths are shaken gently.
10. Remove the gel plate from the water and cover the gel surface with a piece of wet filter paper. The plate can now be left overnight at room temperature, or placed in a incubator at 50–60°C for several hours to allow the gel to dry.
11. When the gel is dry the filter paper is removed. Take care here because the paper tends to stick to the surface of the dried gel and if it is pulled off, the gel film can be torn and lost. To avoid this, moisten the paper with tap water and then gently peel it away from the

gel film. The dried gel will appear as a shiny film fixed to the glass plate. Wash the plate under running water while gently rubbing the gel surface with a finger to remove fragments of adherent paper.

12. Place the plate in the staining solution for 15 min. Shaking is unnecessary. Transfer the plate to the wash/destain solution for 10 min. Destaining is more efficient if the bath is gently shaken.
13. Remove the plate, allow it to dry, and then view over a light bench. The immunoprecipitates appear as dark blue lines between the sample and antiserum wells. A typical result is shown in Fig. 1.

Notes

1. An obvious source of error occurs when the sample and antiserum wells, or the polarity of the electrophoresis tank, are reversed. In either case the reactants will be driven apart rather than together.
2. The gels sometimes have a tendency to float off the glass plate during the washing stage. This can be mini-

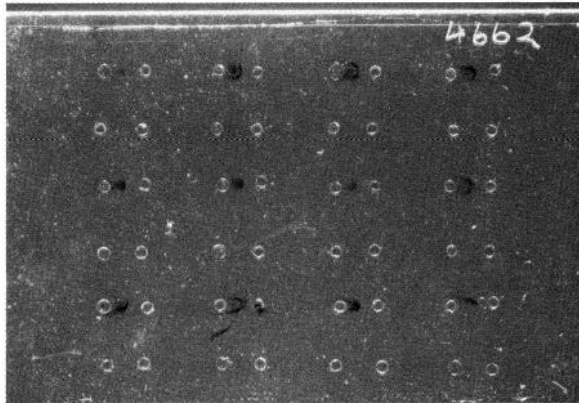


Fig. 1. A crossed immunoelectrophoresis plate used for the species identification of bloodstains in forensic science. The gel has been dried and stained. A dark (stained) precipitin line can be seen between some of the wells indicating a positive reaction between the bloodstain extract and a polyspecific antiserum to human serum proteins.

- mized by precoating the glass plates with 0.2% agar. Simply brush the top surface of the plate with melted 0.2% agar, allow to dry, and proceed from step 3. Alternatively, and less bother, cut the top left hand corner of the gel off before washing. This allows the correct orientation of gel to plate to be made before drying.
3. The sensitivity of the method is ultimately dependent on the titer and avidity properties of the antiserum used. For example, different antisera have been able to detect human blood diluted to between 1 in 15,000 and 1 in 500,000.
 4. The method is absolutely dependent on the agar gel exhibiting electroendosmotic flow. Different agars show different degrees of electroendosmosis and the best agar for any particular system usually has to be found by trial and error.
 5. The formation of an immunoprecipitate and its position between the wells is influenced by the antigen-antibody ratio. Precipitates 'hugging' the sample or antiserum well are to be avoided and it is sometimes necessary to test dilution series of the sample against the antiserum, the sample against a dilution series of the antiserum, or both.

References

- 1 Culliford, B. J. (1964) Precipitin reactions in forensic problems. *Nature* **201**, 1092-1094.

Chapter 35

Rocket Immunoelectrophoresis

John M. Walker

*School of Biological and Environmental
Sciences, The Hatfield Polytechnic
Hatfield, Hertfordshire, England*

Introduction

Rocket immunoelectrophoresis (also referred to as electroimmunoassay) is a simple, quick, and reproducible method for determining the concentration of a specific protein in a protein mixture. The method, originally introduced by Laurell (1) involves a comparison of the sample of unknown concentration with a series of dilutions of a known concentration of the protein, and requires a monospecific antiserum against the protein under investigation. The samples to be compared are loaded side-by-side in small circular wells along the edge of an agarose gel that contains the monospecific antibody. These samples (antigen) are then electrophoresed into the agarose gel where interaction between antigen and antibody takes place. In the presence of excess antigen, the anti-

gen-antibody complex is soluble, but as the antigen moves further into the gel, more antigen combines with antibody until a point of equivalence is reached. At this stage the antigen-antibody complex is insoluble. The end result is a precipitation 'rocket' spreading out from the loading well. Since the height of the peak depends on the relative excess of antigen over antibody, a comparison of the peak heights of the unknown and standard samples allows the unknown protein concentration to be determined. Although particularly useful for measuring the concentration of any given protein in a serum sample, the method may of course be used to measure or compare protein concentrations in any mixture (e.g., urine, cerebrospinal fluid, tissue homogenates, and so on) as long as an antiserum to the protein is available.

Materials

1. 0.06M barbitone buffer, pH 8.4, is prepared from sodium barbitone and barbitone. Dilute this buffer 1:1 with distilled water for use in the electrophoresis tanks
2. Agarose, 2% w/v (aqueous).
3. Protein stain: 0.1% Coomassie Brilliant Blue in 50% methanol/10% acetic acid. (Dissolve the stain in the methanol component first, and then add the appropriate volumes of acetic acid and water). Destain: 10% methanol/7% acetic acid.
4. Glass plates, 5 × 5 cm, 1.0–1.5 mm thick.

Method

1. Melt the 2% agarose by heating to 100°C and then place the bottle in a 52°C water bath. Stand a 5 mL pipet in this solution and allow time for the agarose to cool to 52°C. At the same time the pipet will be warming.
2. Place a test-tube in the water bath and add barbitone buffer (0.06M, 2.8 mL). Leave this for about 3 min to

allow the buffer to warm up and then add agarose solution (2.8 mL). This transfer should be made as quickly as possible to avoid the agarose setting in the pipet. Using a pipet with the end cut off to give a larger orifice allows a more rapid transfer of the viscous agarose solution. Some setting of agarose in the pipet is inevitable, but is of no consequence. Briefly mix the contents of the tube and return to the water bath.

3. Thoroughly clean a 5×5 cm glass plate with methylated spirit. When dry, put the glass plate on a leveling plate or level surface.
4. Add antiserum (50 μ L, but see Note 3) to the diluted agarose solution and *briefly* mix to ensure even dispersion of the antiserum. Briefly return to the water bath to allow any bubbles to settle out.
5. Immediately pour the contents of the tube onto the glass plate. Keep the neck of the tube close to the center of the plate and pour slowly. Surface tension will prevent the liquid running off the edge of the plate. Alternatively, tape can be used to form an edging to the plate. The final gel will be approximately 2 mm thick.
6. Allow the gel to set for 5 min and then make holes (1 mm diameter) at 0.5 cm spacings, 1 cm from one edge of the plate. This is most easily done by placing the gel plate over a predrawn (dark ink) template when well positions can easily be seen through the gel. The wells can be made using a Pasteur pipet or a piece of metal tubing attached to a weak vacuum source (e.g., a water pump).
7. Place the gel plate on the cooling plate of an electrophoresis tank and place the electrode wicks (six layers of Whatman No. 1 paper, prewetted in electrode buffer) over the edges of the gel. Take care not to overlap the sample wells and ensure that these wells are nearest the cathode.
8. Each well is now filled with 2 μ L of sample. Samples should be diluted accordingly with buffer so that 2 μ L samples can be loaded. It is important to completely fill the wells, and also to ensure that all samples are

loaded in the same volume. The loading of wells should be carried out as quickly as possible to minimize diffusion from the wells. Some workers prefer to load wells with a current (1–2 mA) passing through the gel, which should overcome any diffusion problems.

9. Once the samples are loaded, electrophoresis is commenced. For fast runs, a current of 20 mA (~ 200 V, 10 V/cm across the plate) is passed through the gel for about 3–6 h. Alternatively, gels can be run at 2–3 mA overnight. It is important that water cooling be used to dissipate heat generated during electrophoresis.
10. At the end of the run, precipitation rockets can be seen in the gel. These are not always easy to visualize and are best observed using oblique illumination of the gel over a dark background.
11. A more clear result can be obtained by staining these precipitation peaks with a protein stain. Firstly, the antiserum in the gel, which would otherwise stain strongly for protein, must be removed. One way of doing this is to wash the gel with numerous changes of saline (0.14M) over a period of 2–3 d. The washed gel may then be stained directly or dried and then stained to give a permanent record. To dry the washed gel, place it on a clean sheet of glass, wrap a piece of wet filter paper around the gel, and place it in a stream of warm air for about 1 h. Wet the paper again with distilled water and remove it to reveal the dried gel fixed to the glass plate. Any drops of water on the gel can be finally removed in a stream of warm air.
12. A more convenient and quicker method of preparation is to blot the gel dry. With the gel on a sheet of clean glass, place eight sheets of filter paper (Whatman No. 1) on top of the gel and then apply a heavy weight (lead brick) for 1–2 h. After this time, carefully remove the now wet filter papers to reveal a flattened and nearly dry gel. Complete the drying process by heating the gel in a stream of hot air for about 1 min. When completely dry, the gel will appear as a glassy film.

13. Whichever method of gel preparation is used, the gel can now be stained by placing it in stain (with shaking) for 10 min, followed by washing in destain.
14. Peak heights can now be measured and a graph of concentration against peak height plotted. A typical electrophoretic run is shown in Fig. 1.

Notes

1. The size of gel described here (5×5 cm) is suitable for routine use. However, for highly accurate determina-

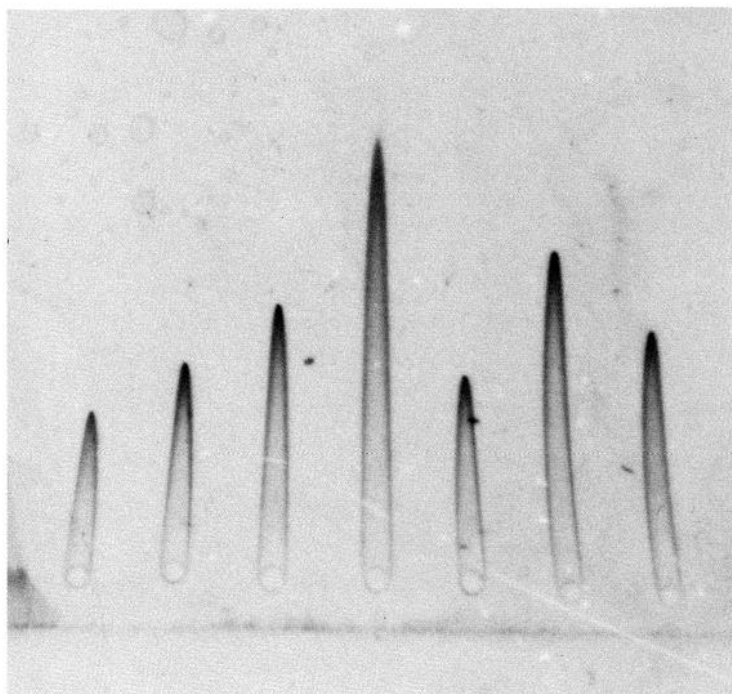


Fig. 1. A rocket immunoelectrophoresis gel (5×5 cm) run at 20 mA for 3h, and stained for protein with Coomassie Brilliant Blue. The gel contained anti-bovine serum albumin ($50 \mu\text{L}$, anti-BSA) and the sample loading ($2 \mu\text{L}/\text{well}$) were (left to right): 1. 40 ng BSA; 2. 67 ng BSA; 3. 100 ng BSA; 4. 200 ng BSA; 5. a 1:1200 dilution of bovine serum; 6. 130 ng BSA; 7. 100 ng BSA.

tions larger plates should be used (7.5×7.5 cm) with larger wells (5–10 μ L volumes) so that sample loadings can be made with greater accuracy, and the correspondingly larger peaks measured more accurately. The volume of agarose and antisera used should of course be increased accordingly.

2. When possible, the gel plates used should be as thin as possible to maximize the effect of the cooling plate. This is particularly important in the case of the shorter run time when considerable heat is evolved, which can cause distortion of rockets.
3. The amount of antiserum to be used in the gel depends of course on the antibody titer, and is best determined by trial and error. The amount quoted in the Method section (50 μ L) is the volume of commercially available anti-bovine serum albumin used in our laboratory when measuring bovine serum albumin levels (see Fig. 1). This is an ideal test system for someone setting up the technique for the first time.
4. Because of the thickness of the electrophoresis wicks used, they are quite heavy when wet and have a tendency to 'slip off' the gel during the course of the run. This can be prevented by placing thick glass blocks on top of the wicks where they join the gel, or by placing a heavy glass sheet across both wicks. This, by also covering the gel, has the added advantage of reducing evaporation from the gel.
5. A uniform field strength over the entire gel is critical in rocket immunoelectrophoresis. For this reason the wick should be exactly the width of the gel.
6. When setting up the electrophoresis wicks, it is important that they be kept well away from the electrodes. Since the buffer used is of low ionic strength, if electrode products are allowed to diffuse into the gel, they can ruin the gel run. Most commercially available apparatus is therefore designed such that the platinum electrodes are masked from the wicks and also separated by a baffle. To minimize these effects, the use of large tank buffer volumes (~ 800 mL per reservoir) is also encouraged.
7. The time quoted for electrophoresis is suitable for most samples. However, for proteins with low electro-

phoretic mobility under the conditions used, these times may not be long enough to allow complete development of rockets. The minimum time necessary for electrophoresis should be determined in initial trial experiments. Do note that it is not possible to 'overrun' the electrophoresis. Once the rockets are formed they are quite stable to further electrophoresis.

8. Rocket immuno-electrophoresis is accurate for samples with protein concentrations as low as 10 $\mu\text{g/mL}$ and require as little as 20 ng of protein to be loaded onto a gel.

References

- 1 Laurell, C-B (1966) Quantitative estimation of proteins by electrophoresis in agarose gel containing antibodies *Anal Biochem* **15**, 45-52

Chapter 36

Radioiodination of Proteins

G. S. Bailey

*Department of Chemistry, University of Essex,
Colchester, Essex, England*

Introduction

Many different substances can be labeled by radioiodination. Such labeled molecules are of major importance in a variety of investigations, e g., studies of intermediary metabolism, determinations of agonist and antagonist binding to receptors, quantitative measurements of physiologically active molecules in tissues and biological fluids, and so on. In most of those studies, it is necessary to measure very low concentrations of the particular substance and that in turn implies that it is essential to produce a radioactively labeled tracer molecule of high specific radioactivity. Such tracers, particularly in the case of polypeptides and proteins, can often be conveniently produced by radioiodination.

Two γ -emitting radioisotopes of iodine are widely available, ^{125}I and ^{131}I . As γ -emitters they can be counted directly in a well-type crystal scintillation counter (com-

monly referred to as a γ counter) without the need for sample preparation in direct contrast to β -emitting radionuclides, such as ^3H and ^{14}C . Furthermore, the count rate produced by 1 g atom of ^{125}I is approximately 75 times and 35,000 times greater than that produced by 1 g-at of ^3H and ^{14}C , respectively. In theory, the use of ^{131}I would result in a further sevenfold increase in specific radioactivity. However, the isotopic abundance of commercially available ^{131}I rarely exceeds 20%, because of ^{127}I and ^{129}I contaminants, and its half-life is only 8 d. In contrast, the isotopic abundance of ^{125}I on receipt in the laboratory is normally at least 90% and its half-life is 60 d. Also, the counting efficiency of a typical well-type crystal scintillation counter for ^{125}I is approximately twice that for ^{131}I . Thus, in most circumstances, ^{125}I is the radionuclide of choice for radioiodination.

Several different methods of radioiodination of proteins have been developed (1). They differ, among other respects, in the nature of the oxidizing agent for converting $^{125}\text{I}^-$ into the reactive species $^{125}\text{I}_2$ or $^{125}\text{I}^+$. In the main, those reactive species substitute into tyrosine residues of the protein, but substitution into other residues, such as histidine, cysteine, and tryptophan, can occur in certain circumstances.

In this chapter two methods of radioiodination of proteins will be described.

The Chloramine-T Method

This method, developed by Hunter and Greenwood (2), is probably the mostly widely used of all techniques of protein radioiodination. It is a very simple method in which the radioactive iodide is oxidized by chloramine-T in aqueous solution. The oxidation is stopped after a brief period of time by addition of excess reductant. Unfortunately, some proteins are denatured under the relatively strong oxidizing conditions, and so other methods of radioiodination that employ more gentle conditions have been devised, e.g., the lactoperoxidase method.

The Lactoperoxidase Method

This method, introduced by Marchalonis (3), employs lactoperoxidase in the presence of a trace of hydrogen peroxide to oxidize the radioactive iodide. The oxidation can be stopped by simple dilution. Although the technique should result in less chance of denaturation of susceptible proteins than the chloramine-T method, it is more technically demanding and is subject to a more marked variation in optimum reaction conditions.

Materials

The Chloramine-T Method

1. Na^{125}I : 1 mCi, concentration 100 mCi/mL.
2. Buffer A: 0.5M sodium phosphate buffer, pH 7.4.
3. Buffer B: 0.05M sodium phosphate buffer, pH 7.4.
4. Buffer C: 0.01M sodium phosphate buffer containing 1M sodium chloride, 0.1% bovine serum albumin, and 1% potassium iodide, final pH 7.4.
5. Chloramine-T solution: A 2 mg/mL solution in buffer B is made just prior to use.
6. Reductant: A 1 mg/mL solution of sodium metabisulfite in buffer C is made just prior to use.
7. Protein to be iodinate: A 0.5–2.5 mg/mL solution is made in buffer B

The Lactoperoxidase Method

1. Na^{125}I : 37 MBq (1 mCi) concentration 3.7 GBq/mL (100 mCi/mL)
2. Lactoperoxidase: Available from various commercial sources.
A stock solution of 10 mg/mL in 0.1M sodium acetate buffer, pH 5.6, can be made and stored at -20°C in small aliquots. A working solution of 20 $\mu\text{g/mL}$ is made by dilution in buffer just prior to use.
3. Buffer A: 0.4M sodium acetate buffer, pH 5.6.
4. Buffer B: 0.05M sodium phosphate buffer, pH 7.4.

5. Hydrogen peroxide. A solution of 10 $\mu\text{g/mL}$ is made by dilution just prior to use
6. Protein to be iodinated: A 0.5–2.5 mg/mL solution is made in buffer B.

It is essential that none of the solutions contain sodium azide as an antibacterial agent since it inhibits lactoperoxidase.

Method

The Chloramine-T Method

1. Into a small plastic test tube (1×5.5 cm) are added successively the protein to be iodinated (10 μg), radioactive iodide (5 μL), buffer A (50 μL), and chloramine-T solution (25 μL)
2. After mixing by gentle shaking, the solution is allowed to stand for 30 s to allow radioiodination to take place.
3. Sodium metabisulfite solution (500 μL) is added to stop the radioiodination and the resultant solution is mixed. It is then ready for purification.

The Lactoperoxidase Method

1. Into a small plastic test tube (1×5.5 cm) are added in turn the protein to be iodinated (5 μg), radioactive iodide (5 μL), lactoperoxidase solution (5 μL), and buffer a (45 μL).
2. The reaction is started by the addition of the hydrogen peroxide solution (10 μL) with mixing
3. The reaction is stopped after 20 min by the addition of buffer B (0.9 mL) with mixing. The resultant solution is then ready for purification

Purification of Radioiodinated Protein

At the end of the radioiodination the reaction mixture will contain the labeled protein, unlabeled protein, radioiodide, mineral salts, enzyme (in the case of the

lactoperoxidase method), and possibly some protein that has been damaged during the oxidation. For most uses of radioiodinated proteins, it is essential to have the labeled species as pure as possible with the constraints, however, that the purification is achieved as rapidly as possible. For that purpose the most widely used of all separation techniques is gel filtration. Various types of Sephadex resin can be employed, e.g., G-50, G-75, and G-100 depending on the differences in sizes of the molecules present in the mixture.

Typically the mixture is applied to a column (1×25 cm) of Sephadex resin and is eluted with 0.05M sodium phosphate buffer of pH 7.4 containing 0.15M sodium chloride and 0.1% bovine serum albumin. Fractions (0.5–1.0 mL) are collected in plastic tubes and aliquots (10 μ L) are counted. Using those results, an elution profile such as that shown in Fig. 1 is drawn.

Notes

- 1 Several parameters can be used to assess the quality of the labeled protein. The *specific radioactivity* of the protein is the amount of radioactivity incorporated per unit mass of protein. It can be calculated in terms of the total radioactivity employed, the amount of the iodination mixture transferred to the gel filtration column, and the amount of radioactivity present in the labeled protein, in the damaged components, and in the residual radioiodine. However, in practice, the calculation does not usually take into account damaged and undamaged protein. The specific activity is thus calculated from the yield of the radioiodination procedure, the amount of radioiodide and the amount of protein used, assuming that there are not significant losses of those two reactants. The *yield of the reaction* is simply the percentage incorporation of the radionuclide into the protein.

For example, consider the results shown in Fig. 1. In terms of elution volumes, it is to be expected that the first peak of radioactivity represents the labeled protein and that the second peak represents

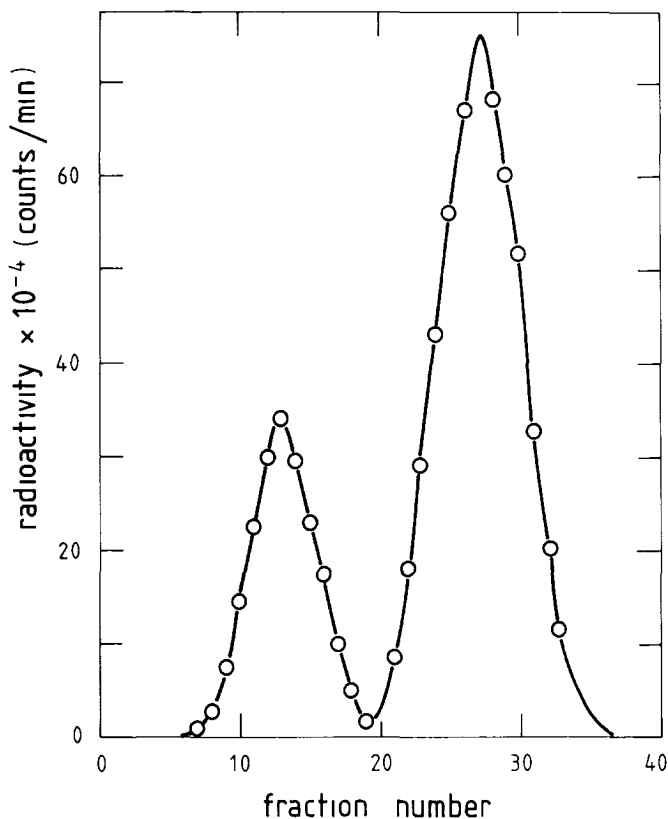


Fig. 1. Gel filtration of radioiodinated kallikrein of rat submandibular gland. The pure enzyme (10 μ g) was iodinated with ^{125}I (18.5 MBq) by the chloramine-T method. It was then purified on a column (1 $\times$ 20 cm) of Sephadex G-75 resin at a flow rate of 20/mL/h and collecting fractions of 0.6 mL. Aliquots (10 μ L) of each fraction were measured for radioactivity. By radioimmunoassay, immunoreactive protein was found only in the first peak and more than 90% of that radioactivity was bound by the antiserum to kallikrein from rat submandibular gland.

unreacted radioiodide. For this case, and, more importantly, for more complicated elution patterns, the nature of the materials giving rise to those peaks can be checked by employing a specific antiserum to the protein being radioiodinated. Aliquots (10 μ L) of different fractions making up the two peaks are diluted

so that each gives the same number of counts (e.g., 5000–10,000 counts/min) per 100 μL . Those samples are incubated with an excess of the antiserum. Only samples containing immunoreactive protein will react with the antiserum. The amount of radioactive protein associated with the antibody molecules can then be measured by radioimmunoassay (*see* Chapter 37).

Having identified the peak or peaks containing the radioiodinated protein, the yield of the radioiodination can be calculated in terms of the ratio of the total counts associated with the protein peak to the sum of the total counts associated with the protein peak and total counts associated with the iodide peak.

It is obviously important that the radioiodinated protein should as far as possible have the same properties as the unlabeled species. Thus the behavior of both molecules can be checked on electrophoresis or ion-exchange chromatography. The ability of the two species to bind to specific antibodies can be assessed by radioimmunoassay.

2. To store the labelled protein, immediately after purification split the sample into small aliquots and then rapidly freeze and store at -20°C . Alternatively the aliquots can be freeze-dried. Each aliquot should be melted and used only once. Radioiodinated proteins differ markedly in their stability. Some can be stored for several weeks (though it must be borne in mind that the half-life of ^{125}I is about 60 d), whereas others can only be kept for several days. If necessary, the labeled protein can be repurified by gel filtration or ion-exchange chromatography prior to use.
3. The pH optimum for iodination of tyrosine residues of a protein by the chloramine-T method is about pH 7.4. Lower yields of iodinated protein are obtained at pH values below about 6.5 and above about 8.5. Indeed, above pH 8.5, the iodination of histidine residues appears to be favored.
4. The total volume of the chloramine-T reaction mix should be as low as practically possible to achieve a rapid and efficient incorporation of the radioactive iodine into the protein. Because of the small volumes of reactants that are employed it is essential to ensure

adequate mixing at the outset of the reaction. Inadequate mixing is one of the commonest reasons for a poor yield of radioiodinated protein by this procedure.

5. If the protein has been seriously damaged by the use of 50 μg of chloramine-T, it may be worthwhile repeating the radioiodination using much less oxidant (10 μg or less). Obviously the minimum amount of chloramine-T that can be used will depend, among other factors, on the nature and amount of protein to be iodinated.
6. It is normal to carry out the chloramine-T method at room temperature. However, if the protein is especially labile, it may be beneficial to run the procedure at a low temperature.
7. For the lactoperoxidase method, the exact nature of buffer A will depend on the properties of the protein to be radioiodinated. Proteins differ markedly in their pH optima for radioiodination by this method (4). Obviously the pH to be used will also depend on the stability of the protein and the optimum pH can be established by trial and error.
8. Other reaction conditions such as amount of lactoperoxidase, amount and frequency of addition of hydrogen peroxide, and so on also markedly affect the yield and quality of the radioiodinated protein. Optimum conditions can be found by trial and error.
9. The longer the time of the incubation the greater is the risk of potential damage to the protein by the radioactive iodide. Thus it is best to keep the time of exposure of the protein to the radioactive iodide as short as possible, but commensurate with a good yield of radioactive product.
10. Some of the lactoperoxidase itself may become radioiodinated, which may result in difficulties in purification if the enzyme is of a similar size to the protein being labeled. Thus it is best to keep the ratio of the amount of protein being labeled to the amount of lactoperoxidase used as high as possible.
11. Some of the problems may be overcome by the use of solid-phase lactoperoxidase systems. Such a system is commercially available in which immobilized glucose

oxidase is used to generate a small, steady stream of hydrogen peroxide from added glucose. The hydrogen peroxide is utilized by immobilized lactoperoxidase to oxidize the radioactive iodide.

References

- 1 Bolton, A E (1977) "Radioiodination Techniques" Amersham International, Amersham, Bucks, England
- 2 Hunter, W M., and Greenwood, F C. (1962) Preparation of iodine-131 labeled human growth hormone of high specific activity *Nature* **194**, 495-496.
- 3 Marchalonis, J J (1969) An enzymic method for trace iodination of immunoglobulins and other proteins *Biochem J* **113**, 299-305
- 4 Morrison, M., and Bayse, G S (1970) Catalysis of iodination by lactoperoxidase *Biochemistry* **9**, 2995-3000

Chapter 37

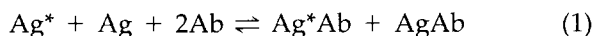
Radioimmunoassay

G. S. Bailey

*Department of Chemistry, University of Essex,
Colchester, Essex, England*

Introduction

Radioimmunoassay is often described in terms of the competition between a radiolabeled antigen (Ag^*) and its unlabeled counterpart (Ag) for binding to a limited amount of specific antibody (Ab) (1). In most radioimmunoassays the reaction is allowed to proceed to equilibrium and thus, can be represented by Eq. (1)



The concentration of the antibody is limited such that the labeled antigen, although present in a trace amount, is in relative excess over the antibody. Thus, even in the absence of unlabeled antigen, only some of the radioactive antigen will be associated with the antigen-antibody complex while the remainder will be free in solution. In the radioimmunoassay, the total amounts of antibody and radiolabeled antigen are kept constant. The presence of unlabeled antigen will result in less of the labeled species

being able to bind to the antibody. The greater the amount of unlabeled antigen (Ag) present, the lower will be the amount of radiolabeled antigen combined to the antibody (Ag*Ab). Thus, on suitable calibration, the amount of the unlabeled species can be accurately measured in terms of the amount of radioactivity associated with the antigen-antibody complex.

The widespread use of radioimmunoassays results from several significant advantages that the method has compared to many other quantitative assays, particularly for substances that are otherwise measured by pharmacological assays. Those advantages include very high sensitivity and high specificity. A wide variety of exogenous and endogenous substances can be measured in biological tissues and fluids at concentrations of pg/mL by the use of suitable radioimmunoassays. Such high sensitivity requires the use of antisera of high avidity and the use as tracers of antigens labeled to high specific radioactivities. The production of antisera and the radioiodination of proteins are dealt with in other chapters.

1. Incubation of Antigens and Antiserum. The incubation conditions have to ensure the stability of all reagents as antigen binds to antibody and allow equilibrium to be reached.
2. Separation of Free and Bound Antigen. When equilibrium has been achieved, the antigen bound to antibody is quickly and efficiently separated from free antigen so that the radioactivity associated with either or both components can be counted. Many different separation procedures have been reported (2). This chapter will deal with two which are widely used.

Fractional Precipitation with Polyethyleneglycol

The basis of the method is the ability of relatively low concentrations of polyethyleneglycol to bring about the precipitation of antibody molecules, presumably by removal of the attendant hydration shell of water molecules, without the precipitation of the smaller antigen molecules (3). The method is not efficient in all cases, but is worth trying because of its simplicity and very low cost.

Double (Second) Antibody Method

This procedure is very widely used and can achieve an efficient separation of free and bound antigen in more or less all radioimmunoassays. The basis of the most common type of this method is to use an antiserum (the second antibody), raised to antibodies of the antiserum (first antibody) employed in the incubation, to precipitate the antigen-antibody complex (4). The addition of non-immune γ -globulin of the species in which the first antiserum was raised increases the bulk of material that can interact with the second antiserum and so enables a precipitate to be formed.

Materials

Incubation of Antigens and Antiserum

1. Antiserum. The antiserum to be used in the assay should be of high avidity, high titer, and high specificity for the antigen to be measured.
2. Radiolabeled Antigen. Pure antigen must be used for labeling. Purification of the labeled species must also be carried out. Radioiodination is the method of choice for labeling proteins (*see* Chapter 36) to high specific radioactivity.
3. Unlabeled Antigen. Pure antigen is used as the standard in the assay. It is essential that the standard and the antigen to be measured show identical behavior towards the antiserum.
4. Buffer. Many different buffers can be used for radioimmunoassay. Whatever buffer is chosen must ensure the stability of all of the reagents. In practice, in order to achieve equilibrium in a reasonable time, as well as maintain stability, the buffer employed has a pH within the range pH 6–8.6 and a molarity within the range 0.01–0.1M. A preservative such as sodium azide or sodium ethyl mercurithiosalicylate (0.01–0.1%) has to be included. Proteins such as bovine serum albumin

or ovalbumin (0.1–1.0%) are normally added as a carrier.

5. Disposable Plastic Tubes Tubes of various shapes and sizes can be used, depending on the volume of the incubation mixture and the separation procedure employed.

Separation of Free and Bound Antigen

Fractional Precipitation Using Polyethyleneglycol

- 1 Polyethyleneglycol of molecular weight 6000.
- 2 Bovine γ -globulin (Cohn Fraction II)

Double (Second) Antibody Method

1. Antiserum to γ -globulins of the species in which the first antiserum (i.e., that used in the incubation) was raised. Such antisera are available from several commercial sources.
2. Non-immune γ -globulin (Cohn Fraction II) or serum of the species in which the first antiserum was raised.

Method

Incubation of Antigens and Antiserum

1. At the outset, the amount of labeled antigen to be employed in the incubation is chosen to be of the same order of magnitude as the smallest amount of unlabeled antigen that is required to be measured. In practice the amount of tracer used is often simply that which gives a specified number of counts per min per unit volume (usually 5000–10,000 counts/min/100 μ L).
2. The optimal dilution (titer) of the antiserum to be used in the assay is that which will bind 30–50% of the labeled antigen. That dilution is chosen with the aid of an antiserum dilution curve. The curve is constructed from the results of incubations, carried out at least in

duplicate, of serial dilutions of the antiserum with the chosen amount of tracer. The incubations, for example, consisting of diluted antiserum (100 μL), tracer (100 μL) and buffer (200 μL), are allowed to proceed to equilibrium. The time required to reach equilibrium obviously depends on the particular circumstances of the assay. For a completely new radioimmunoassay, it may be necessary to run pilot experiments varying temperature from 4 to 37°C and time of incubation from 4 to 72 h to establish the optimal conditions. The bound and free forms of the radioactive labeled tracer are then separated and counted. A typical antiserum dilution curve is shown in Fig. 1. In that particular case, the dilution of antiserum to be used in the assay was chosen as 1:60,000 (i.e., final dilution 1:240,000), corresponding to 37% binding of the tracer. The slope of the dilution curve in its descending portion can be

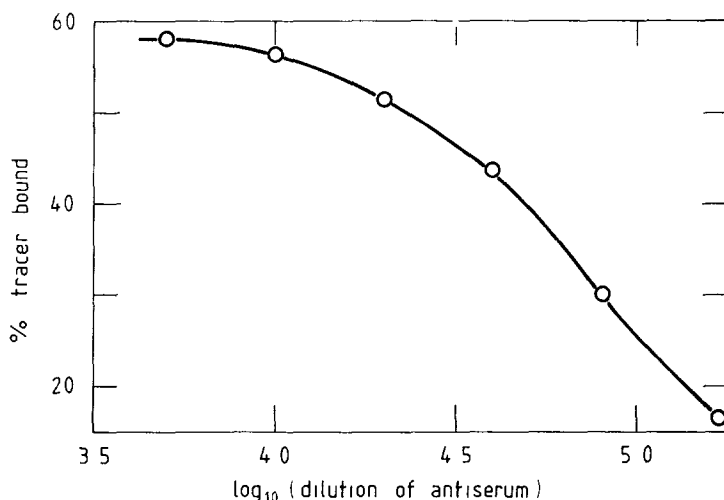


Fig. 1 Antiserum Dilution Curve Radioiodinated kallikrein (100 μL , 7000 counts/min) was incubated with serial dilutions (1:5000–1:160,000) of antiserum (100 μL) to kallikrein from rat submandibular gland in a total volume of 400 μL for 20 h at 25°C. The bound antigen was separated from the free antigen by incubation at 37°C for 2 h with a solid-phase second antibody. The bound radioactivity was measured and was expressed as a percentage of the total radioactivity that was used.

used as a measure of the avidity of the antiserum (the greater the slope, the greater is the avidity).

3. Having chosen the amount of tracer and dilution of antiserum to be used, the radioimmunoassay can then be set up to measure the amount of antigen in unknown solutions with the aid of standard solutions of antigen
4. Standard solutions of antigen are made by dilution of a master solution of accurately known concentration. The working solutions are made just prior to use and are kept at 4°C until required.
5. The tubes to be used in the incubation are numbered.
6. Buffer is added to all of the tubes apart from the first three tubes, which will be used to measure the total counts in the assay. To each of those three tubes that will be used to measure nonspecific binding is added 300 μL of buffer. To each of the three tubes that will represent the zero standard is added 200 μL of buffer, while 100 μL of buffer is added to every other tube. The standard solutions (100 μL) of antigen or unknown solutions are then added to the relevant numbered tubes in duplicate. Next the antiserum (100 μL) at the chosen dilution is added to all tubes except for the three tubes to be used to measure total counts and the three tubes to be used to measure nonspecific binding. Finally the radioactive tracer (100 μL) is added to every tube. The contents of each tube are thoroughly mixed with the aid of a vortex mixer. Each tube is then left at a constant temperature until equilibrium is reached (often 24–48 h at 4°C).
7. When equilibrium has been reached the free and bound antigen are separated.
8. Although it is possible for either or both phases to be counted, it is normal for the radioactivity associated with the antibody to be measured. There should be good agreement between the counts of each tube in a particular pair or set of three.
9. The counts associated with the three tubes representing nonspecific binding are averaged and in turn are subtracted from the average counts for each set of tubes.

10. The resultant, average specific counts associated with the three zero-standard tubes can be expressed as a percentage of the average specific counts of the three tubes containing just tracer. If the conditions of the assay have been satisfactory, the counts of the zero standard should be 30–50% of the total counts of the tracer
11. If so, the average specific counts of each set of duplicates can then be expressed as a percentage of the average specific counts of the three zero standards.
12. A standard curve is constructed from the calculated results.

Separation of Free and Bound Antigen

Fractional Precipitation Using Polyethylene Glycol

1. A solution (25–30%) of polyethylene glycol 6000 is prepared in 0.05M sodium phosphate buffer, pH 7.4, with thorough mixing. A separate solution (1%) of bovine γ -globulin is made in the same buffer.
2. To each of the tubes requiring separation of bound and free antigen is added at 4°C γ -globulin solution (400 μ L) and polyethylene glycol solution (800 μ L). Each tube is vortexed and is allowed to remain at 4°C for 15 min.
3. Each tube is then centrifuged at 4°C at 5000g for 30 min.
4. The supernatants are carefully removed by aspiration at a water pump. The precipitate at the bottom of each tube is then counted. If necessary, the precipitates can be washed at 4°C with 15% polyethylene glycol solution (800 μ L) followed by vortexing and centrifugation.

Double (Second) Antibody Method

1. Firstly, the optimal dilution of the second antiserum has to be carefully assessed. Serial dilutions of the second antiserum are made in 0.05M sodium phosphate buffer, pH 7.4, containing 0.5% non-immune serum and 50 mM EDTA.

2. Samples containing tracer (100 μL), first antiserum at optimal dilution (100 μL), and buffer (200 μL) are allowed to come to equilibrium at constant temperature
3. Aliquots (100 μL) of the different dilutions of the second antiserum are added. The tubes are vortexed and are kept at constant temperature for a further 18–24 h. The tubes are then centrifuged at 5000g for 30 min. The supernatants are aspirated at a water pump and the tubes are counted.
4. The highest dilution of the second antiserum that results in the precipitation of a maximum percentage of the tracer represents the minimum amount of second antiserum that should be employed in the separation
5. Aliquots (100 μL) of the chosen dilution of the second antiserum are added to each of the sample tubes requiring separation of free and bound antigen. The tubes are then treated as in point 3 above

Notes

1. The purpose of a radioimmunoassay is to measure the concentration of a particular antigen in an unknown sample by comparison with standard solutions of the antigen. From the measurements of the bound or free radioactivity in the presence of various known amounts of antigen is constructed a standard curve. The amount of antigen in the test sample is then found from that curve by simple interpolation. The standard curve can be represented in several different ways. One straightforward method is to plot the proportion of tracer bound expressed as a percentage of that in the zero standard against the corresponding concentration of standard antigen (*see* Fig. 2). However, to avoid the problems associated with the subjectivity of drawing a curve, it is better to construct a linear plot. One such linear transformation that is very widely used is shown in Fig. 3, where the logit of the tracer bound (y) is plotted against log concentration.

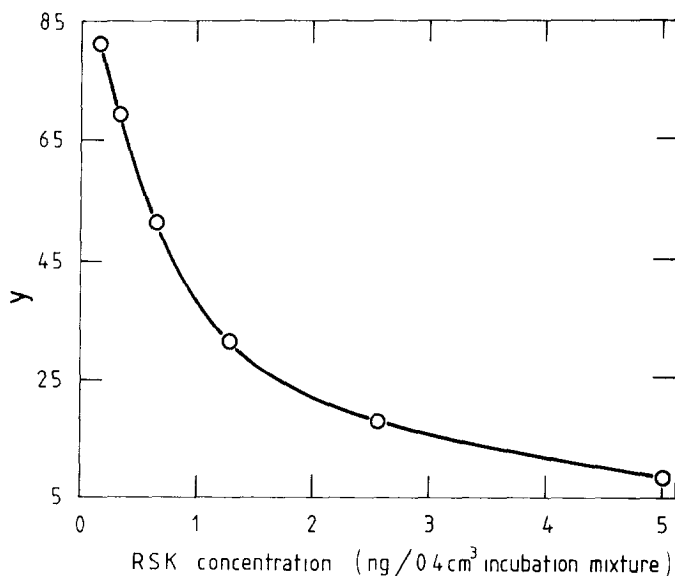


Fig 2 Standard Curve for Radioimmunoassay. Radioiodinated kallikrein (100 μ L, 7000 counts/min) was incubated with standard solutions of kallikrein from rat submandibular gland (200 μ L containing 0.16–5.12 ng kallikrein) and antiserum to the enzyme (100 μ L of 1:60,000 dilution) for 20 h at 25°C. The bound radioactivity was separated from the free radioactivity by use of a solid-phase second antibody on incubation from 2 h at 37°C and was counted

$$y = \frac{B - N}{B_0 - N} \times 100$$

where

B = counts associated with standard solution

B_0 = counts associated with zero standard

N = counts associated with nonspecific binding

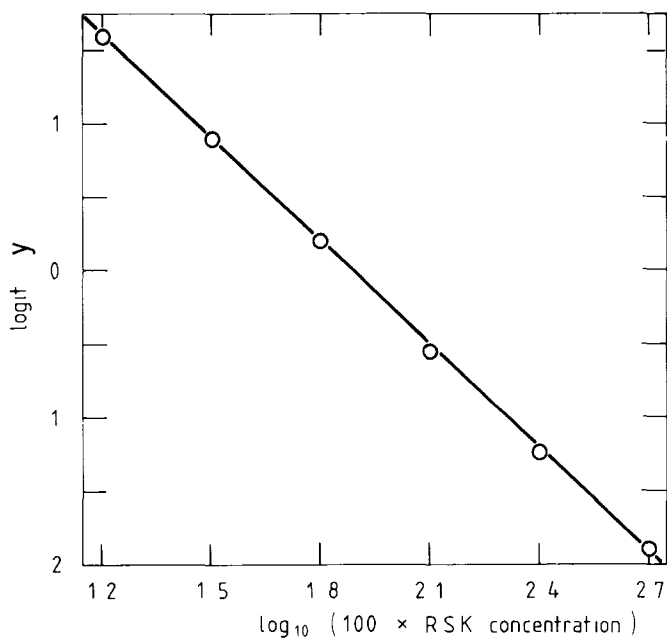


Fig. 3 Logit-log plot of the standard curve. For details see legend to Fig. 2.

$$\text{logit } y = \log_e \left(\frac{y}{100 - y} \right)$$

and

$$y = \frac{B - N}{B_o - N} \times 100$$

where B = counts associated with a certain concentration of antigen

B_o = counts associated with the zero standard

N = counts associated with nonspecific binding

Often the logit plot becomes nonlinear near its extremes and in that case the points in those regions are not employed in construction of the straight line

- 2 There are a very large number of variations to the procedure outlined. The best set of conditions for a particular case can be worked out by trial and error. Of particular importance is the volume of the unlabeled

antigen in the incubation mixture. In theory, an increase in that volume relative to the other components should lead to an increase in sensitivity of the assay, i.e., detection of a lower amount of antigen. In practice, there may actually be a decrease in sensitivity because of interference in the antigen-antibody binding by other substances present in the sample.

- 3 Other factors that often increase the sensitivity of the assay include
 - (a) 'Late' addition of the tracer. The labeled antigen is added a considerable time after the unlabeled species and antiserum have been allowed to interact, but before the attainment of equilibrium.
 - (b) Decrease in the amount of antiserum. However, a limit will be reached beyond which a further reduction in antiserum will be counter-productive because of a loss of precision in the assay.
 - (c) Decrease in the amount of tracer. Again, there is a limit beyond which a further reduction will produce no significant change in sensitivity.

It should be noted, however, that very high sensitivity may not be required of a radioimmunoassay. What is far more important is that the assay should be able to accurately, precisely and reproducibly measure antigen in the range of concentrations found in biological tissues and fluids, preferably without the need for sample dilutions. However, the following note should be borne in mind.

4. One major problem of radioimmunoassay, particularly when applied to undiluted biological samples, is that substances in the samples interfere with antigen-antibody interaction. High concentrations of salts and plasma proteins decrease antigen-antibody binding. Thus it may be necessary to dilute the samples to reduce such effects or to include such substances in the standard antigen solutions.
5. Another problem may be encountered if the biological samples contain proteolytic enzymes that will degrade the antigen. In those cases, it is necessary to include a broad-spectrum inhibitor in the assay, e.g., aprotinin.

6. Precipitation by polyethylene glycol is very sensitive to fluctuations in temperature. Hence it is essential to keep all solutions and tubes at 4°C during the whole procedure.
7. One problem that is often associated with the method of fractionation using polyethylene glycol is the occurrence of significant nonspecific binding, i.e., high blanks. Washing of the precipitate with polyethylene glycol solution may reduce the nonspecific binding.
8. When carrying out the double (second) antibody method, normal serum or non-immune γ -globulin from the species in which the first antiserum was raised can be included in the incubation mixture rather than be added with the second antiserum.

However, the presence of components of the complement system in serum samples can affect the formation of the immune precipitate. Thus it is usual to include EDTA (0.05–0.1M) in the incubation and separation buffers to inactivate those components and avoid any such difficulties.

9. If the concentrations of reactants are low, it can take a considerable time for complete immunoprecipitation to occur. The use of higher amounts of γ -globulin or serum will greatly speed up the process, but of course higher amounts of the second antiserum will also be needed.
10. Many of the difficulties associated with the double antibody method can be avoided through the use of a second antibody that is covalently linked to a suitable, inert, solid matrix. Because of the solid-phase nature of such a second antibody, immune precipitation occurs relatively quickly and a carrier non-immune γ -globulin is not required. Of course, considerable effort, second antiserum, and a suitable solid support are necessary to produce the solid-phase second antiserum (5). Certain preparations are commercially available, but are expensive, e.g., antibodies raised in sheep against rabbit γ -globulins, coupled to cellulose, antibodies raised in goat against rabbit γ -globulins and coupled to polyacrylamide.
11. One particularly useful modification of the double antibody method is to include a small amount of ammo-

num sulfate or polyethylene glycol with the second antiserum (6). The inclusion of the chemical precipitant speeds up the immune precipitation and enables a lower amount of second antiserum to be used.

References

- 1 Felber, J P (1975) Radioimmunoassay of Polypeptide Hormones and Enzymes in *Methods of Biochemical Analysis* 22 (ed. Glick, D) pp 1-94 Wiley, New York
- 2 Chard, T (1978) Requirements for a binding assay—separation of bound and free ligand, in *Laboratory Techniques in Biochemistry and Molecular Biology* 6 (eds , Work, T S , and Work, E) pp 401-426. Elsevier/North Holland Biomedical Press, Amsterdam
- 3 Desbuquois, B , and Aurbach, G D (1971) Use of Polyethylene Glycol to Separate Free and Antibody-Bound Peptide Hormones in Radioimmunoassay *J Clin Endocrinol Metab* 33, 732-738
- 4 Midgley, A R , and Hepburn, M R (1980) Use of the double-antibody method to separate antibody bound from free ligand in radioimmunoassay in *Methods in Enzymology* 70 (eds Van Vunakis, H , and Langone, J J) pp 266-274 Academic Press, New York.
- 5 Koninckx, P , Bouillon, R , and De Moor, P (1976) Second antibody chemically linked to cellulose for the separation of bound and free hormone An improvement over soluble second antibody in gonadotrophin radioimmunoassay *Acta Endocrinol* 81, 43-53.
6. Peterson, M A , and Swerdloff, R. S (1979) Separation of bound from free hormone in radioimmunoassay of lutropin and follitropin. *Clin Chem* 25, 1239-1241

Chapter 38

Enzyme-Linked Immunosorbant Assay (ELISA)

Wim Gaastra

*Department of Microbiology, The Technical
University of Denmark, Lyngby,
Denmark*

Introduction

In general, immunological methods are not very well suited for a quantitative determination of the antigen to be studied. The ELISA technique, however, can be used for a quantitative or at least semiquantitative determination of the concentration of a certain antigen. The method was first introduced by Engvall and Perlmann (1). The principle of ELISA (see Fig. 1), also called the double antibody sandwich technique, is the following: Antibodies against the antigen to be measured are adsorbed to a solid support, in most cases a polystyrene microtiter plate. After coating the support with antibody and washing, the antigen is added and will bind to the adsorbed antibodies

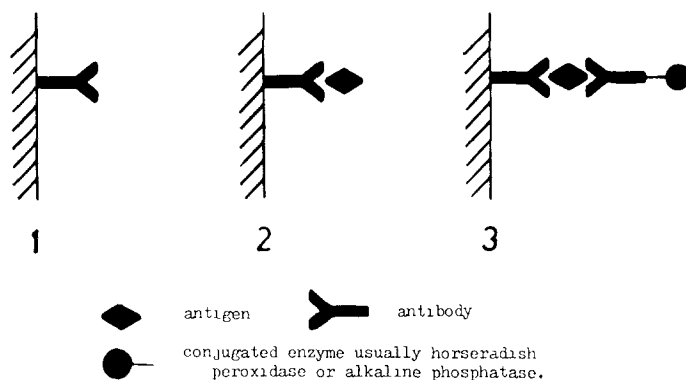


Fig. 1 The main steps in a (noncompetitive) ELISA test. (1) The antibody to the antigen being quantitated is adsorbed onto a solid phase, usually polystyrene (2) The sample containing the antigen being measured is then added (3) Following the incubation and washing steps, a second enzyme-labeled antibody is then added. After further incubation and washing steps, enzyme substrate is added. (A substrate is chosen that will give a colored product) The amount of color produced is therefore proportional to the amount of antigen bound to the original antibody.

Next, a conjugate that will also bind to the antigen is added. Conjugates are antibody molecules to which an enzyme is covalently bound.

After addition of a chromogenic substrate for the enzyme, the intensity of the colored reaction products generated will be proportional to the amount of conjugated enzyme and thus indirectly to the amount of bound antigen molecules. Since the intensity of the developed color is proportional to the amount of antigen molecules present, determination of the intensity of the color produced by a standard series of antigens will allow the calculation of the amount of antigen in an unknown sample. This chapter describes the double antibody sandwich technique together with a method for preparing conjugates.

Materials

1. Microtiter plates, 96 wells, flat-bottomed polystyrene
2. Multichannel pipet: 0–250 μ L (8-channel), used for pipetting of all reagents.

3. Gilson or Finnpipettes: ranging from 0 to 250 μL used for pipeting of blanks, standards, and samples into wells.
4. Titertek multiscanphotometer or equivalent if available.
5. 0.1M carbonate buffer (pH 9.6). (0.1M Na_2CO_3 brought to pH 9.6 with NaOH.) This buffer should be made of the purest chemicals available and double-distilled H_2O .
6. Wash solution. 90 mL Tween 80 added to 910 mL H_2O .
7. BST: 0.2% (w/v) BSA (bovine serum albumin), 0.01% Tween 80 and 0.9% (w/v) NaCl in distilled water.
8. Substrate solutions. Depending on which enzyme is coupled to the conjugate (*see* Notes), different substrate solutions have to be used. The substrate solutions given here have to be used when horseradish peroxidase (HRP) is the enzyme coupled to the conjugate. There are two substrate solutions in use for HRP. One containing 5-amino-salicylic acid (purple-redbrown color) and one containing ortho-phenylene diamine (yellow color). Both compounds are light-sensitive and the solutions containing them must be freshly made and protected against light. Both substrate solutions are possibly carcinogenic.

Solution 1: Dissolve 80 mg 5-amino salicylic acid in 100 mL 0.05M potassium phosphate buffer (pH 6.0) containing 0.001M EDTA. Add 20 mL H_2O_2 (30%) and mix.

Solution 2 Mix: 24.3 mL 0.1M citric acid
25.7 mL 0.2M Na_2HPO_4
50 mL H_2O
40 mg ortho-phenylene diamine
40 μL H_2O_2 (30%)

In the case of solution 1, 0.3M NaOH is used to stop the reaction, when solution 2 is used the reaction is stopped with 1M H_2SO_4 .

9. A diluted solution of IgG against the antigen to be measured. Depending on the quality of the IgG (titer), the dilution is usually 1500–2500-fold. Dilutions are made in 0.1M carbonate buffer.

10. Antigen solutions to be tested and standard antigen solutions.
11. HRP-labeled diluted IgG solution As a rule the conjugate solution has to be diluted 500–2000 times in BST (but *see also* Notes section).

Method

The Double Antibody Sandwich Technique

1. Add 150 μL of the diluted IgG solution to each of the wells of a microtiter plate (or plates) using the 8-channel Titertek pipet. Cover the plate(s) and incubate for 16 h at room temperature.
2. Wash the plate(s). The wells are emptied by flicking the plate over a sink. Residual liquid is removed by “beating” the plate upside-down against a filterpaper. If kept covered and cool, coated plates can be kept for a substantial period of time, up to several months). An appropriate amount of wash solution is pipetted into the wells and left for a couple of minutes. The wells are then emptied as described above. This procedure is then repeated once. If a special washing device is used (i.e., a Titertek Microplate Washer or a homemade device that can be connected to a waterpump), the plates are washed three times for 15 s with the washing solution, after having been emptied each time as described above. Good washing is essential. One should rather wash for 20 than 10 s.
3. After washing add 100 μL BST to every well, using the multichannel pipet.
4. Add 100 μL of an antigen solution to be tested to the first well of each row. Mix carefully by taking the solution up and down with the multichannel pipet several times. Care should be taken to avoid air bubbles or splashing of small drops.
5. Take 100 μL from the first wells and transfer them to the second wells with the multichannel pipet. Repeat the mixing procedure. Take 100 μL from the second wells and add to the third and so on. In this way a

- two-fold dilution series from wells 1–12 is created. Finally, remove the 100 μL excess from the last wells.
6. Incubate the plate for 2 h at 37°C, to let the antigen bind to the coated antiserum, and then wash thoroughly
 7. Add 100 μL of the diluted conjugate solution to all wells and incubate for 2 h at 37°C, then wash the plate thoroughly.
 8. Add 100 μL of substrate solution to all wells and incubate for 1–2 h at 37°C in the dark.
 9. Stop the reaction with 100 μL of stop solution.
 10. Read the titer of the antigen solutions. This can be done by either using a Titertek multichannel photometer or by determining the last well that still gives some coloring observable with the naked eye. The incubations with the antigen and conjugate solutions should be done while the plates are continuously gently shaken. This is not absolutely necessary for the incubation with the substrate solution.

Preparation of a Conjugate for ELISA

As stated above, a conjugate is the covalent complex of IgG and an enzyme. In the procedure below, HRP is the enzyme that is coupled. See Notes section for other enzymes that can be used.

1. Dissolve 5 mg of horseradish peroxidase in 1 mL 0.3M Na_2CO_3 (pH 8.1). This solution should be prepared fresh.
2. Add 0.1 mL of 1% fluorodinitrobenzene in pure ethanol. (**N.B.** Fluorodinitrobenzene is suspected to be carcinogenic.) If the HRP used is not pure, a precipitate may be formed that must be removed by centrifugation (10 min, 18,000 rpm).
3. Mix thoroughly and incubate for 1 h at room temperature
4. Add 1 mL of 0.16M ethylene glycol, mix, and incubate for another hour at room temperature. The total volume is now 2.1 mL.

5. Dialyze the mixture against 0.01M sodium carbonate buffer (pH 9.5) for 25 h. The carbonate buffer should be changed at least three times.
6. Add IgG (*see below*) dissolved in 0.01M sodium carbonate buffer (pH 9.5) to the peroxidase-aldehyde solution in the following ratio: one volume IgG solution to one volume activated peroxidase-aldehyde, or 5 mg purified IgG (protein) to 3 mL peroxidase solution.
7. Mix well and incubate 2–3 h, but not longer at room temperature. In the case of precipitate formation, the precipitate should be removed by centrifugation (10 min, 10,000 rpm).
8. Dialyze extensively against 0.01M phosphate buffer (pH 7.2) containing 0.9% NaCl at 4°C. Store the conjugate in a refrigerator or freezer.
N.B. Repeated freezing and thawing of IgG and conjugate solutions diminishes the activity of the proteins. Thawed solutions that are not completely used can be very well kept at 4°C.

Preparation of a Purified IgG Fraction from Whole Serum

1. Take 100 mL serum and add 200 mL of 0.06M sodium acetate (pH 4.6). The pH of the mixture should be 4.8
2. Add 8.2 mL caprylic acid dropwise at room temperature (but *see also* Notes section).
3. Stir for 30 min and remove the precipitate (10 min, 10,000 rpm).
4. Dialyze the IgG fraction against 0.9% sodium chloride and lyophilize.

Notes

1. The amount of caprylic acid needed for purification of an IgG fraction varies from sera to sera as indicated in this table.

100 mL serum	Caprylic acid, mL
Rabbit	8.2
Horse	7.6
Human	7.6
Bovine	6.8
Goat	8.0

2. The procedure described above can be varied in many ways. For example, one can add a fixed amount of enzyme labeled antigen to the sample. This will then compete with the unlabeled antigen and from the degree of competition in regard to a standard antigen solution, the concentration of the antigen can be evaluated. In some cases, the antigen can be coated to the wall of the polystyrene plate and then incubated with the conjugate directly. In these cases then, there is no sandwich of the antigen. The principle of the method however, remains the same.
3. Alkaline phosphatase is the other enzyme used in ELISA tests to prepare conjugates. In this case the substrate is *p*-nitrophenylphosphate that the enzyme converts to the yellow *p*-nitrophenol that is measured at 405 nm.
4. Occasionally an ELISA test will fail to give the desired result and in all wells the same amount of colour will develop. In most cases this problem can be overcome by preparing fresh solutions.

References

- 1 Engvall, E., and Perlmann, P. (1971) *Immunochemistry* 8, 871-874

